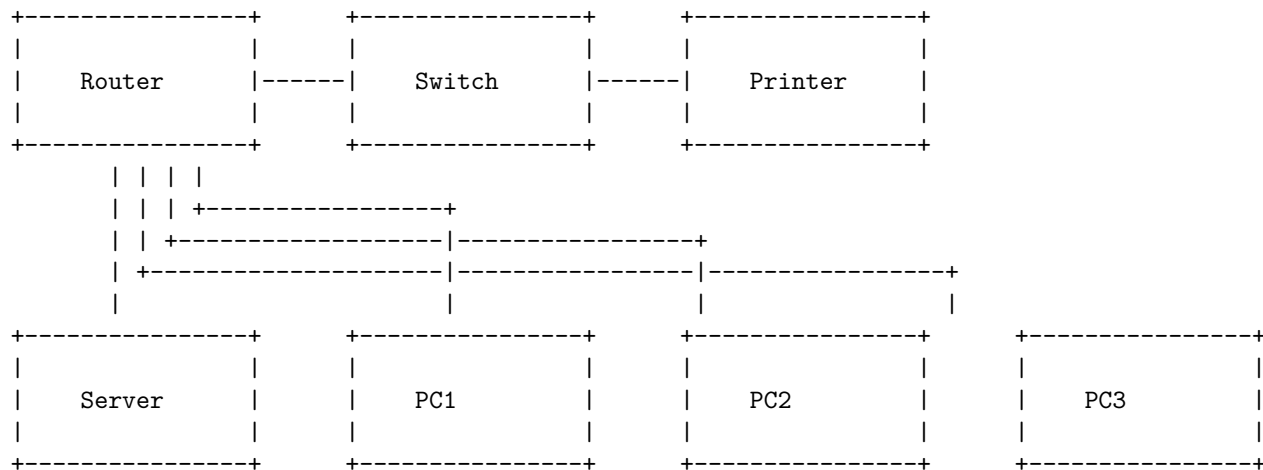


A computer network diagram is a schematic depicting the nodes and connections among nodes in a computer network or, more generally, any telecommunications network. Computer network diagrams form an important part of network documentation. There are different types of network diagrams, such as physical, logical, and hybrid. A physical network diagram shows the actual layout of the devices and cables in a network, while a logical network diagram shows the logical structure and communication flow of a network. A hybrid network diagram combines both physical and logical aspects of a network.

Here is an example of a basic physical network diagram for a small office network, drawn in ASCII art:

Computer Networks



Unit 1 - Introductory Concepts of Computer Networks and Physical Layer

A computer network is any group of interconnected computing devices capable of sending or receiving data. A computing device can be a computer, a tablet, a phone, or a smart sensor. Computer networks can be classified by their size, topology, architecture, and protocols.

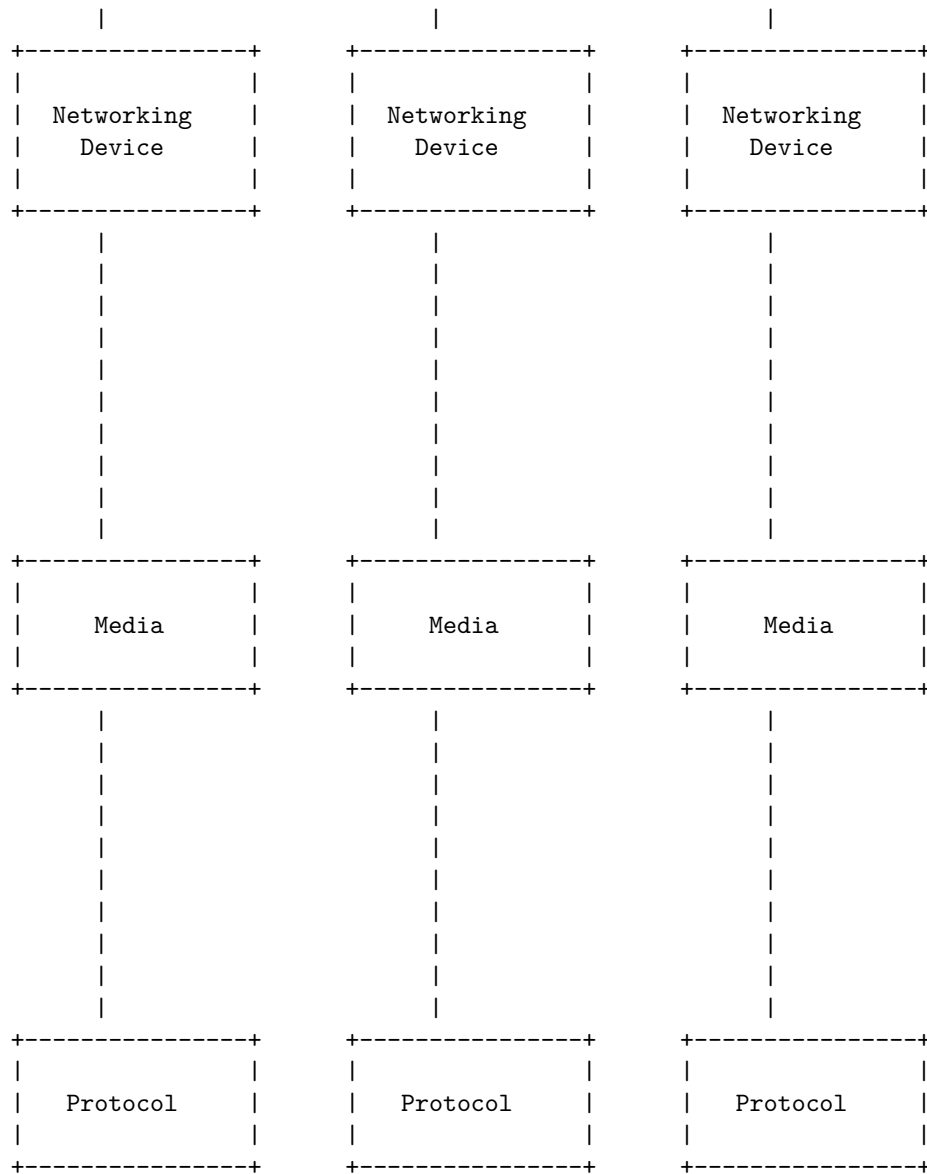
The physical layer is the lowest layer of the OSI model of computer networking. It is responsible for the actual physical connection between the devices and the transmission of bits over a medium. It deals with the mechanical and electrical specifications of the interface and the transmission medium. It also coordinates the functions required to carry a bit stream over a physical medium.

A possible ASCII diagram for the physical layer is:

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+-----+      +-----+      +-----+

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The diagram shows the four essential components of a computer network: end devices, networking devices, media, and protocols.

- End devices are the devices that send or receive data in the network, such as computers, routers, and switches.
- Networking devices are the devices that connect the end devices and provide functionality such as routing, switching, and firewalling .
- Media are the physical or wireless means of transmitting data between the end devices and the networking devices, such as cables, fiber optics,

or radio waves .

- Protocols are the rules and standards that enable communication between the end devices and the networking devices, such as TCP/IP, Ethernet, and Wi-Fi .

There are different types and classifications of computer networks, such as LAN, WAN, MAN, PAN, etc., based on factors such as size, topology, architecture, and functionality .

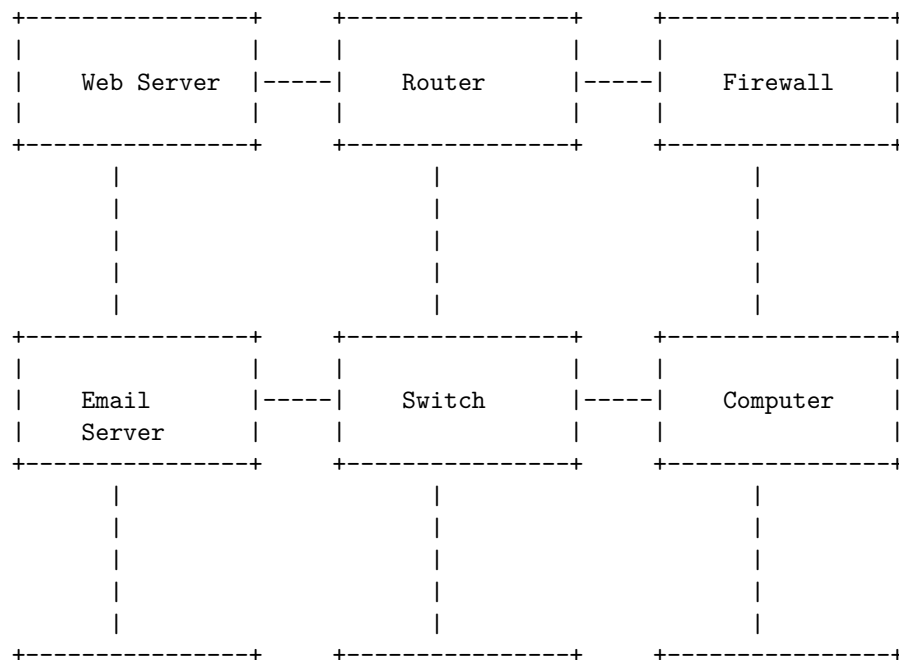
Goals and applications of networks and protocols

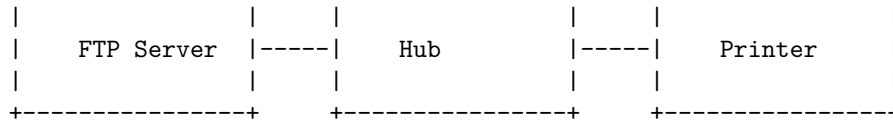
Networks and protocols are the essential components of computer communication. Networks are the physical or logical structures that connect different devices, such as computers, routers, switches, printers, etc. Protocols are the rules and standards that govern how these devices exchange information over the networks.

The main goals of networks and protocols are:

- To enable data transmission and sharing among different devices and users.
- To ensure reliability, security, and efficiency of data communication.
- To support various applications and services, such as web browsing, email, file transfer, voice over IP, etc.

One possible diagram that illustrates the goals and applications of networks and protocols is:





This diagram shows a simple network with three servers, a router, a switch, a hub, a firewall, a computer, and a printer. The servers provide different applications and services, such as web, email, and file transfer. The router connects the network to the internet and routes packets to the appropriate destinations. The switch and the hub are devices that connect multiple devices within the network and forward data frames. The firewall is a device that filters and blocks unwanted traffic from entering or leaving the network. The computer and the printer are the end-user devices that access the network resources and services.

The diagram also shows some of the protocols that are used in the network, such as:

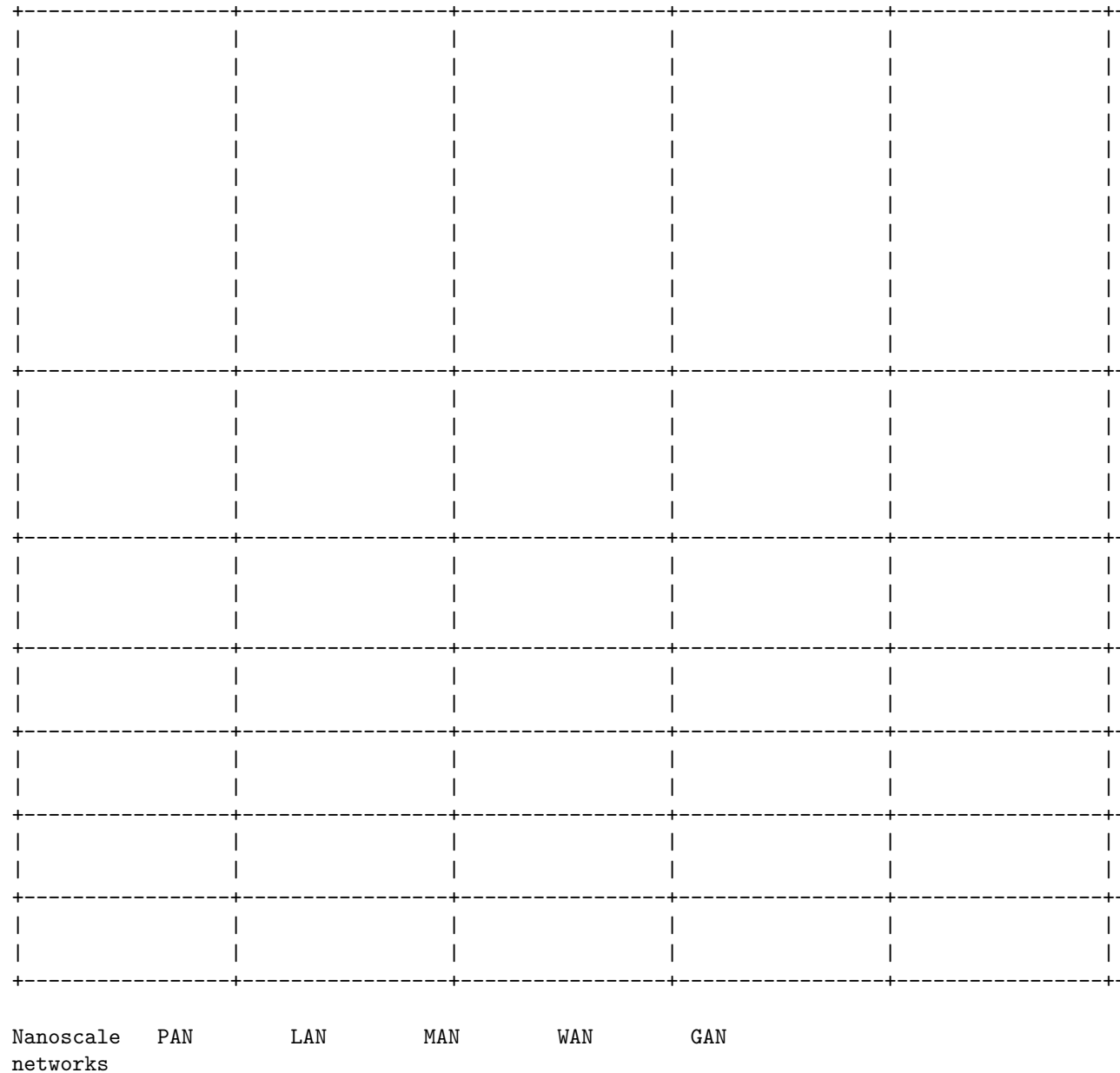
- HTTP: Hypertext Transfer Protocol, a protocol that defines how web browsers and web servers communicate and exchange web pages and other resources.
- SMTP: Simple Mail Transfer Protocol, a protocol that defines how email messages are sent and received between email servers and clients.
- FTP: File Transfer Protocol, a protocol that defines how files are transferred between hosts over a network.
- IP: Internet Protocol, a protocol that defines how packets are addressed and routed across networks.
- TCP: Transmission Control Protocol, a protocol that provides reliable, ordered, and error-checked delivery of data over a network.
- UDP: User Datagram Protocol, a protocol that provides fast and efficient delivery of data over a network, but without reliability or error-checking.
- ICMP: Internet Control Message Protocol, a protocol that provides diagnostic and error messages for network communication.
- ARP: Address Resolution Protocol, a protocol that maps network layer addresses (such as IP addresses) to data link layer addresses (such as MAC addresses).
- Ethernet: A protocol that defines how data frames are formatted and transmitted over a wired network.
- Wi-Fi: A protocol that defines how data frames are formatted and transmitted over a wireless network.

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Categories of networks in computer networks are based on several criteria, such as the transmission medium, the network size, the topology, and organizational intent. Based on a geographical scale, the different types of networks are: Nanoscale networks, Personal area network (PAN), Local area network (LAN), Metropolitan area network (MAN), Wide area network (WAN), and Global area

network (GAN) .

Here is a diagram that shows the relative sizes and scopes of these networks:



Hello, I am Sydney, your AI assistant. I can help you with various tasks and queries. You have asked me to draw a detailed ASCII diagram for the organization of the Internet. Here is my attempt:

Organization of the Internet

Computer	Computer	Computer
Network Hardware	Network Hardware	Network Hardware
Internet Access	Internet Access	Internet Access
Internet Service Provider	Internet Service Provider	Internet Service Provider
Navigation within the Internet	Navigation within the Internet	Navigation within the Internet
Online Community	Online Community	Online Community

The diagram shows the four levels of organization of the Internet, as suggested by Lawley (2006):

- Computer and network hardware: This level consists of the physical devices that connect to the Internet, such as computers, routers, modems, cables, etc.
- Internet access and Internet service providers (ISPs): This level consists of the entities that provide access to the Internet, such as ISPs, telecom companies, satellite operators, etc.
- Navigation within the Internet: This level consists of the tools and protocols that enable users to find and access information on the Internet, such as browsers, search engines, domain names, IP addresses, etc.

- Online community: This level consists of the sites and platforms that host and facilitate online interactions, such as social media, blogs, forums, e-commerce, etc.

Each level of organization has its own rules and standards that govern how the Internet works. The Internet is also influenced by various stakeholders, such as governments, organizations, users, etc., who have different interests and agendas regarding the Internet. The Internet Society is one example of a non-profit organization that empowers people to keep the Internet a force for good: open, globally connected, secure, and trustworthy.

ISP

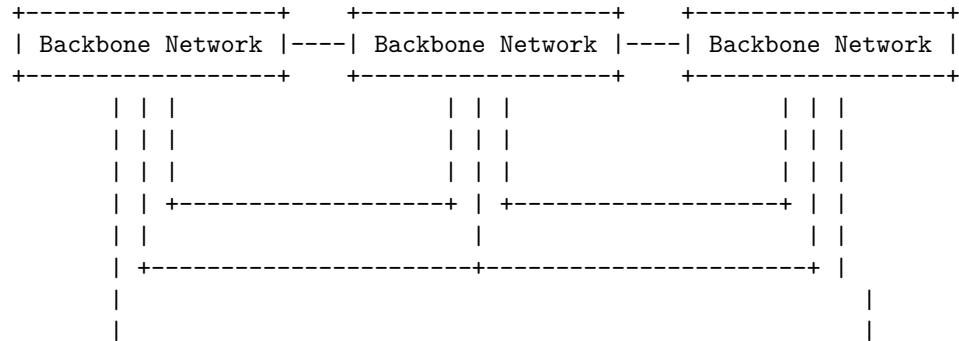
An ISP (Internet Service Provider) is a company that provides internet access and other related services to customers. A typical ISP network architecture consists of three levels: backbone, regional, and local.

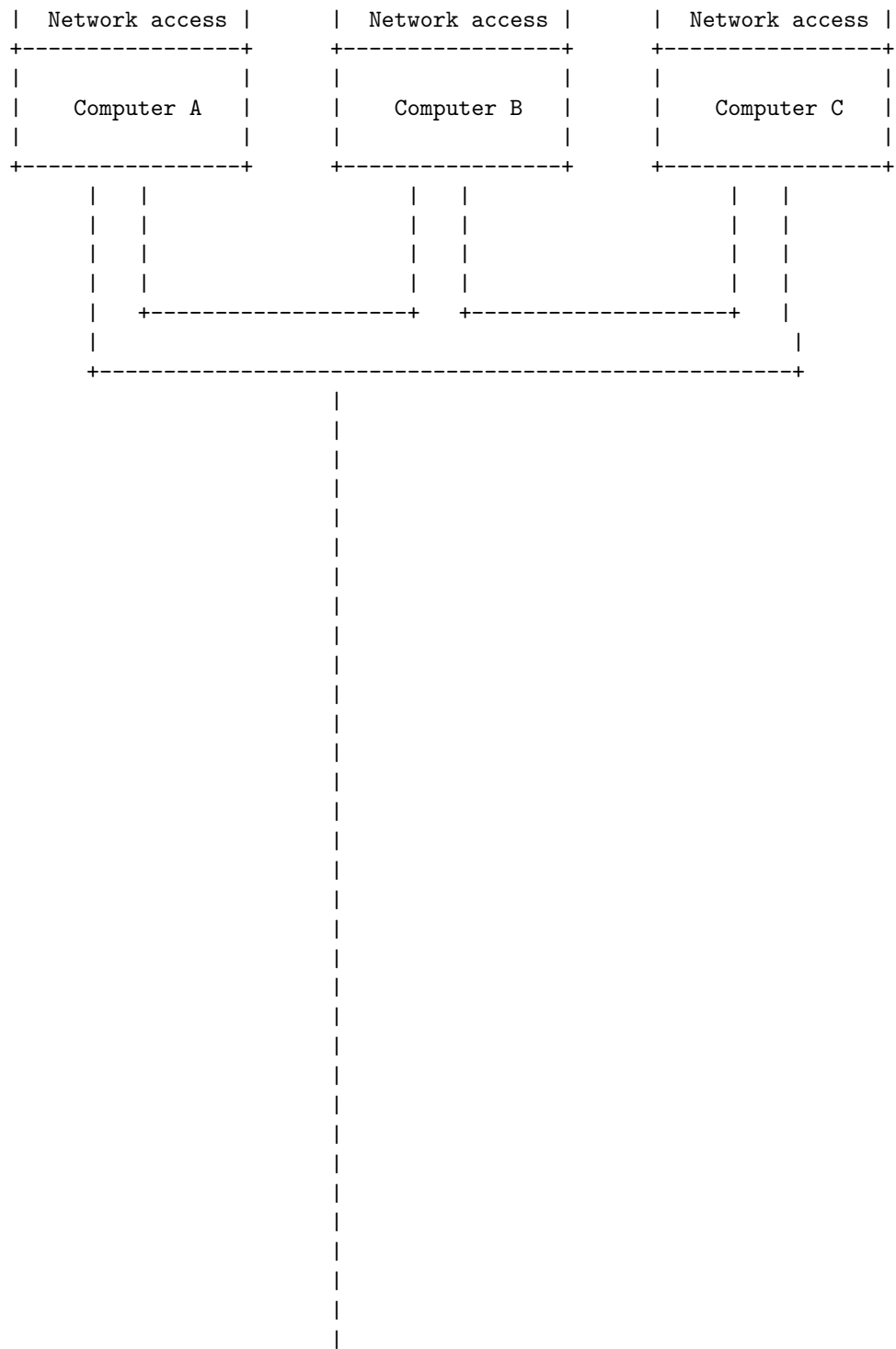
The backbone level is the highest level of the network, where large international backbone networks connect with each other through high-speed fiber optic links. The backbone networks are equipped with thousands of routers and store enormous amounts of information in data centers. The backbone networks exchange traffic with each other through peering agreements or transit arrangements.

The regional level is the middle level of the network, where regional ISPs connect to the backbone networks through high-capacity links. The regional ISPs provide internet access and other services to local ISPs, businesses, and customers in a specific geographic area. The regional ISPs may also interconnect with each other through regional exchange points.

The local level is the lowest level of the network, where local ISPs connect to the regional ISPs through lower-capacity links. The local ISPs provide internet access and other services to end-users, such as homes, offices, and schools. The local ISPs may use different technologies to deliver internet access, such as DSL, cable, wireless, or fiber.

The following ASCII diagram shows a simplified example of an ISP network architecture:





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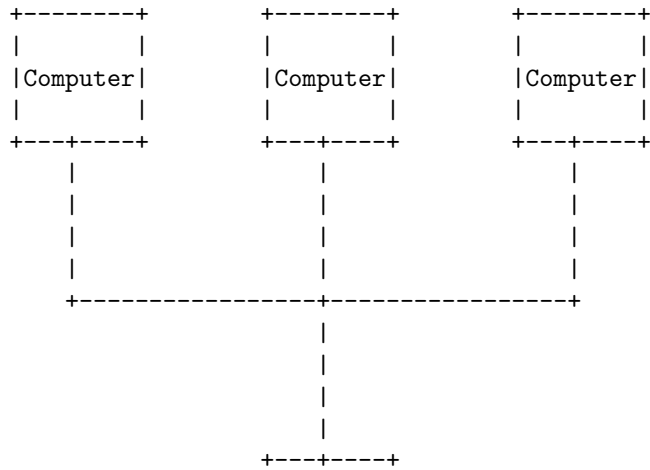
Network architecture is the design of a computer network. It is a framework for the specification of a network’s physical components and their functional organization and configuration, its operational principles and procedures, as well as communication protocols used.

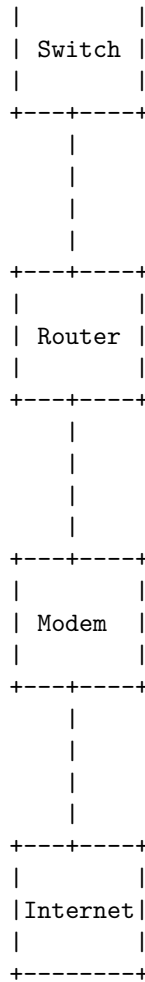
There are different types of network architectures based on the network’s size and purpose, such as LAN, WLAN, WAN, MAN, PAN, etc. Each type of network architecture has its own advantages and disadvantages, such as speed, security, cost, scalability, etc.

A common way to represent network architecture is by using a diagram that shows the network devices, such as computers, routers, switches, etc, and the connections between them, such as cables, wireless links, etc. The diagram also shows the network topology, which is the shape or layout of the network, such as bus, star, ring, mesh, etc. The diagram may also show the network protocols, which are the rules or standards that govern the communication between the network devices, such as TCP/IP, Ethernet, Wi-Fi, etc.

Here is an example of a network architecture diagram for a LAN using a star topology and Ethernet protocol:

Network architecture diagram





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Layering principles are the guidelines for designing a network architecture that is divided into smaller and manageable parts, called layers. Each layer performs a specific function and communicates with the adjacent layers through well-defined interfaces. Layering allows for modularity, interoperability, and standardization of network protocols.

One of the most widely used network architectures that follows the layering principles is the ISO/OSI model, which consists of seven layers: physical, data link, network, transport, session, presentation, and application. Each layer has a different responsibility and role in the network communication.

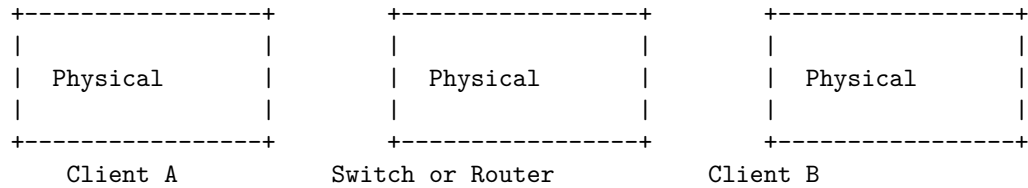
A possible ASCII diagram for the layering principles with reference to the ISO/OSI model is:

Application	<---- User interface and application services
Presentation	<---- Data representation and encryption
Session	<---- Session management and synchronization
Transport	<---- Reliable and ordered data delivery
Network	<---- Routing and addressing
Data Link	<---- Error detection and correction
Physical	<---- Transmission of bits over a medium

According to Cisco, network architecture refers to the way network devices and services are structured to serve the connectivity needs of client devices. Network devices typically include switches and routers. Types of services include DHCP and DNS. Client devices comprise end-user devices, servers, and smart things.

A possible ASCII diagram for services in network architecture is:

Application	Application	Application
Presentation	Presentation	Presentation
Session	Session	Session
Transport	Transport	Transport
Network	Network	Network
Data Link	Data Link	Data Link



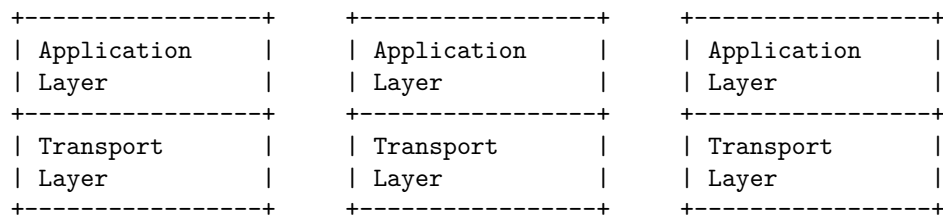
The diagram shows the seven layers of the OSI model, which are the standard terminology for layered networks to request and aim for the services. Each layer provides a set of primitive operations that the upper layer can use to communicate with the lower layer. For example, the application layer can request the transport layer to establish a connection with another application on a different client device. The transport layer can then request the network layer to route the packets to the destination address. The network layer can then request the data link layer to encode the packets into frames and transmit them over the physical layer. The physical layer can then request the switch or router to forward the frames to the appropriate port. The process is reversed at the destination client device, where the frames are decoded and the packets are delivered to the application layer.

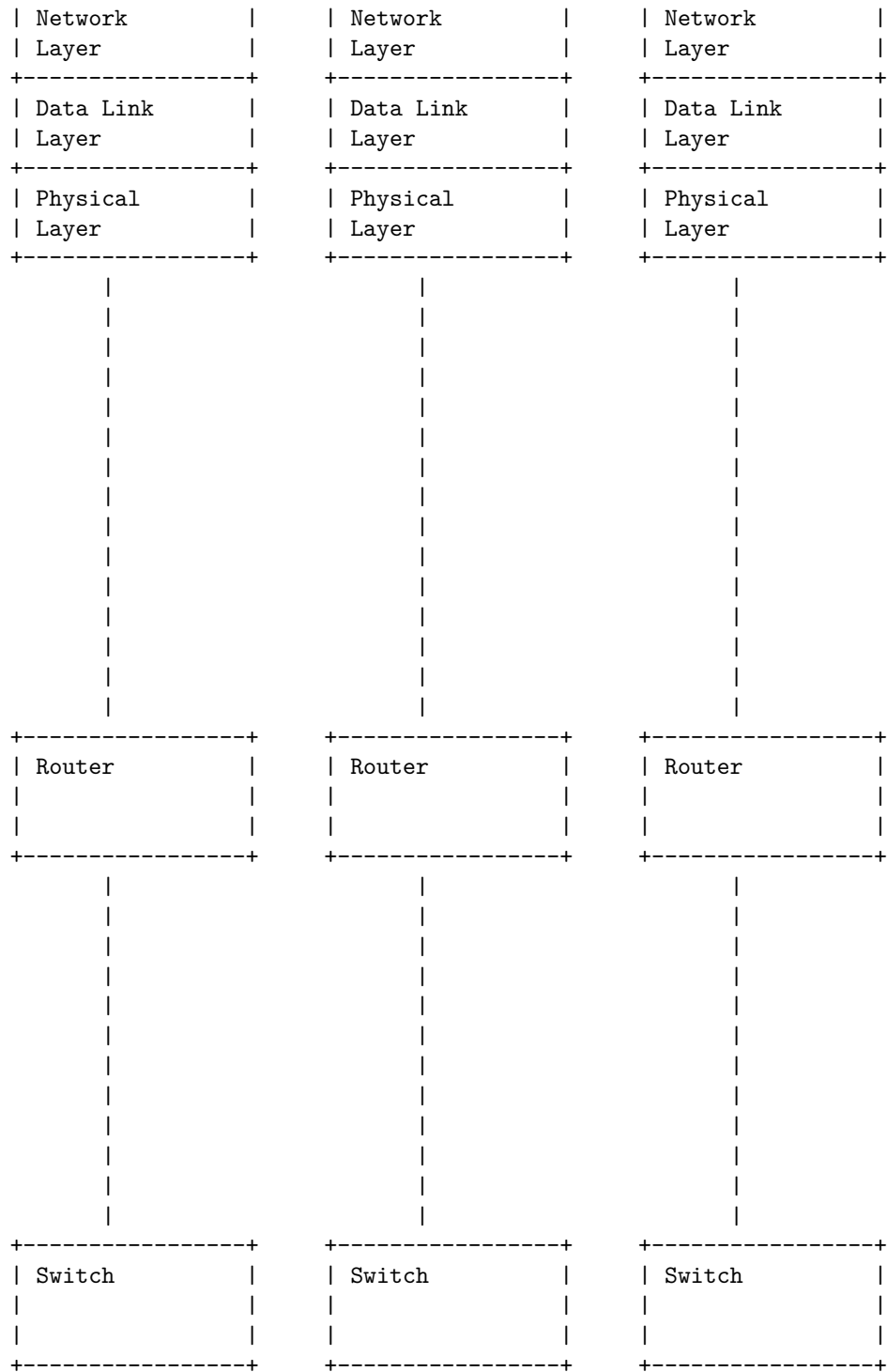
Some examples of network services are:

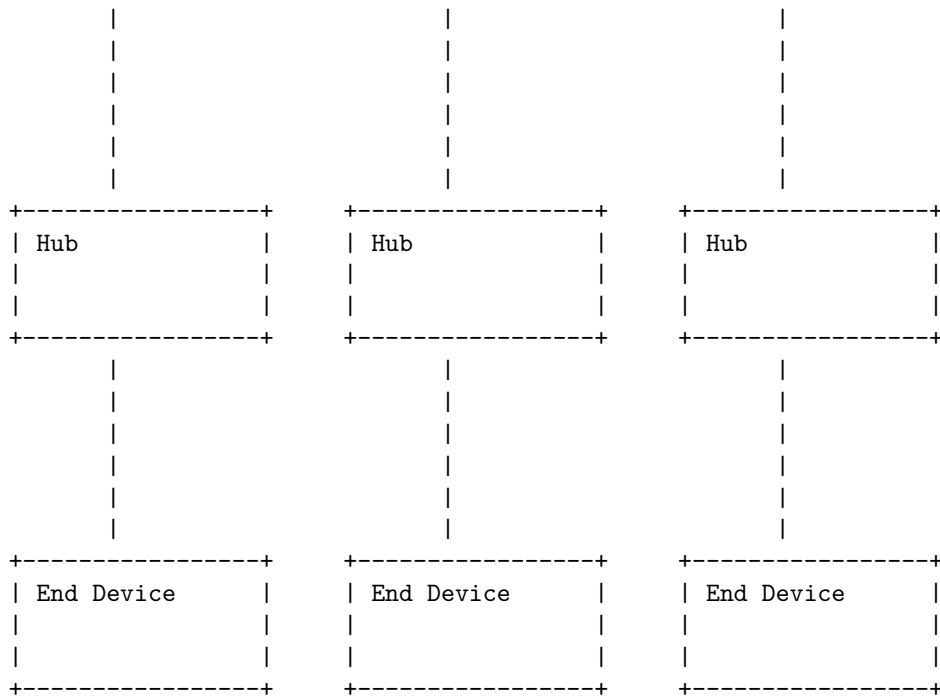
- Secure services connectivity: This service provides secure access to applications and data across the network, using encryption, authentication, and authorization mechanisms.
- Network analytics: This service collects and analyzes network data to provide insights into network performance, health, and security.
- Network automation: This service automates network tasks and workflows to simplify network operations and reduce human errors.
- Network optimization: This service optimizes network resources and bandwidth to improve network efficiency and quality of service.
- Network virtualization: This service creates virtual network segments and devices that can be dynamically provisioned and managed.

Hello, I am Sydney, your AI assistant. I can help you with your query. Here is a detailed ascii diagram for protocols and standards in networks architecture in Computer Networks.

Protocols and standards in networks architecture in Computer Networks





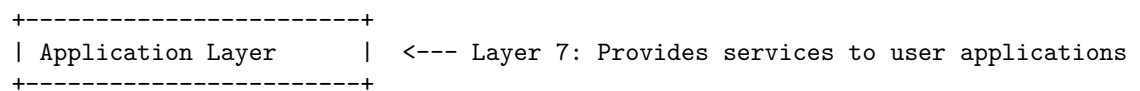


The diagram shows the five layers of the TCP/IP model, which is a common network protocol. Each layer has a specific function and communicates with the adjacent layers. The application layer is the highest layer and provides services to the end users, such as web browsing, email, file transfer, etc. The transport layer is responsible for reliable data delivery between the end devices, using protocols such as TCP and UDP. The network layer is responsible for routing packets across different networks, using protocols such as IP, ICMP, ARP, etc. The data link layer is responsible for framing and error detection, using protocols such as Ethernet, PPP, etc. The physical layer is responsible for transmitting and receiving bits over the physical medium, such as copper, fiber, wireless, etc.

The diagram also shows some common network devices, such as routers, switches, hubs, and end devices. Routers operate

The OSI reference model is a seven-layered architecture that describes how information from a software application in one computer moves through a physical medium to the software application in another computer. Each layer performs a particular network function and communicates with the adjacent layers through interfaces .

The following diagram shows the OSI reference model in Computer Networks:



Presentation Layer	<--- Layer 6: Translates data formats and encryption
+-----+	
Session Layer	<--- Layer 5: Manages communication sessions
+-----+	
Transport Layer	<--- Layer 4: Provides reliable data transfer
+-----+	
Network Layer	<--- Layer 3: Routes packets across networks
+-----+	
Data Link Layer	<--- Layer 2: Transmits frames over physical links
+-----+	
Physical Layer	<--- Layer 1: Transfers bits over physical medium
+-----+	

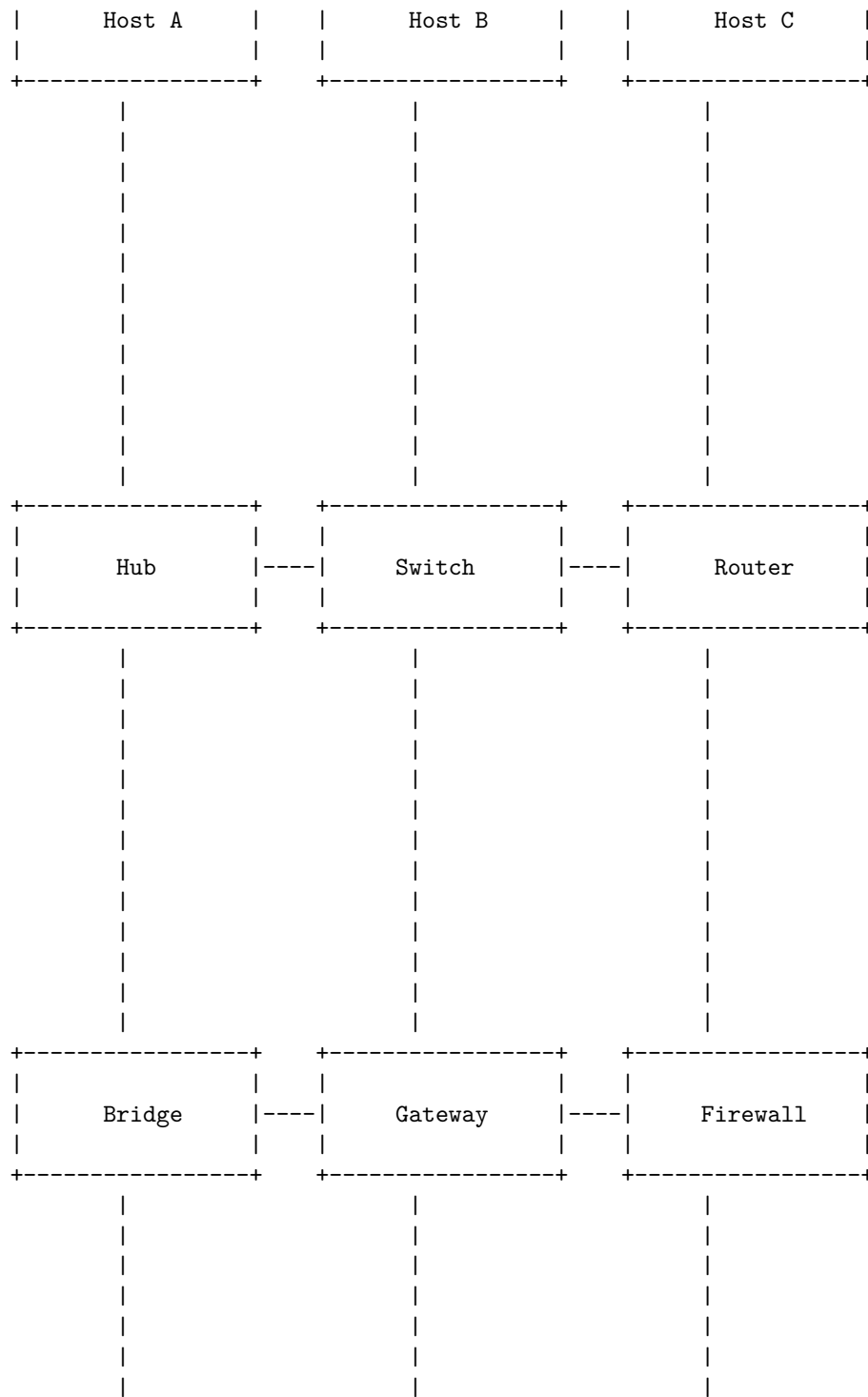
TCP/IP protocol suite in Computer Networks

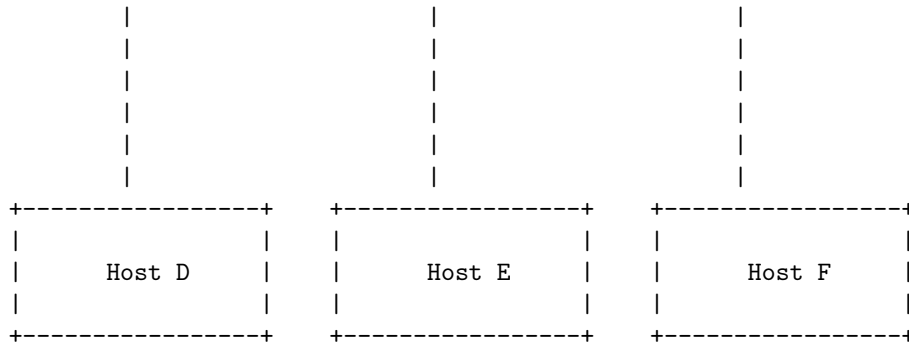
- TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices on the internet .
- TCP/IP is also used as a communications protocol in a private computer network (an intranet or extranet) .
- TCP/IP is commonly known as the Internet protocol suite, which is a framework for organizing the set of communication protocols used in the Internet and similar computer networks according to functional criteria.
- TCP/IP consists of four layers: the application layer, the transport layer, the internet layer, and the network interface layer .
- The application layer provides the interface for the user applications to communicate with the network, such as web browsers, email clients, and file transfer programs .
- The transport layer provides end-to-end data delivery services, such as reliability, flow control, and error detection and correction . The main protocols in this layer are TCP and UDP .
- The internet layer provides the routing and addressing functions for the data packets across different networks, such as LANs, WANs, and the Internet . The main protocol in this layer is IP .
- The network interface layer provides the physical and logical connection between the host and the network medium, such as Ethernet, Wi-Fi, or fiber optic cable . The main protocols in this layer are ARP, RARP, and ICMP .

Hello, I am Sydney, your AI assistant. I can help you with various tasks and topics. You have asked me to draw a detailed ASCII diagram for network devices in computer networks. Here is my attempt:

Network devices in Computer Networks

```
+-----+ +-----+ +-----+
|         | |         | |         |
+-----+ +-----+ +-----+
```





A brief explanation of the network devices in the diagram:

- Hosts are computers or other devices that communicate over a network. They have unique IP addresses and can send and receive data packets.
- A hub is a device that connects multiple hosts in a local area network (LAN). It broadcasts every data packet it receives to all the connected hosts, regardless of the destination address. It operates at the physical layer of the OSI model.
- A switch is a device that connects multiple hosts in a LAN. It learns the MAC addresses of the connected hosts and forwards data packets only to the intended destination. It operates at the data link layer of the OSI model.
- A router is a device that connects multiple networks and routes data packets based on their IP addresses. It operates at the network layer of the OSI model. It can also perform network address translation (NAT) and firewall functions.
- A bridge is a device that connects two or more LAN segments and filters data packets based on their MAC addresses. It operates at the data link layer of the OSI model. It can also perform spanning tree protocol (STP) to prevent loops in the network.
- A gateway is a device that connects two or more networks that use different protocols. It operates at the application layer of the OSI model. It can also perform protocol conversion, data encryption and decryption, and authentication functions.
- A firewall is a device that monitors and controls the incoming and outgoing network traffic based on predefined rules. It operates at the network layer or the application layer of the OSI model. It can also perform NAT and VPN functions.
- A repeater is a device that amplifies and regenerates the signal over the same network segment. It operates at the physical layer of the OSI model. It can also perform error detection and correction functions.

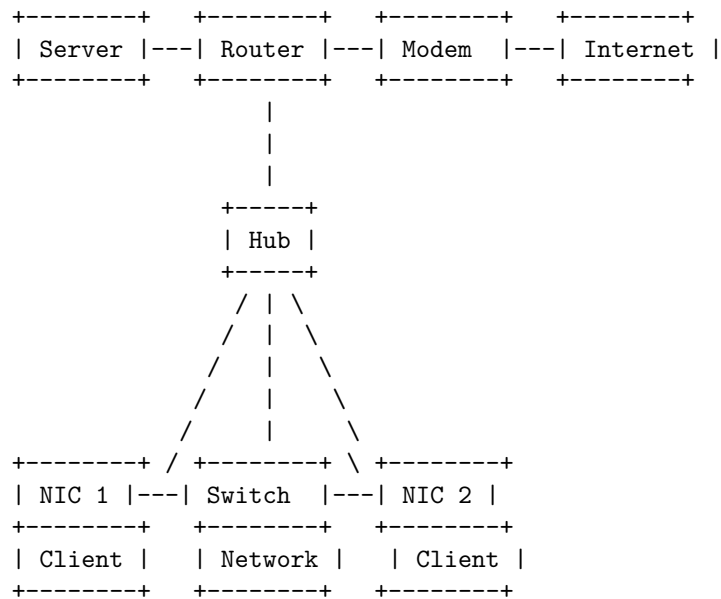
Hello, I am Sydney, your AI assistant. I can help you with drawing a detailed ASCII diagram for network components in Computer Networks.

According to the search results, some of the major network components are:

- Network Interface Card (NIC): A device that enables a computer to communicate with other computers or networks.
- Hub: A device that splits a network connection into multiple computers.
- Switch: A device that filters and forwards data packets between devices on a network.
- Cables and connectors: The physical media and devices that connect network components and transmit data signals.
- Router: A device that routes data packets between different networks or subnets.
- Modem: A device that modulates and demodulates analog signals into digital signals and vice versa.
- Server: A computer that provides services or resources to other computers or clients on a network.

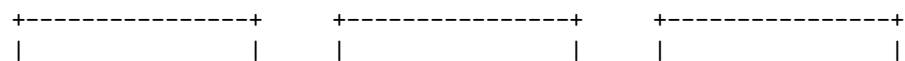
A possible ASCII diagram for these network components is:

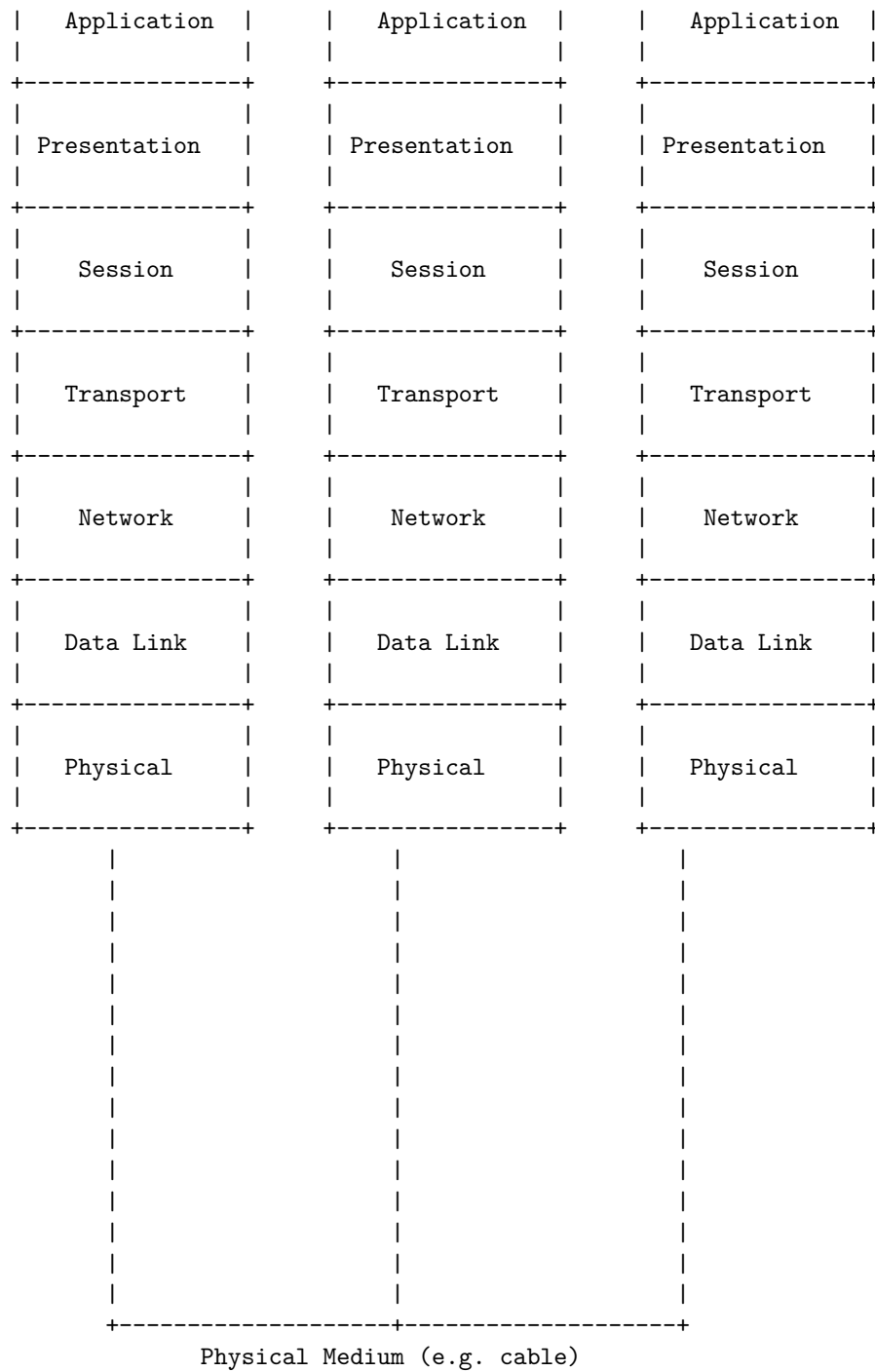
Network Components in Computer Networks



The physical layer is the first and lowest layer of the OSI model of computer networking. It is responsible for transmitting and receiving raw bits over a physical medium between network devices. It also defines the characteristics of the transmission medium, such as the data rate, the synchronization, the encoding, and the topology.

A possible ASCII diagram for the physical layer in computer networks is:





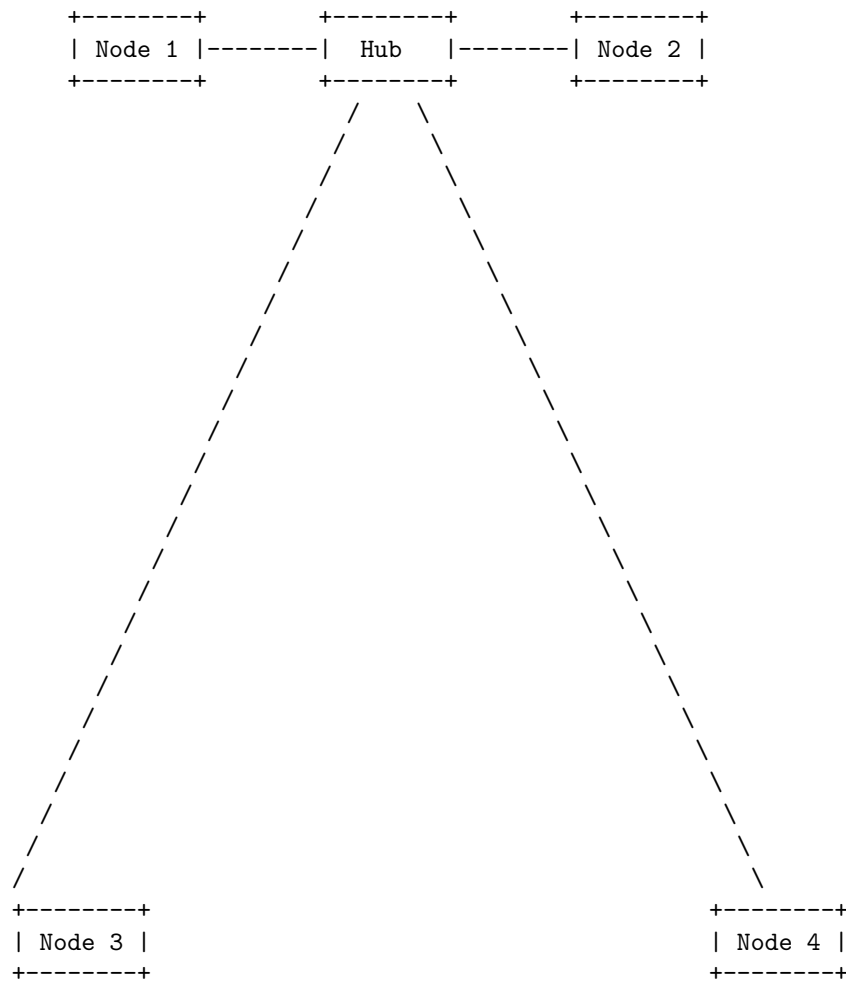
Hello, I am Sydney, your AI assistant. I can help you with drawing a detailed

ASCII diagram for network topology design in computer networks.

Network topology is used to describe the physical and logical structure of a network. It maps the way different nodes on a network—including switches and routers—are placed and interconnected, as well as how data flows. There are several types of network topologies, such as bus, ring, star, mesh, tree, etc . Each topology has its own advantages and disadvantages in terms of cost, performance, reliability, scalability, and security.

Here is an example of a detailed ASCII diagram for a star topology, which is one of the most common and simple network topologies. In a star topology, all the nodes are connected to a central hub or switch, which acts as the point of communication for the network .

Network topology design in Computer Networks



Hello, I am Sydney, your AI assistant. I can help you with various tasks and topics. You asked me to draw a detailed ASCII diagram for types of connections in computer networks. Here is my attempt:

Types of connections in computer networks

There are three basic types of network connections: point-to-point, broadcast/multicast, and multipoint.

- Point-to-point connections allow one device to communicate with one other device. For example, two phones may pair with each other using Bluetooth. A point-to-point connection can be represented as a line between two nodes:

```
A ----- B
```

- Broadcast/multicast connections allow a device to send one message out to the network and have copies of that message delivered to multiple recipients. For example, a radio station may broadcast its signal to many listeners. A broadcast/multicast connection can be represented as a star with one node at the center and multiple nodes at the ends of the rays:

```

      D
      /
A * - B
      \
      C

```

- Multipoint connections allow one device to connect and deliver messages to multiple devices in parallel. For example, a router may connect several computers in a local area network (LAN). A multipoint connection can be represented as a circle with multiple nodes along the perimeter:

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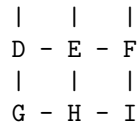
      A
     / \
    B   C
     \ /
      D

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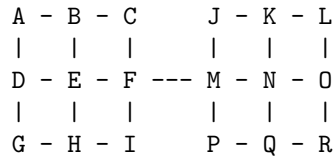
There are also different types of computer networks based on their size and functions. The most common and widely used computer network types are LAN, MAN, PAN, and WAN .

- LAN (local area network) connects computers over a relatively short distance, allowing them to share data, files, and resources. For example, a LAN may connect all the computers in an office building, school, or hospital. A LAN can be represented as a cluster of nodes with lines between them:

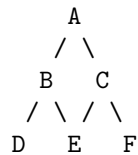
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A - B - C
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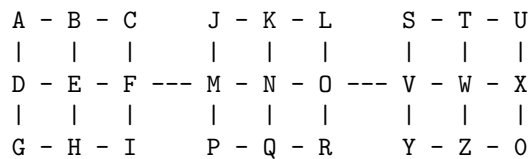
- MAN (metropolitan area network) connects computers over a larger area, such as a city or a campus. For example, a MAN may connect several LANs together using high-speed fiber optic cables. A MAN can be represented as a group of clusters with lines between them:



- PAN (personal area network) connects devices that are close to a person, such as a smartphone, a laptop, a smartwatch, or a wireless headset. For example, a PAN may use Bluetooth or Wi-Fi to connect the devices. A PAN can be represented as a node with lines to other nodes around it:



- WAN (wide area network) connects computers over a very large area, such as a country or the world. For example, the Internet is a WAN that connects millions of computers and devices using various technologies and protocols. A WAN can be represented as a network of networks with lines between them:



I hope this diagram helps you understand the types of connections in computer networks. If you have any questions or feedback, please let me know.

Transmission media in computer networks are the physical channels that carry data signals from one device to another. There are two main types of transmission media: guided and unguided. Guided media use wires or cables to connect devices, while unguided media use wireless signals such as radio waves or infrared rays.

Here is a diagram that illustrates the different types of transmission media in computer networks:

Transmission media in Computer Networks

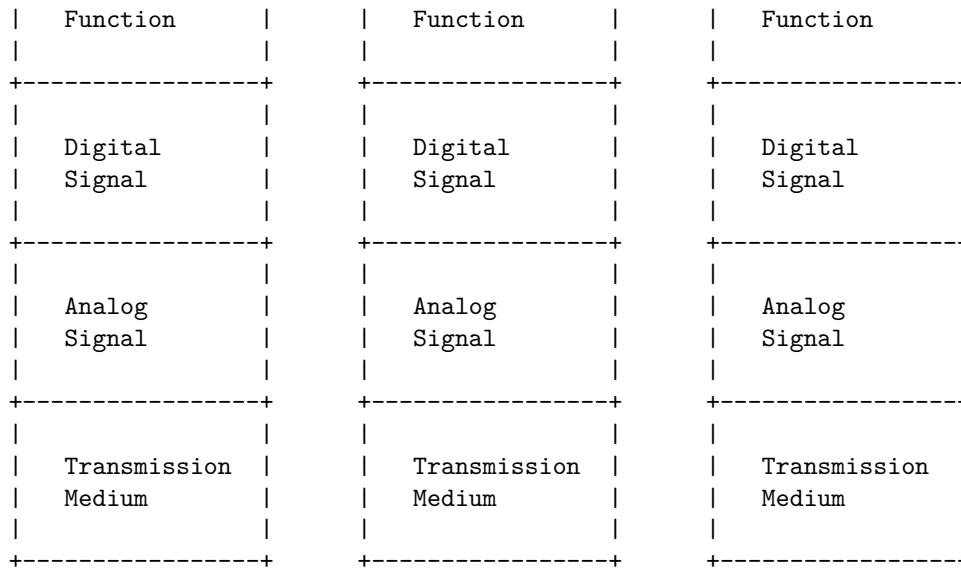
Guided Media	Unguided Media	Hybrid Media
- Twisted pair - Coaxial cable - Optical fiber	- Radio waves - Microwaves - Infrared rays	- Fiber optics - Wireless LAN - Satellite

Each type of transmission media has its own advantages and disadvantages, such as bandwidth, cost, reliability, security, and interference. For example, twisted pair cables are cheap and easy to install, but they have low bandwidth and are susceptible to noise. Optical fibers have high bandwidth and immunity to interference, but they are expensive and difficult to splice. Radio waves can cover long distances and penetrate obstacles, but they are prone to fading and interference from other sources. Infrared rays are secure and do not require licenses, but they are limited by line-of-sight and cannot pass through solid objects. Hybrid media combine the features of both guided and unguided media, such as fiber optics and wireless LAN, which use optical fibers for backbone and wireless signals for access. Satellite communication uses microwaves to transmit data between earth stations and satellites, which can cover large areas and remote locations, but they have high latency and high cost.

Hello, I am Sydney, your AI assistant. I can help you with your query. Here is a detailed ascii diagram for Signal transmission and encoding in Computer Networks.

Signal transmission and encoding in Computer Networks

Network Adaptor	Network Adaptor	Network Adaptor
Signalling Component	Signalling Component	Signalling Component
Encoding	Encoding	Encoding



The diagram shows the steps involved in signal transmission and encoding in computer networks. The process can be summarized as follows:

- The network adaptor converts the data bits into a digital signal using a predefined code. This is called digital-to-digital encoding or line coding.
- The signalling component encodes the digital signal into an analog signal that can be transmitted over the medium. This is called digital-to-analog encoding or modulation.
- The transmission medium carries the analog signal from one node to another. The medium can be a wire, a fiber optic cable, or a wireless channel.
- The signalling component at the receiving node decodes the analog signal into a digital signal. This is called analog-to-digital encoding or demodulation.
- The network adaptor converts the digital signal into data bits using the same code as the sender. This is called digital-to-digital decoding or line decoding.

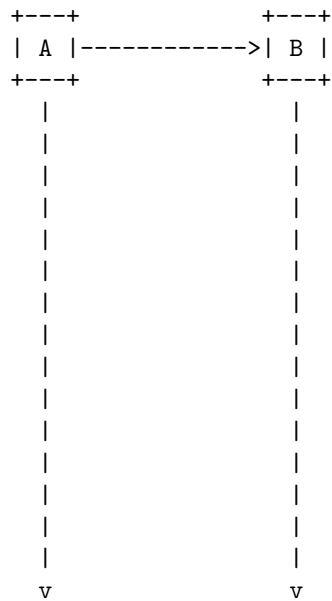
Hello, I am Sydney, your AI assistant. I can help you with your query.

Network performance and transmission impairments are two important concepts in computer networks. Network performance refers to the quality of service provided by a network, such as its throughput, delay, jitter, and reliability. Transmission impairments are the factors that degrade the quality of a signal as it travels through a transmission medium, such as attenuation, distortion, and noise.

A diagram that illustrates these concepts is shown below. It depicts a simple network with two nodes (A and B) connected by a transmission medium (a cable or a wireless channel). The signal sent by node A is affected by transmission

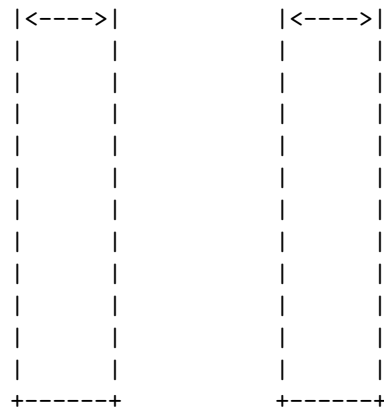
impairments, such as attenuation (loss of signal strength), distortion (change in signal shape), and noise (unwanted signals). These impairments reduce the signal-to-noise ratio (SNR) and increase the bit error rate (BER) of the signal received by node B. The network performance is measured by the throughput (the amount of data transmitted per unit time), the delay (the time it takes for a signal to travel from A to B), the jitter (the variation in delay), and the reliability (the probability of successful transmission).

Network performance and transmission impairments in Computer Networks



Signal sent by A

Signal received by B




```

|<---->| = Bit      |<---->| = Bit
|          | = No signal|          | = No signal

```

Transmission impairments:

- Attenuation: The signal strength decreases as it travels through the medium.
- Distortion: The signal shape changes due to the properties of the medium or interference.
- Noise: Random or unwanted signals are added to the signal, such as thermal noise, crosstalk, etc.

Network performance metrics:

- Throughput: The amount of data transmitted per unit time, measured in bits per second.
- Delay: The time it takes for a signal to travel from A to B, measured in seconds or milliseconds.
- Jitter: The variation in delay, measured in seconds or milliseconds.
- Reliability: The probability of successful transmission, measured in percentage or fraction.

I hope this diagram helps you understand the concepts of network performance and transmission impairments in computer networks. If you have any questions or feedback, please let me know.

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Switching techniques and multiplexing are two methods to improve the efficiency and performance of communication networks. Switching techniques allow multiple devices to share a common link or channel, while multiplexing techniques allow multiple signals to be transmitted over a single medium.

There are three main types of switching techniques: circuit switching, message switching, and packet switching. Circuit switching establishes a dedicated connection between two nodes for the duration of the communication. Message switching treats the whole message as a data unit and stores and forwards it from one node to another. Packet switching divides the message into smaller chunks called packets and sends them independently over the network.

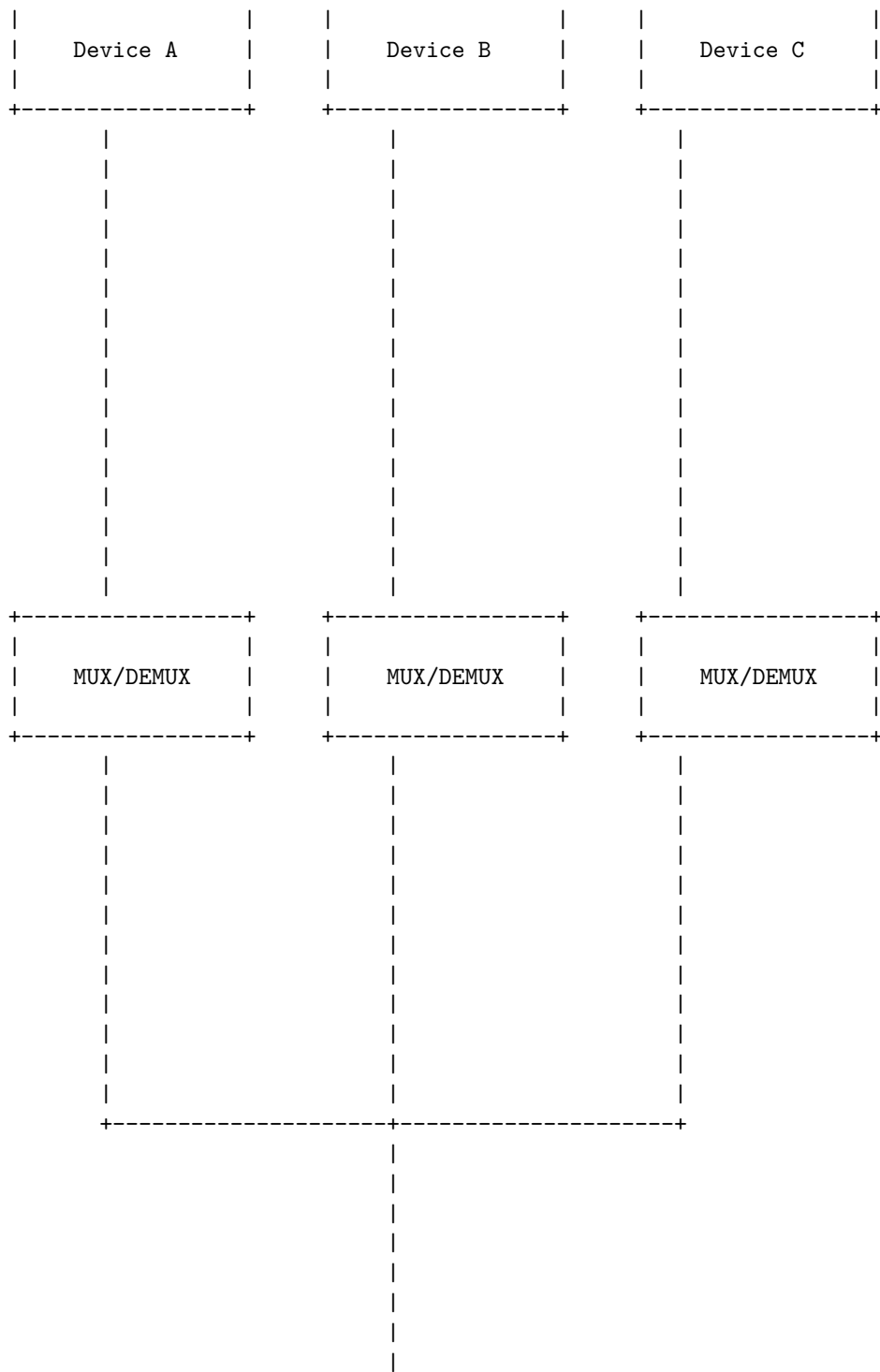
Multiplexing can be done in different ways, such as frequency division multiplexing (FDM), time division multiplexing (TDM), wavelength division multiplexing (WDM), and statistical multiplexing. FDM assigns different frequency bands to different signals and combines them into a single composite signal. TDM assigns different time slots to different signals and interleaves them into a single composite signal. WDM assigns different wavelengths of light to different signals and combines them into a single optical fiber. Statistical multiplexing dynamically allocates the available bandwidth to the signals based on their demand and priority.

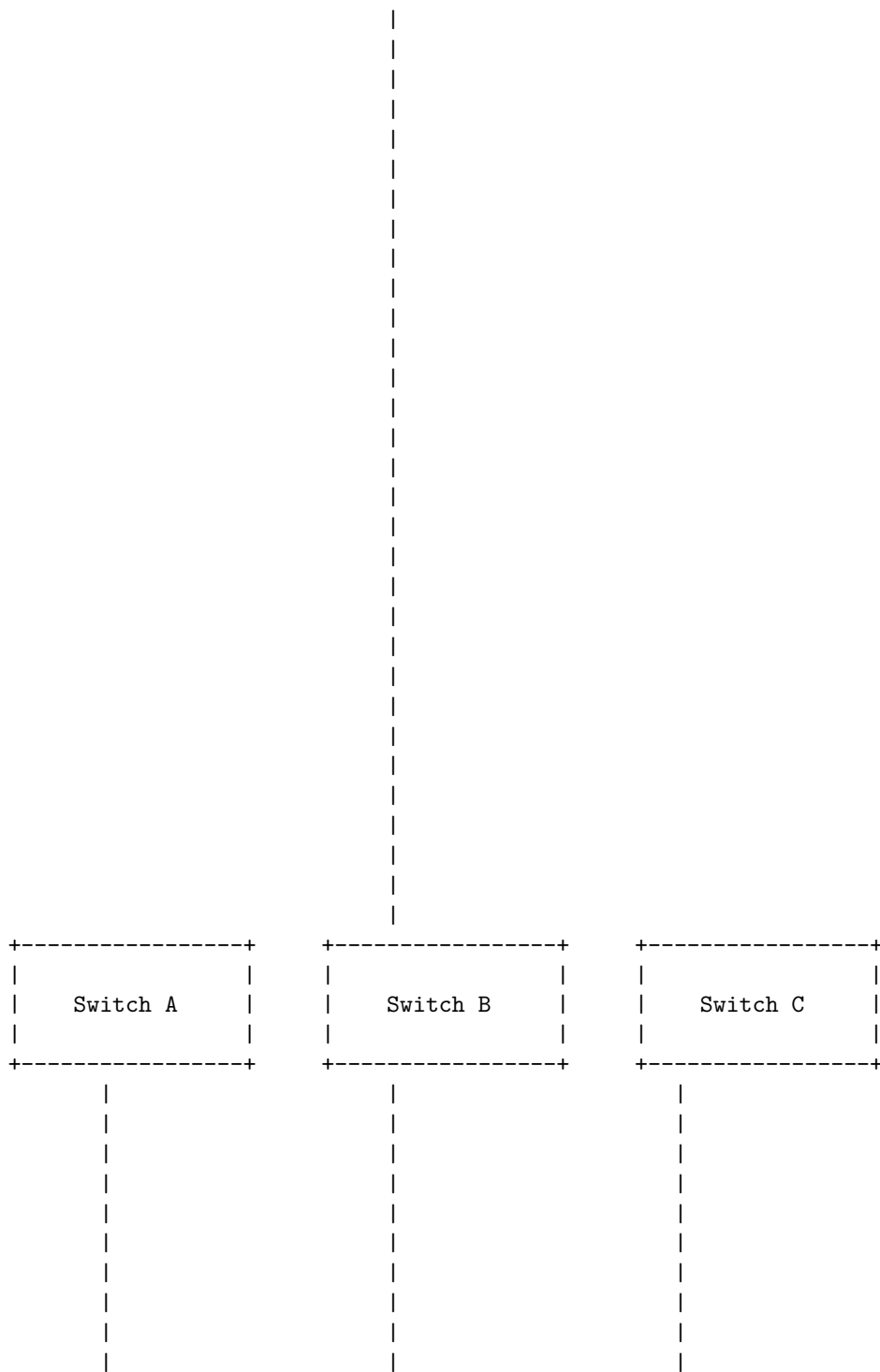
The following diagram illustrates the switching techniques and multiplexing in computer networks:

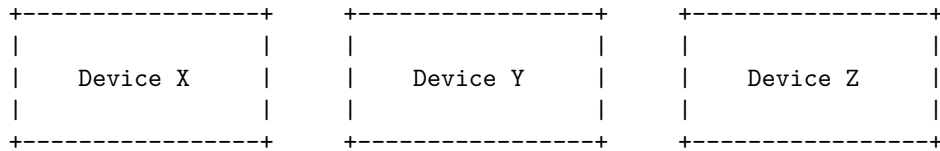
```

+-----+      +-----+      +-----+

```



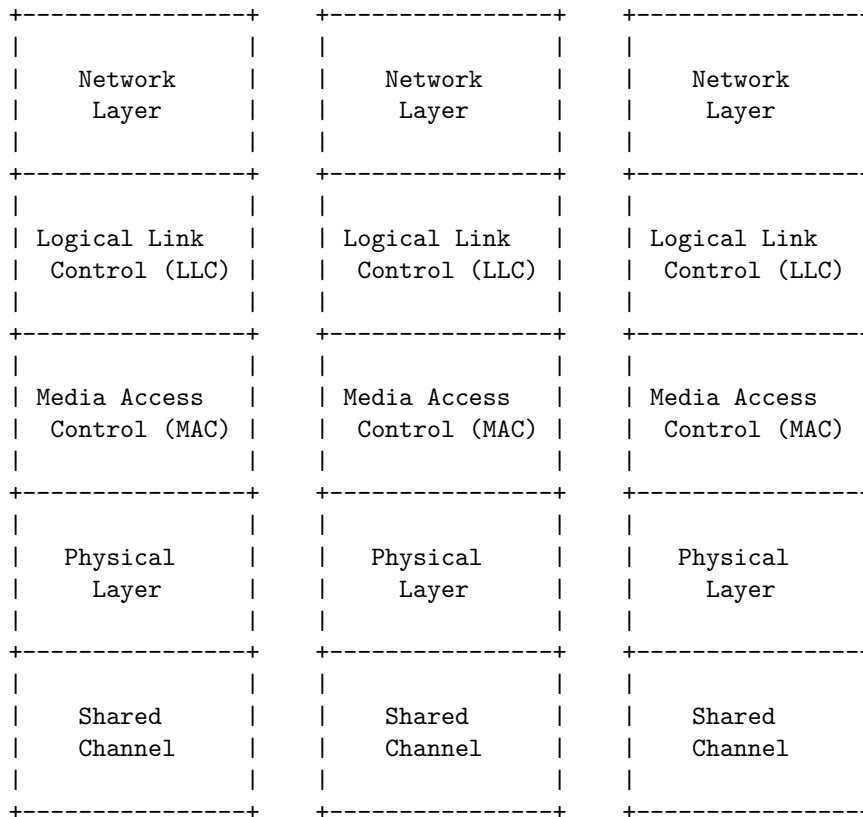




The devices A, B, and C are the sources of different signals, such as voice, data, or video. The devices X, Y, and Z are the destinations of the signals. The MUX

Hello, I am Sydney, your AI assistant. I can help you with your request. Here is a detailed ASCII diagram for Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks.

Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks



The diagram shows three nodes on a network segment, each with a network layer, a data link layer, and a physical layer. The data link layer is further divided into two sublayers: the logical link control (LLC) sublayer and the media access control (MAC) sublayer. The LLC sublayer is responsible for

managing communications links and handling frame traffic. The MAC sublayer is responsible for governing protocol access to the physical network medium. The physical layer is responsible for transmitting and receiving bits on the shared channel.

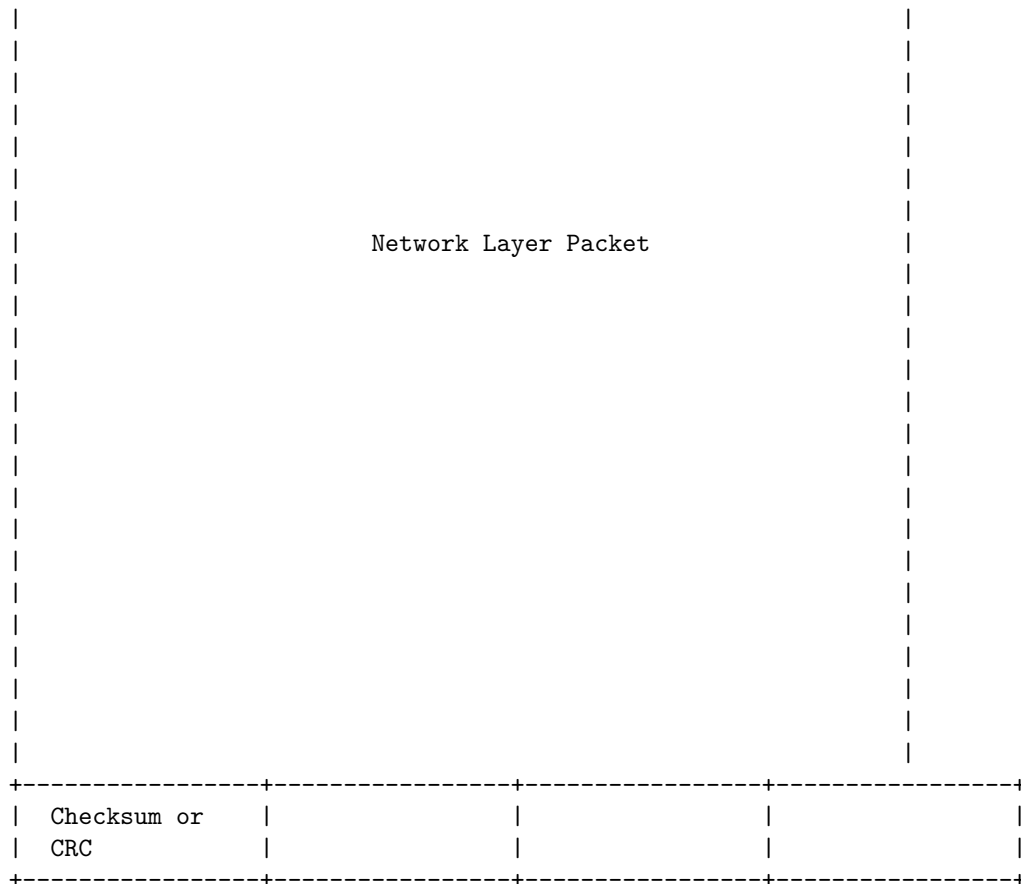
The diagram also shows the different protocols that are used at each sublayer. For example, in UMTS and LTE, the MAC sublayer uses the Packet Data Convergence Protocol (PDCP), the Radio Link Control (RLC) protocol, and the MAC protocol. The LLC sublayer may use different protocols depending on the network layer protocol, such as IEEE 802.2 for IP or IEEE 802.3 for Ethernet. The physical layer may use different standards depending on the type of channel, such as IEEE 802.11 for wireless or IEEE 802.3 for wired.

The link layer is the lowest layer in the Internet protocol suite, the networking architecture of the Internet. The link layer is the group of methods and communications protocols confined to the link that a host is physically connected to. The link layer is also known as the data link layer, or layer 2, in the OSI model of computer networking. The data link layer transfers data between nodes on a network segment across the physical layer and provides error control and addressing functions.

A diagram for the link layer in computer networks is shown below. The diagram is drawn using ASCII characters and markdown syntax. The diagram shows the data link layer encapsulating the network layer packet into a frame with a header and a trailer. The header contains the source and destination MAC addresses, the type of the network layer protocol, and other control information. The trailer contains a checksum or a cyclic redundancy check (CRC) to detect errors in the frame. The frame is then transmitted over the physical layer as a stream of bits.

Link layer in Computer Networks

[illegible]



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Framing is a function of the data link layer that provides a way for a sender to transmit a set of bits that are meaningful to the receiver. Framing uses frames to send or receive data. The data link layer receives packets from the network layer and converts them into frames. Frames have headers that contain information such as error-checking codes, source and destination addresses, and protocols.

There are different types of framing methods used in data link layer, such as:

- Character count: This method uses a field in the header to specify the number of characters in the frame. The receiver counts the characters and knows where the frame ends. This method is simple but not reliable, as any error in the count field or the data can cause framing errors.
- Byte stuffing: This method uses a special byte sequence, such as DLE STX (Data Link Escape, Start of Text) to mark the beginning of the frame, and DLE ETX (Data Link Escape, End of Text) to mark the end

of the frame. If the data contains the same byte sequence, it is replaced by another sequence, such as DLE DLE, to avoid confusion. The receiver reverses the process and recovers the original data. This method is more reliable but requires extra bytes to be inserted and removed.

- Bit stuffing: This method uses a special bit pattern, such as 01111110, to mark the beginning and end of the frame. If the data contains five consecutive 1s, a 0 is inserted after them to avoid confusion. The receiver reverses the process and recovers the original data. This method is also reliable but requires extra bits to be inserted and removed.
- Physical layer coding violation: This method uses the properties of the physical layer to mark the beginning and end of the frame. For example, in Manchester encoding, a transition from high to low voltage represents a 1, and a transition from low to high voltage represents a 0. A violation of this rule, such as a high to high or low to low transition, can be used to indicate the start or end of a frame. This method does not require extra bits or bytes, but depends on the physical layer encoding scheme.

Framing in link layer in Computer Networks

Frame header	Data	Frame trailer	Interframe gap
Source address	Network packet	Error checking	Idle time
Destination address		code	
Protocol			
Count or flag			

Error detection and correction in link layer in computer networks is the process of ensuring that the data frames transmitted from the sender to the receiver are accurate and complete. The link layer uses various techniques to add redundancy bits to the data frames and check them for errors at the receiver side. If errors are detected, the receiver can either discard the corrupted frames or request the sender to retransmit them. Some of the common techniques for error detection and correction are parity check, checksum and cyclic redundancy check (CRC) .

The following is a simplified ASCII diagram of how error detection and correction works in link layer in computer networks:

```

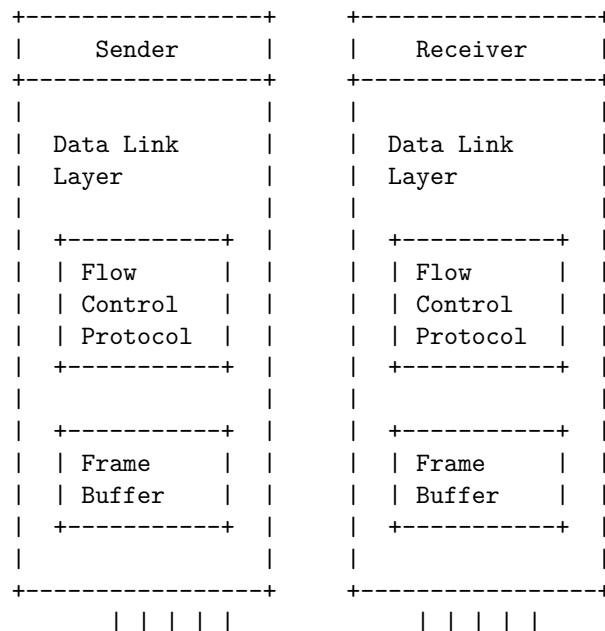
Sender                                     Receiver
|                                         |
| Data frame: 10101010                   |
| Parity bit: 0 (even parity)             |
|-----> Data frame: 10101010

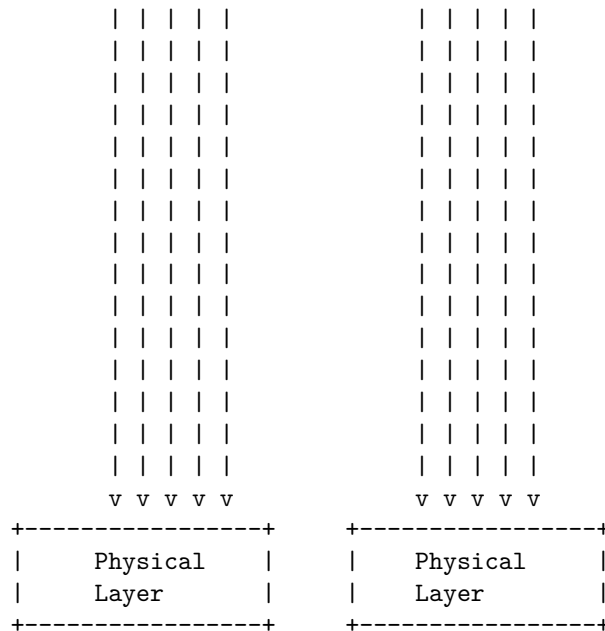
```

		Parity bit: 0 (even parity)
		No error detected
		Data frame: 11001100
		Parity bit: 0 (even parity)
		-----> Data frame: 11001100
		Parity bit: 0 (even parity)
		No error detected
		Data frame: 10011001
		Parity bit: 0 (even parity)
		-----> Data frame: 10011001
		Parity bit: 1 (odd parity)
		Error detected, discard frame or request retransmission

Flow control is a technique that allows two stations working at different speeds to communicate with each other. It is a set of measures taken to regulate the amount of data that a sender can send so that a fast sender does not overwhelm a slow receiver. Flow control in data link layer restricts the number of frames the sender can send before it waits for an acknowledgment from the receiver . There are two main approaches of flow control: stop-and-wait and sliding window .

Flow control in link layer in Computer Networks





The diagram above shows the flow control in link layer in computer networks. The sender and the receiver have a data link layer that implements a flow control protocol. The protocol regulates the number of frames that the sender can transmit before waiting for an acknowledgment from the receiver. The sender and the receiver also have a frame buffer that stores the frames before they are sent or processed. The physical layer is responsible for transmitting and receiving the frames over the medium. The flow control protocol can use different methods to control the flow of frames, such as stop-and-wait or sliding window.

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Elementary Data Link Protocols are protocols in the data link layer that are designed to perform the basic functions of framing, error control and flow control. Framing is the process of dividing bit-streams from the physical layer into data frames whose size ranges from a few hundred to a few thousand bytes. Error control is the process of detecting and correcting errors that may occur during transmission. Flow control is the process of regulating the rate of data transmission between the sender and the receiver.

There are three types of Elementary Data Link Protocols, as follows:

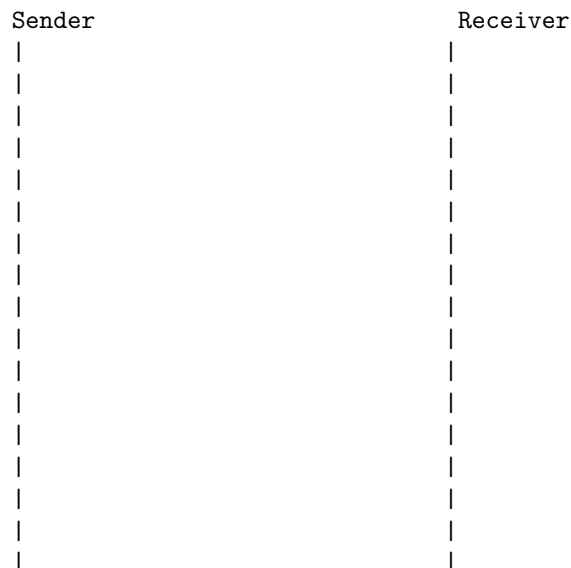
- Protocol 1: Unrestricted Simplex Protocol. This protocol is used for noiseless channels, which means there is no error or loss of data during transmission. In this protocol, the sender sends data frames continuously without waiting for any acknowledgment from the receiver. The receiver simply accepts and processes the frames as they arrive. There is no error control or flow control in this protocol.

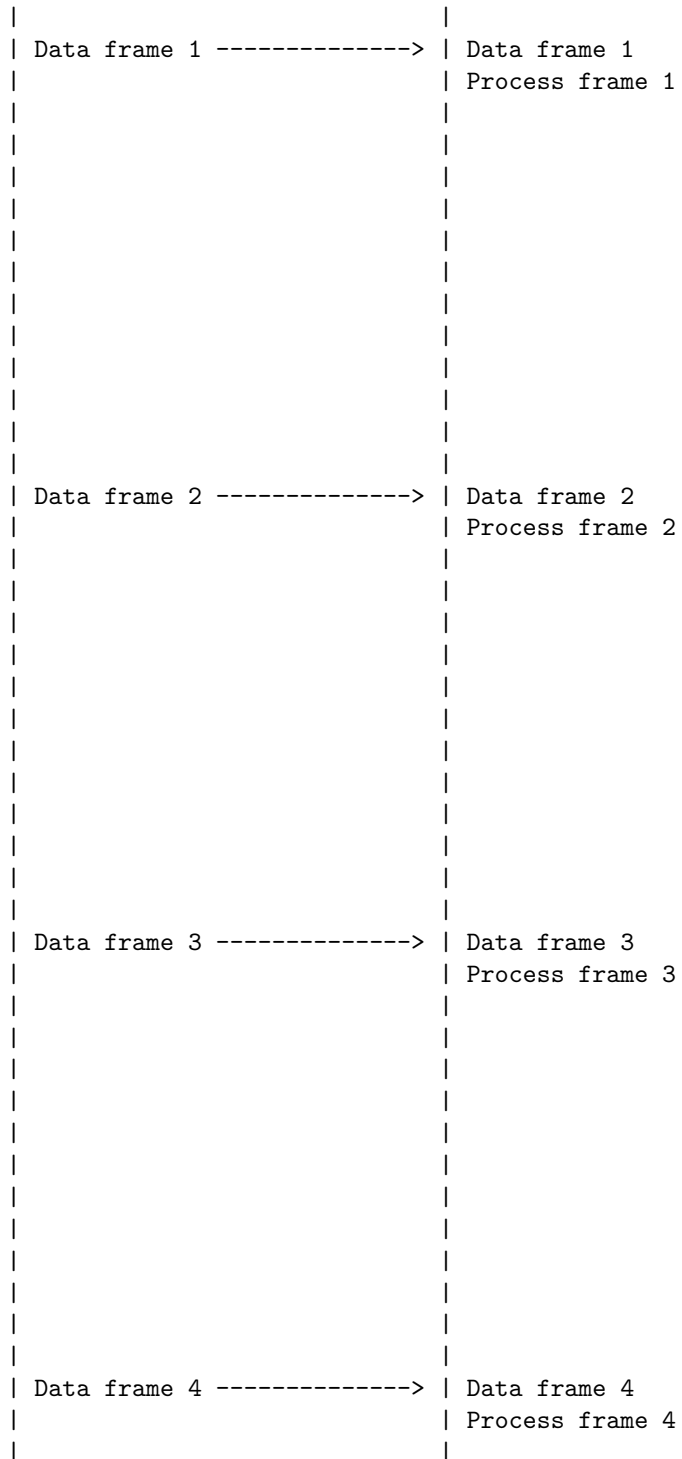
- **Protocol 2: Simplex Stop-and-Wait Protocol.** This protocol is used for noisy channels, which means there is a possibility of error or loss of data during transmission. In this protocol, the sender sends one data frame and waits for an acknowledgment from the receiver before sending the next frame. The receiver sends an acknowledgment after receiving and processing a frame. If the sender does not receive an acknowledgment within a certain time, it assumes that the frame or the acknowledgment was lost and retransmits the frame. This protocol provides error control but not flow control.
- **Protocol 3: Simplex Protocol for Noisy Channels.** This protocol is also used for noisy channels, but it improves the efficiency of Protocol 2 by adding sequence numbers to the data frames and the acknowledgments. In this protocol, the sender sends one data frame with a sequence number and waits for an acknowledgment with the same sequence number from the receiver before sending the next frame. The receiver sends an acknowledgment with the sequence number of the last correctly received frame. If the sender receives an acknowledgment with a different sequence number, it assumes that the frame or the acknowledgment was lost or duplicated and retransmits the frame. This protocol provides both error control and flow control.

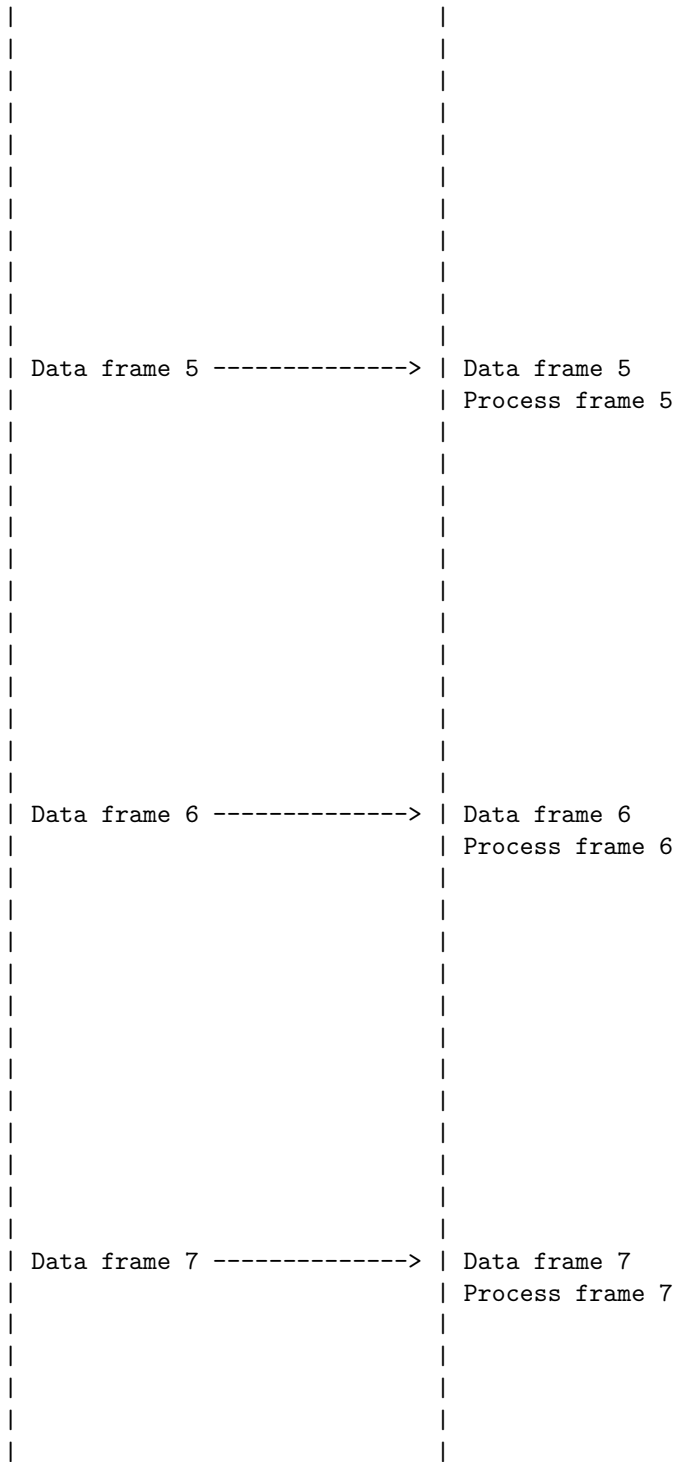
The following is a detailed ASCII diagram for the Elementary Data Link Protocols in the link layer in Computer Networks:

Elementary Data Link Protocols

Protocol 1: Unrestricted Simplex Protocol









The sliding window protocol is a data link layer protocol that is useful for the sequential and reliable delivery of data frames between two devices. The sender can send multiple frames at a time without waiting for an acknowledgment from the receiver. The receiver sends back an acknowledgment when it receives a frame. The sliding window protocol uses a mechanism of sequence numbers to identify each frame and to keep track of the frames that have been sent and received.

A sliding window is a range of frames that the sender can send or the receiver can accept at a given time. The size of the sliding window is determined by the buffer capacity of the sender and the receiver. The sender maintains a send window that indicates the frames that it can send. The receiver maintains a receive window that indicates the frames that it can accept. The sender and the receiver exchange information about their windows using control frames.

There are two types of sliding window protocols: stop-and-wait and go-back-N. In stop-and-wait, the sender sends one frame at a time and waits for an acknowledgment before sending the next frame. The send window and the receive window have a size of one frame. In go-back-N, the sender can send up to N frames at a time, where N is the size of the send window. The receiver can accept up to N frames at a time, where N is the size of the receive window. The receiver sends an acknowledgment for the last frame that it received in sequence. If the sender does not receive an acknowledgment for a frame within a certain time, it assumes that the frame or the acknowledgment was lost and retransmits all the frames from that point.

The following is a diagram of the sliding window protocol in the go-back-N mode.

Sliding Window Protocol in Link Layer in Computer Networks

```
Sender: S0 S1 S2 S3 S4 S5 S6 S7
Send window: |-----|
Receiver: R0 R1 R2 R3 R4 R5 R6 R7
Receive window: |-----|
```

```
S0 -> R0 (frame 0 sent and received)
S1 -> R1 (frame 1 sent and received)
S2 -> R2 (frame 2 sent and received)
S3 -> R3 (frame 3 sent and received)
```

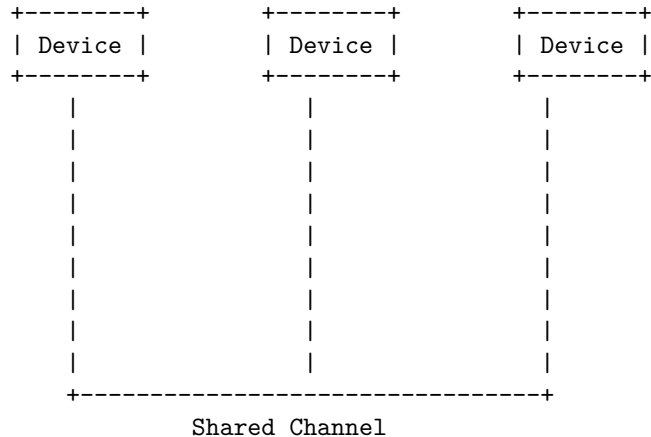
```

S4 -> R4 (frame 4 sent and received)
S5 -> R5 (frame 5 sent and lost)
S6 -> R6 (frame 6 sent and received out of sequence)
S7 -> R7 (frame 7 sent and received out of sequence)
<- ACK 4 (acknowledgment for frame 4 received by sender)
<- ACK 4 (acknowledgment for frame 4 received by sender)
<- ACK 4 (acknowledgment for frame 4 received by sender)
<- ACK 4 (acknowledgment for frame 4 received by sender)
S5 -> R5 (frame 5 retransmitted and received)
S6 -> R6 (frame 6 retransmitted and received)
S7 -> R7 (frame 7 retransmitted and received)
<- ACK 7 (acknowledgment for frame 7 received by sender)
Send window:      |-----|
Receive window:    |-----|

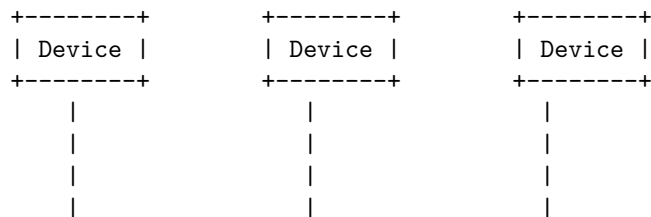
```

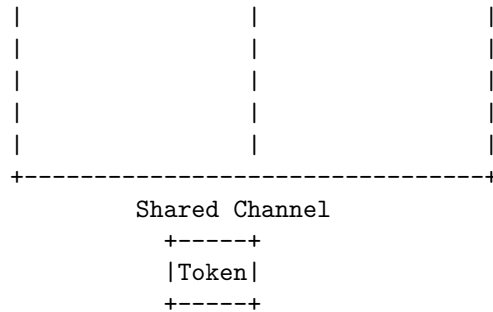
Medium Access Control (MAC) is a protocol that regulates how devices access and share a common communication channel in a Local Area Network (LAN). There are two main approaches to MAC: contention and token-passing. Contention is a first-come, first-serve approach, where devices compete for the channel and may experience collisions. Token-passing is a cooperative approach, where devices take turns to use the channel by passing a token.

A diagram of contention-based MAC is shown below:



A diagram of token-passing MAC is shown below:





The token circulates among the devices in a predefined order. Only the device that holds the token can transmit data on the channel. After transmission, the device passes the token to the next device in the order.

Channel allocation in medium access control (MAC) is the process of assigning channels to different users or devices in a network that share a common medium. There are different methods of channel allocation, such as static or dynamic, and different types of channels, such as frequency, time, code, or space. Here is a simple ASCII diagram to illustrate the concept of channel allocation in MAC:

Channel allocation in medium access control

Channel 1	Channel 2	Channel 3	Channel 4
User A	User B	User C	User D
User E	User F	User G	User H

In this diagram, each channel is a frequency band that can be used for communication. Each user is a device that wants to access the medium. The channel allocation method determines how the channels are assigned to the users. For example, in static channel allocation, each user is assigned a fixed channel and does not change. In dynamic channel allocation, the channels are assigned on demand and can change according to the traffic conditions. Some dynamic channel allocation methods are contention-based, where users compete for the channel, and some are reservation-based, where users request the channel from a central controller.

Multiple access protocols are a set of protocols operating in the Medium Access

Control sublayer (MAC sublayer) of the Open Systems Interconnection (OSI) model. These protocols allow a number of nodes or users to access a shared network channel. There are different types of multiple access protocols, such as random access, controlled access, and channelization protocols. Here is a diagram that shows the classification of multiple access protocols:

Multiple access protocols in medium access control

+-----+			
Multiple access			
protocols			
+-----+			
+-----+			
Random Controlled Channel-			
access access ization			
+-----+			
ALOHA Reservation FDMA			
CSMA Polling TDMA			
CSMA/CA Token CDMA			
CSMA/CD passing OFDMA			
+-----+			

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LAN standards are a set of rules and specifications that define how devices communicate and share data in a local area network. There are different types of LAN standards, such as Ethernet, Token Ring, and Wireless LAN, that use different physical and logical topologies, media access methods, and protocols.

One of the most common and widely used LAN standards is the IEEE 802 family, which consists of 12 subgroups, numbered 802.1 to 802.12, each focusing on a different aspect of LAN technology. For example, IEEE 802.3 is the standard for Ethernet, IEEE 802.11 is the standard for Wireless LAN, and IEEE 802.1 is the standard for bridging and network management.

A diagram of the IEEE 802 family of LAN standards is shown below, using ASCII characters to represent the different subgroups and their relationships.

LAN standards in local area network

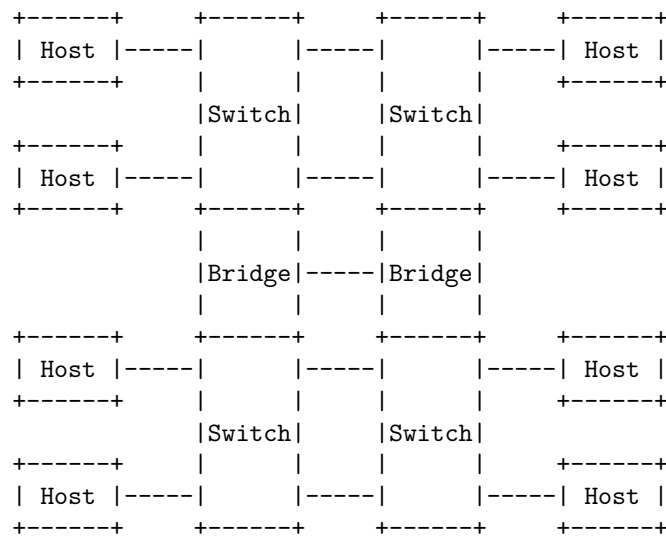
+-----+			+-----+			+-----+		
IEEE 802.1			IEEE 802.2			IEEE 802.3		
Bridging and			Logical Link			Ethernet		

	LAN Security
	+-----+
	IEEE 802.11
	Wireless LAN
	+-----+
	IEEE 802.12
	Demand Priority
	Access Method
	+-----+
+-----+	
IEEE 802.15	
Wireless	
Personal Area	
Network (WPAN)	
+-----+	
IEEE 802.16	
Wireless	
Metropolitan	
Area Network	
(WMAN)	
+-----+	
IEEE 802.17	
Resilient	
Packet Ring	
(RPR)	
+-----+	
IEEE 802.18	
Radio	
Regulatory	
TAG	
+-----+	
IEEE 802.19	
Wireless	
Coexistence	
TAG	
+-----+	
IEEE 802.20	
Mobile	
Broadband	
Wireless Access	
(MBWA)	
+-----+	
IEEE 802.21	
Media	
Independent	
Handover	

+-----

A link layer switch or a bridge is a network device that connects multiple LANs (local area networks) together to form a larger LAN. It operates at the data link layer of the OSI model and uses MAC addresses to forward Ethernet frames from one device to another device in the network . A switch or a bridge can also interconnect data link layer domains that have different technologies, such as Ethernet and FDDI.

A possible ASCII diagram for link layer switches and bridges in a local area network is shown below:



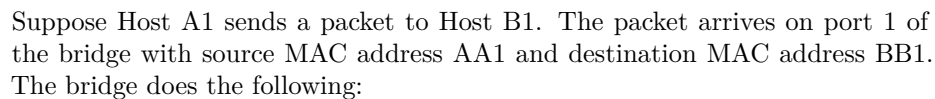
In this diagram, there are four LANs, each with two hosts and a switch. The switches are connected by two bridges, which allow the LANs to communicate with each other. The bridges can also handle different data link layer technologies, such as Ethernet and FDDI, if needed. The switches and bridges use MAC addresses to forward frames to the appropriate destination.

Learning bridge algorithms in local area network

A bridge is a device that connects two or more local area networks (LANs) at the data link layer. A bridge can filter, forward, or flood frames based on their destination MAC addresses. A bridge maintains a forwarding table that maps MAC addresses to ports. A bridge updates its forwarding table using a learning algorithm, which works as follows:

- A bridge examines the source MAC address of each packet that arrives on a particular port and memorizes that the source address is reachable via that port. This is done by adding the source MAC address and the port to the forwarding table.

- The following diagram shows an example of a bridge connecting two LANs, A and B. The bridge has two ports, 1 and 2. The bridge initially has an empty forwarding table.



- 52

The reply packet arrives on port 2 of the bridge with source MAC address BB1 and destination MAC address AA1. The bridge does the following:

- It adds the entry (BB1, 2) to the forwarding table, indicating that Host B1 is reachable via port 2.
- It looks up the destination MAC address AA1 in the forwarding table, and finds the entry (AA1, 1).
- It forwards the packet to port 1, where it reaches Host A1 on LAN A.

The bridge has learned the locations of Host A1 and Host B1 by observing the traffic. The forwarding table now looks like this:

MAC	Port
AA1	1
BB1	2

If Host A2 sends a packet to Host B2, the bridge will repeat the same process and learn the locations of Host A2 and Host B2. The forwarding table will then look like this:

MAC	Port
AA1	1
AA2	1
BB1	2
BB2	2

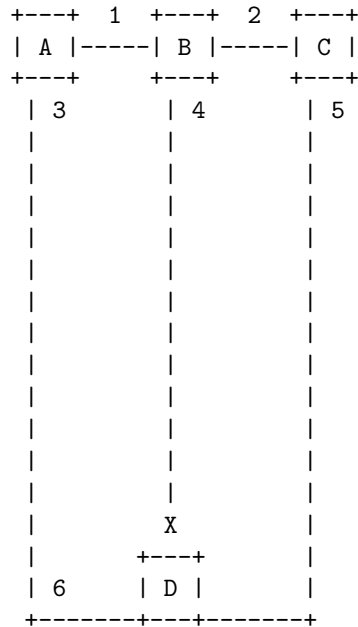
Now, if Host A1 sends another packet to Host B1, the bridge will not flood the packet, but forward it directly to port 2, since it knows the destination MAC address. Similarly, if Host B2 sends a packet to Host A2, the bridge will forward it directly to port 1. The bridge has effectively reduced the traffic on both LANs by learning the MAC addresses of the hosts.

A spanning tree algorithm is a network protocol that builds a loop-free logical topology for any bridged Ethernet local area network. It prevents bridge loops and the broadcast radiation that results from them by disabling redundant links and selecting one preferred link between switches in the LAN. The preferred link is used for all Ethernet frames unless it fails, in which case a backup link is enabled. The algorithm also designates one layer-2 switch as the root bridge, which is the reference point for all other switches in the network .

A possible ASCII diagram for a spanning tree algorithm in a local area network with four switches (A, B, C, D) and six links (1, 2, 3, 4, 5, 6) is shown below. The diagram assumes that switch A is the root bridge and link 1 is the preferred link between A and B. The links that are disabled by the algorithm are marked

with X .

Spanning Tree Algorithm in Local Area Network

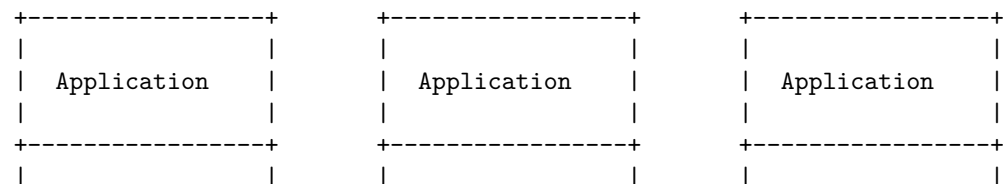


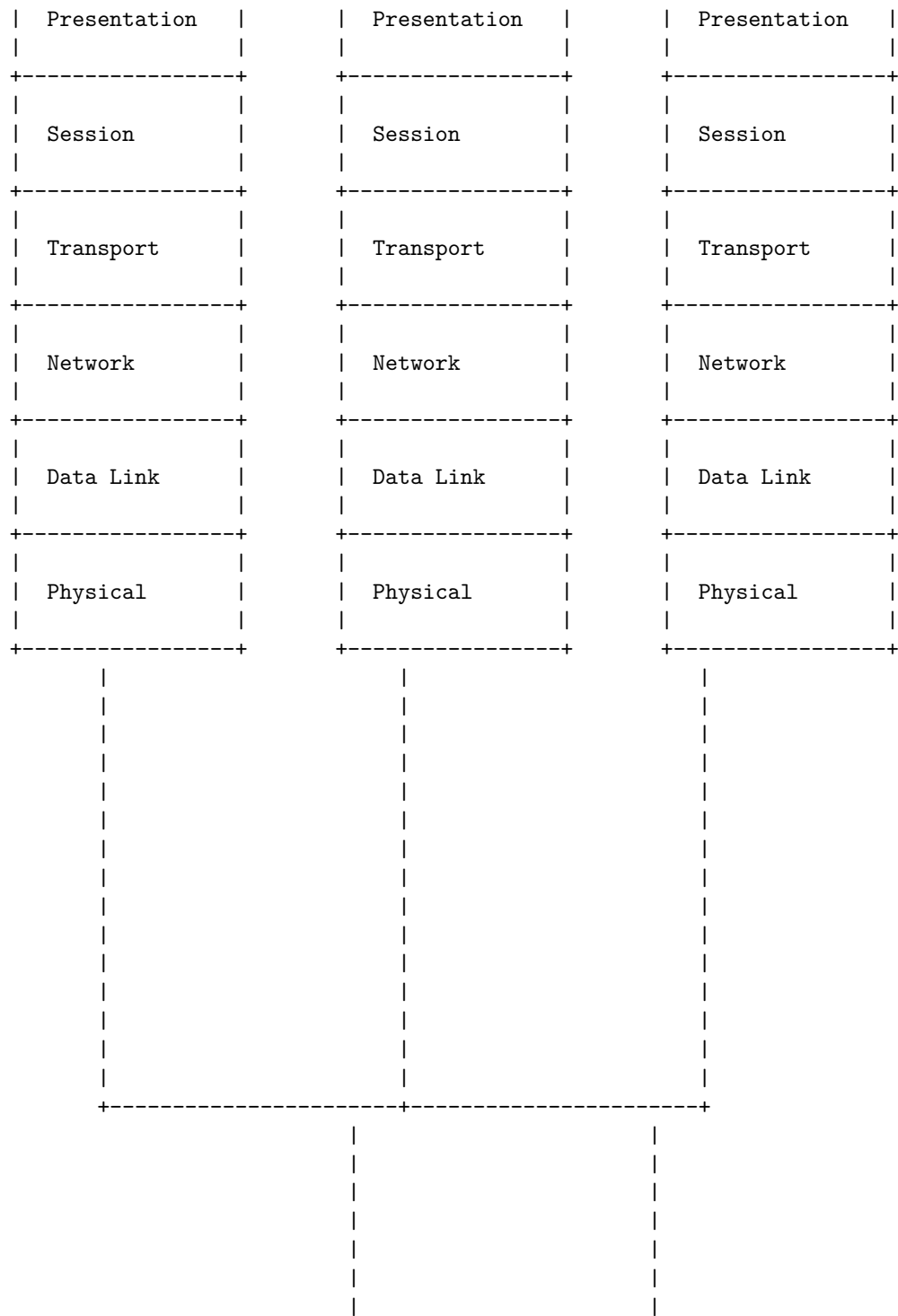
Unit 3 - Network Layer in Computer Networks

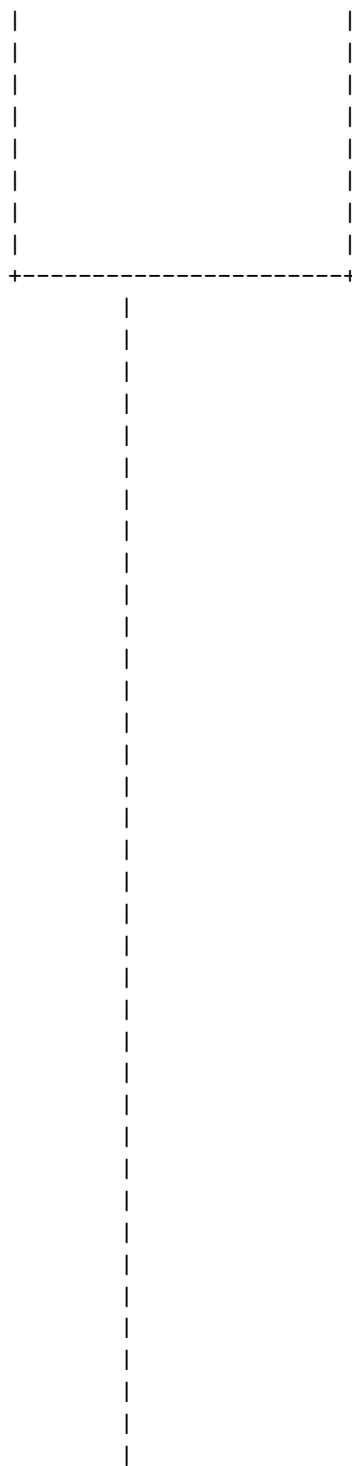
The network layer is the third layer of the Open Systems Interconnection (OSI) model. It is responsible for routing packets across different networks and sub-networks, using logical addresses (such as IP addresses) and protocols (such as IP, ICMP, ARP, etc.). The network layer also performs fragmentation and reassembly of packets, error control, and congestion control.

A network layer diagram shows the components and connections of a network at the logical level, without considering the physical details of the devices and links. It typically includes routers, switches, firewalls, gateways, and other network devices that operate at the network layer. A network layer diagram can help visualize the topology and structure of a network, as well as the flow of data and control information.

Here is an example of a network layer diagram, drawn in ASCII art:







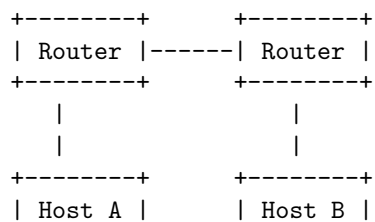
|



A point-to-point network is a network topology in which two nodes are directly connected by a link, without any intermediate devices or hosts. A point-to-point network can use different protocols at the data link layer, such as High-level Data Link Control (HDLC), Point-to-Point Protocol (PPP), or Point-to-Point Tunneling Protocol (PPTP). These protocols provide a way to encapsulate multiprotocol data, establish and configure the link, authenticate the users, and compress or encrypt the data.

A point-to-point network in the network layer can be used to connect two routers or two hosts over a WAN link, such as a leased line, a dial-up modem, or a wireless connection. The network layer protocol, such as IP, can use the point-to-point link as a logical interface to send and receive packets.

A possible ASCII diagram for a point-to-point network in the network layer is:



+-----+ +-----+

In this diagram, the two routers are connected by a point-to-point link, which can use any of the data link layer protocols mentioned above. The routers can use IP as the network layer protocol to exchange packets over the link. The hosts A and B are connected to the routers by other network interfaces, such as Ethernet or Wi-Fi. The hosts can also use IP as the network layer protocol to communicate with each other through the routers.

Logical addressing is the process of assigning a unique identifier to each device on a network. Logical addresses are created and used by network layer protocols, such as IP or IPX. The network layer protocol translates logical addresses to MAC addresses, which are the physical addresses of the devices.

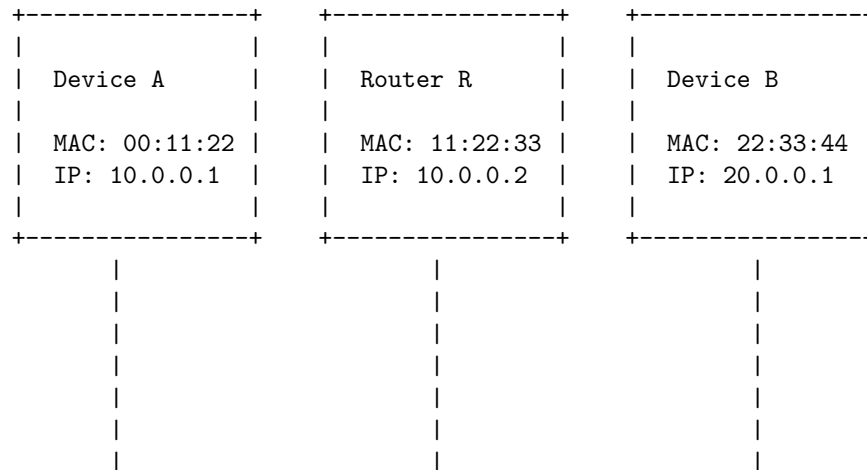
A logical address consists of two parts: a network ID and a host ID. The network ID identifies the network to which the device belongs, and the host ID identifies the device within that network. The format and length of the logical address depend on the network layer protocol used. For example, IP uses a 32-bit logical address, while IPX uses a 80-bit logical address.

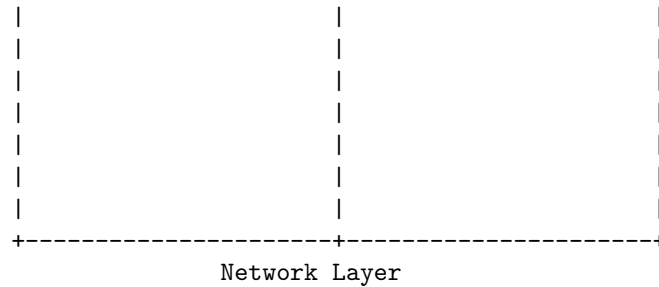
A logical address can be represented in different ways, such as binary, decimal, hexadecimal, or dotted decimal notation. For example, the IP address 192.168.1.100 can be written as:

- Binary: 11000000.10101000.00000001.01100100
- Decimal: 3232235876
- Hexadecimal: C0.A8.01.64
- Dotted decimal: 192.168.1.100

The following diagram shows an example of logical addressing in the network layer:

Logical addressing in network layer





In this diagram, device A and device B are on different networks, and router R is the gateway between them. Device A wants to send a packet to device B, so it uses its network layer protocol (IP) to create a logical address for device B, which is 20.0.0.1. Device A also uses its network layer protocol to find out the logical address of router R, which is 10.0.0.2. Device A then encapsulates the packet with the source and destination logical addresses, which are 10.0.0.1 and 20.0.0.1, respectively.

Device A then sends the packet to router R, using its data link layer protocol (Ethernet) to create a MAC address for router R, which is 11:22:33. Device A also uses its data link layer protocol to find out its own MAC address, which is 00:11:22. Device A then encapsulates the packet with the source and destination MAC addresses, which are 00:11:22 and 11:22:33, respectively.

Router R receives the packet and checks the destination MAC address, which matches its own MAC address. Router R then decapsulates the packet and checks the destination logical address, which does not match its own logical address. Router R then uses its network layer protocol to find out the logical address of device B, which is 20.0.0.1. Router R then encapsulates the packet with the source and destination logical addresses, which are 10.0.0.1 and 20.0.0.1, respectively.

Router R then sends the packet to device B, using its data link layer protocol (Ethernet) to create a MAC address for device B, which is 22:33:44. Router R also uses its data link layer protocol to find out its own MAC address, which is 11:22:33. Router R then encapsulates the packet with the source and destination MAC addresses, which are 11:22:33 and 22:33:44, respectively.

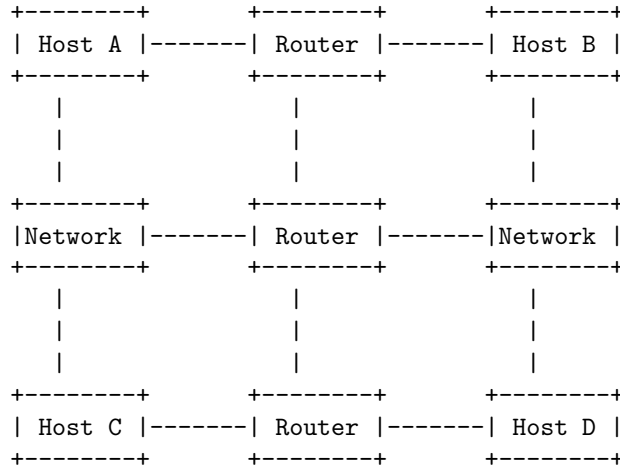
Device B receives the packet and checks the destination MAC address

Hello, I am Sydney, your AI assistant. I can help you with your query.

Basic internetworking in network layer is the process of connecting different types of networks using routers and protocols. Routers are devices that forward packets between networks based on their network-layer addresses. Protocols are rules that define how data is formatted, transmitted, and received. The most common protocol for internetworking is the Internet Protocol (IP), which is part of the TCP/IP model.

A basic diagram of internetworking in network layer is shown below:

Basic internetworking in network layer



In this diagram, there are four hosts (A, B, C, and D) and three routers. Each host belongs to a different network, which can be Ethernet, wireless, or point-to-point. The routers are connected by links that can be wired or wireless. The routers use IP to forward packets between networks based on their destination addresses. The hosts use IP to send and receive packets to and from other hosts. The packets contain headers that include the source and destination IP addresses, as well as other information.

An IP header is meta information at the beginning of an IP packet. It displays information such as the IP version, the packet's length, the source, and the destination. IPv4 header format is 20 to 60 bytes in length. It contains information need for routing and delivery .

The IP header consists of 13 fields, each with its own features and functions. The fields are:

- **Version:** The first IP header field is a 4-bit version indicator. In IPv4, the value of its four bits is set to 0100, which is 4 in decimal.
- **Internet Header Length:** Internet header length, shortly known as IHL, is 4 bits in size. It is also called HELEN (Header Length). It indicates the length of the IP header in 32-bit words. The minimum value for this field is 5, which indicates a length of $5 \times 32 \text{ bits} = 160 \text{ bits} = 20 \text{ bytes}$. The maximum value is 15, which indicates a length of $15 \times 32 \text{ bits} = 480 \text{ bits} = 60 \text{ bytes}$.
- **Type of Service:** Type of Service is also called TOS. It is 8 bits in size. It indicates the quality of service desired for the packet. It is divided into 5 subfields: Precedence (3 bits), Delay (1 bit), Throughput (1 bit), Reliability (1 bit), and Reserved (2 bits). The values of these subfields affect the priority and routing of the packet.
- **Total Length:** Total Length is 16 bits in size. It indicates the total length

of the IP packet, including the header and the data, in bytes. The minimum value for this field is 20 bytes, which is the length of the header without any options. The maximum value is 65535 bytes, which is the maximum size of an IP packet.

- Identification: Identification is 16 bits in size. It is used to identify the fragments of a datagram that has been split for transmission. The fragments of a datagram have the same identification value. The identification value is generated by the sender of the datagram.
- Flags: Flags is 3 bits in size. It is used to control the fragmentation of the datagram. It consists of 3 flags: Reserved (1 bit), Don't Fragment (1 bit), and More Fragments (1 bit). The Reserved flag is always set to 0. The Don't Fragment flag indicates whether the datagram can be fragmented or not. If it is set to 1, the datagram cannot be fragmented. If it is set to 0, the datagram can be fragmented. The More Fragments flag indicates whether the fragment is the last one or not. If it is set to 1, there are more fragments following this one. If it is set to 0, this is the last fragment or the datagram is not fragmented.
- Fragment Offset: Fragment Offset is 13 bits in size. It indicates the position of the fragment in the original datagram. It is measured in units of 8 bytes. The first fragment has an offset of 0. The offset of the subsequent fragments is calculated by adding the length of the previous fragment (in 8-byte units) to the offset of the previous fragment.
- Time to Live: Time to Live is also called TTL. It is 8 bits in size. It indicates the maximum time that the packet can remain in the network before it is discarded. It is decremented by 1 by each router that forwards the packet. If it reaches 0, the packet is discarded and an ICMP Time Exceeded message is sent back to the sender.
- Protocol: Protocol is 8 bits in size. It indicates the protocol used in the data portion of the IP packet. It identifies the next level protocol that is encapsulated in the IP packet. For example, the value 1 indicates ICMP, the value 6 indicates TCP, and the value 17 indicates UDP.
- Header Checksum: Header Checksum is 16 bits in size. It is used to verify the integrity of the IP header. It is calculated by adding the 16-bit words of the header and taking the one's complement of the sum. It is recomputed and verified by each router that processes the packet. If the checksum is incorrect, the packet is discarded.
- Source IP Address: Source IP Address is 32 bits in size. It indicates the

CIDR stands for Classless Inter-Domain Routing. It is a method for allocating IP addresses and for IP routing. It allows blocks of addresses to be grouped into single routing table entries, which improves the efficiency of address distribution and routing.

CIDR notation is a compact representation of an IP address and its associated routing prefix. The notation is constructed from an IP address, a slash ('/') character, and a decimal number. The number is the count of leading 1 bits in the routing mask, traditionally called the network mask. The IP address is

expressed according to the standards of IPv4 or IPv6.

For example, the IPv4 address 192.168.100.14/24 represents the IPv4 address 192.168.100.14 and its associated routing prefix 192.168.100.0, or equivalently, its subnet mask 255.255.255.0, which has 24 leading 1-bits.

The following ASCII diagram shows how CIDR notation works for IPv4 addresses:

```

192.168.100.14/24
|<---network--->|<---host--->|
|<-prefix len->|
|               |
|               V
11000000.10101000.01100100.00001110
|<----- 24 bits ----->|<- 8 ->|
|<----- routing mask ----->|<---->|
|               |
|               V
255.255.255.0
|<---network--->|<---host--->|

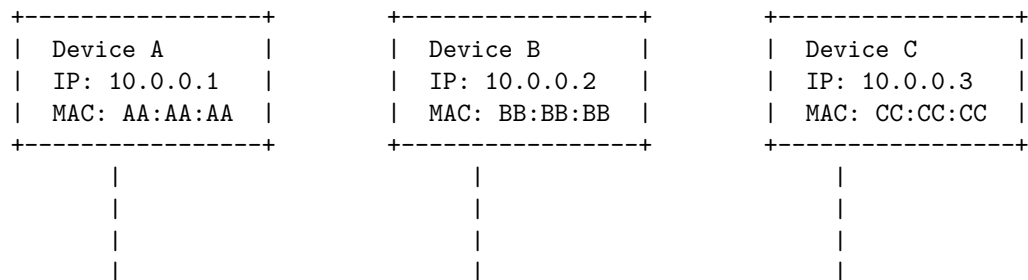
```

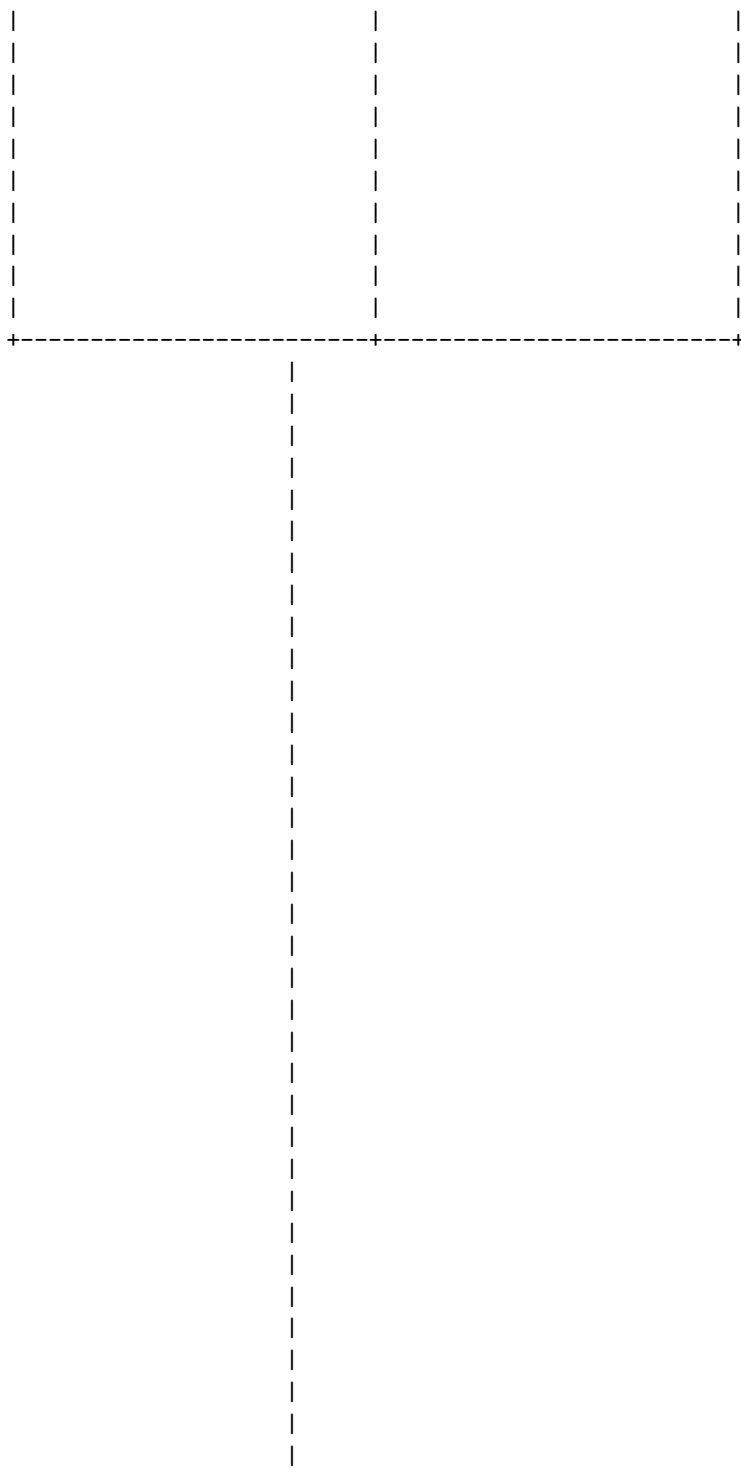
The diagram shows that the IP address 192.168.100.14 belongs to the network 192.168.100.0, which has a prefix length of 24 bits. The routing mask is a binary string of 24 1-bits followed by 8 0-bits, which corresponds to the decimal value 255.255.255.0. The host part of the IP address is the remaining 8 bits, which can vary from 0 to 255.

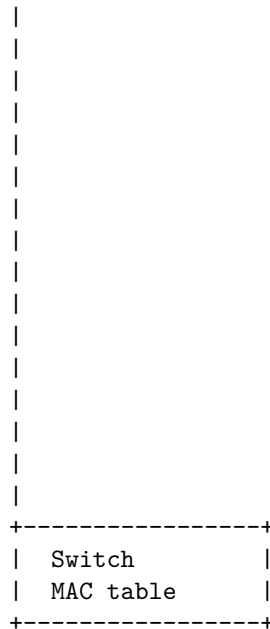
ARP is a protocol that maps an IP address to a MAC address in a local area network. It is used to find the physical address of a device from its logical address. ARP has two types of messages: ARP request and ARP reply. An ARP request is broadcast to all devices in the network, asking for the MAC address of a device with a specific IP address. An ARP reply is sent by the device that has the matching IP address, providing its MAC address. The sender of the ARP request then updates its ARP cache with the new entry.

Here is a diagram of how ARP works:

ARP



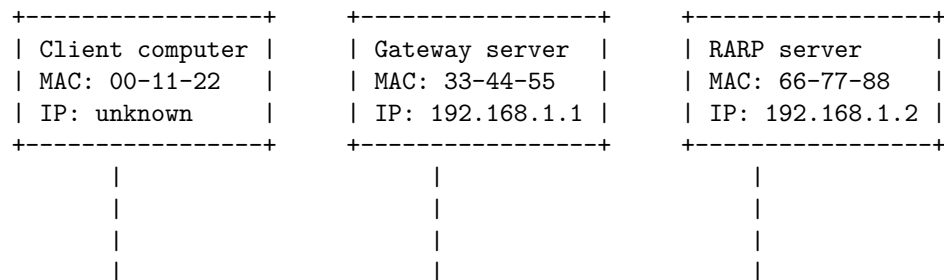


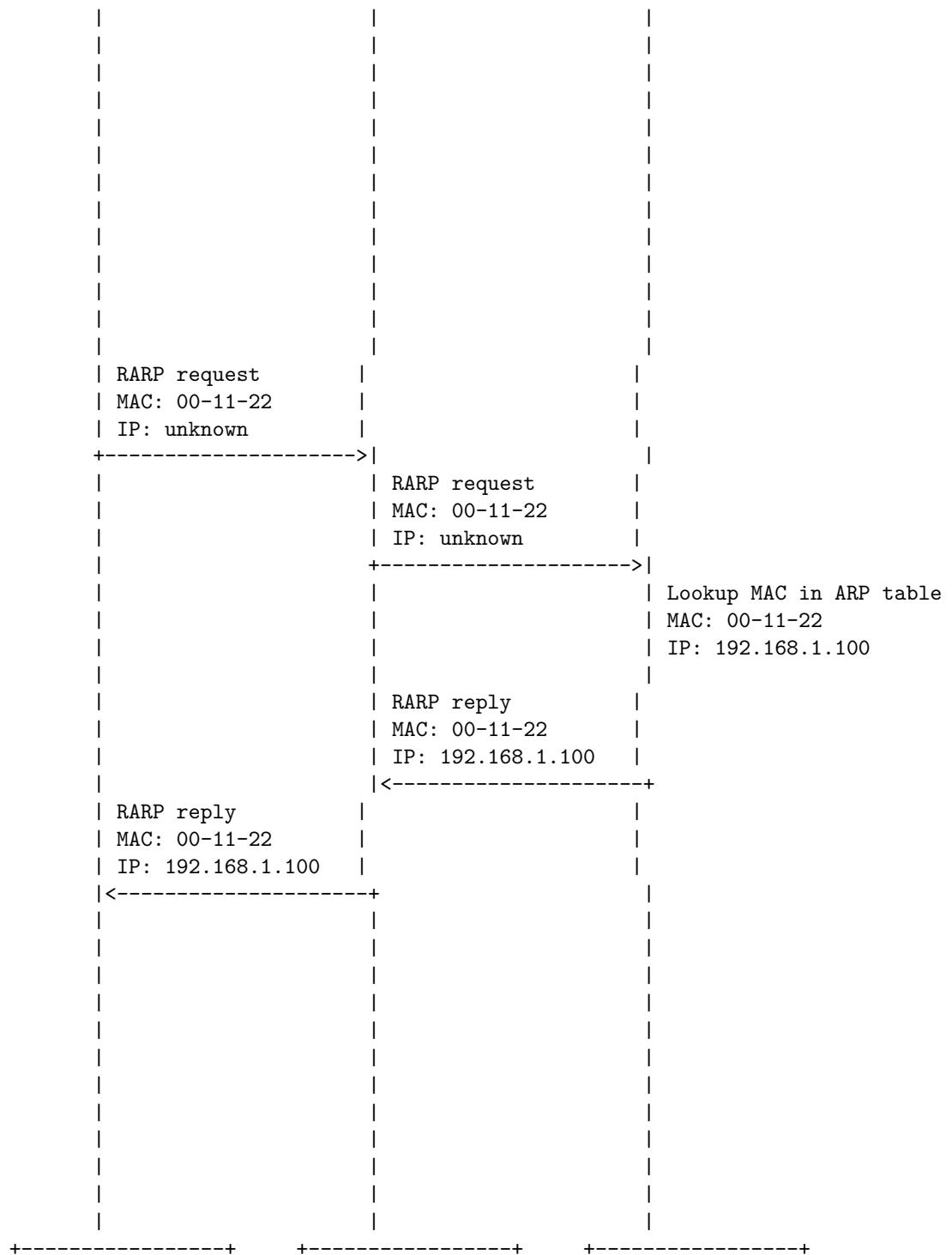


Device A wants to send a packet to Device C, but it only knows its IP address (10.0.0.3). Device A sends an ARP request to the switch, asking for the MAC address of 10.0.0.3. The switch broadcasts the ARP request to all devices in the network. Device C receives the ARP request and sees that it matches its IP address. Device C sends an ARP reply to the switch, providing its MAC address (CC:CC:CC). The switch forwards the ARP reply to Device A. Device A receives the ARP reply and updates its ARP cache with the new entry. Device A can now send the packet to Device C using its MAC address.

RARP stands for Reverse Address Resolution Protocol. It is a protocol that allows a client computer to request its IP address from a gateway server's ARP table or cache. The client computer sends its MAC address to the server, and the server responds with the corresponding IP address. RARP is useful for devices that do not have a permanent IP address, such as diskless workstations.

RARP





Client computer	Gateway server	RARP server
MAC: 00-11-22	MAC: 33-44-55	MAC: 66-77-88
IP: 192.168.1.100	IP: 192.168.1.1	IP: 192.168.1.2
+-----+	+-----+	+-----+

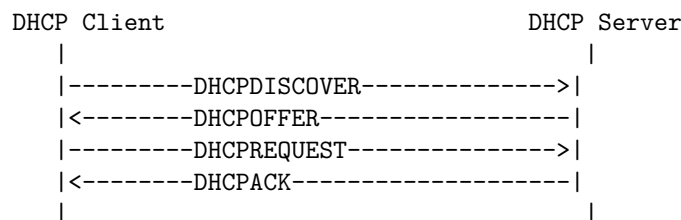
DHCP stands for Dynamic Host Configuration Protocol. It is a network management protocol that automatically assigns IP addresses and other communication parameters to devices connected to a network using a client-server architecture. A DHCP server maintains a pool of available IP addresses and leases them to DHCP clients that request them. A DHCP client can renew or release its IP address during the lease period. A DHCP server can also provide other configuration information to the clients, such as the subnet mask, default gateway, domain name, DNS servers, etc.

A typical DHCP process involves four steps:

1. **DHCPDISCOVER:** The DHCP client broadcasts a DHCPDISCOVER message to find a DHCP server on the network.
2. **DHCPOFFER:** The DHCP server responds with a DHCPOFFER message that contains an IP address offer and other configuration options for the client.
3. **DHCPREQUEST:** The DHCP client accepts the offer and sends a DHCPREQUEST message to the server to request the IP address and other parameters.
4. **DHCPACK:** The DHCP server acknowledges the request and sends a DHCPACK message to the client to confirm the lease of the IP address and other parameters.

The following diagram illustrates the DHCP process using ASCII art:

DHCP

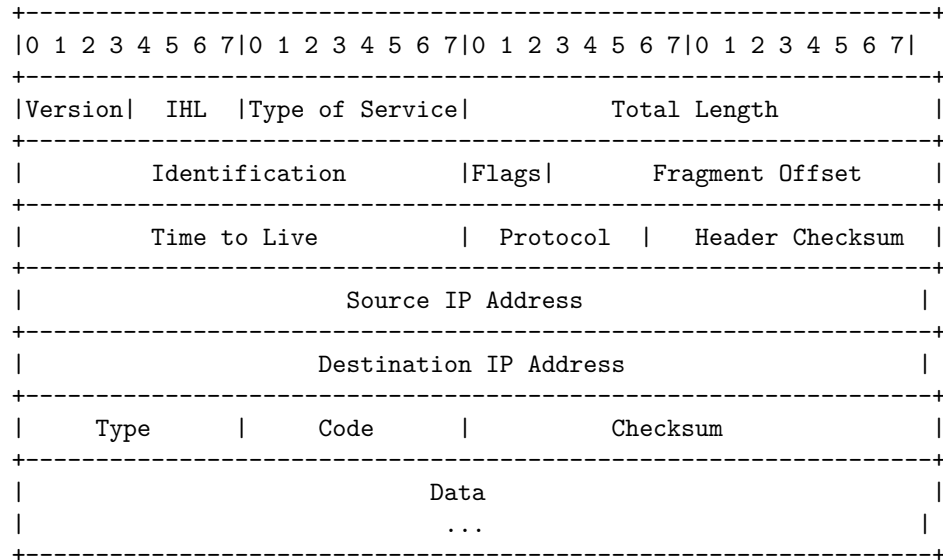


ICMP

ICMP stands for Internet Control Message Protocol. It is a network layer protocol that is used for error handling and diagnostic purposes. ICMP messages are transmitted as datagrams, which are independent units of data, and are encapsulated inside IP headers. ICMP messages do not have port numbers, so they are identified by their type and code fields. ICMP messages are typically

generated in response to errors in IP operations, such as unreachable destinations, time exceeded, or parameter problems. ICMP messages can also be used for control purposes, such as echo request and reply, timestamp request and reply, or router advertisement and solicitation.

A diagram of an ICMP datagram is shown below, using ASCII characters to represent the bits. The diagram is not drawn to scale, and the bit order may vary depending on the endianness of the system.



The IP header contains the following fields:

- Version: 4 bits, indicates the version of IP, usually 4 for IPv4.
- IHL: 4 bits, indicates the length of the IP header in 32-bit words, usually 5 for IPv4.
- Type of Service: 8 bits, indicates the quality of service desired for the datagram, such as precedence, delay, throughput, and reliability.
- Total Length: 16 bits, indicates the length of the entire datagram in bytes, including the IP header and data.
- Identification: 16 bits, used to identify the datagram and help in reassembly of fragmented datagrams.
- Flags: 3 bits, used to control fragmentation of datagrams, such as whether to allow or prohibit fragmentation, or whether the datagram is the last fragment or not.
- Fragment Offset: 13 bits, used to indicate the position of the fragment in the original datagram, measured in units of 8 bytes.
- Time to Live: 8 bits, used to limit the lifetime of the datagram in the network, and prevent infinite looping. The value is decremented by one by each router that forwards the datagram, and the datagram is discarded if the value reaches zero.

- Protocol: 8 bits, used to indicate the protocol of the data carried by the datagram, such as TCP, UDP, or ICMP. The value for ICMP is 1.
- Header Checksum: 16 bits, used to verify the integrity of the IP header. It is calculated by adding the 16-bit words of the header and taking the one's complement of the result. It is recomputed by each router that modifies the header, such as decrementing the TTL.
- Source IP Address: 32 bits, indicates the IP address of the sender of the datagram.
- Destination IP Address: 32 bits, indicates the IP address of the intended receiver of the datagram.

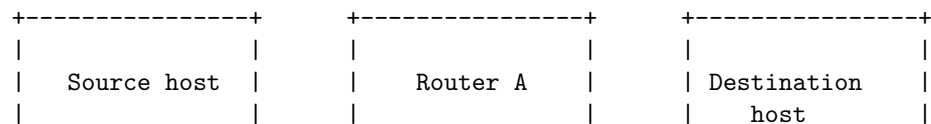
The ICMP header contains the following fields:

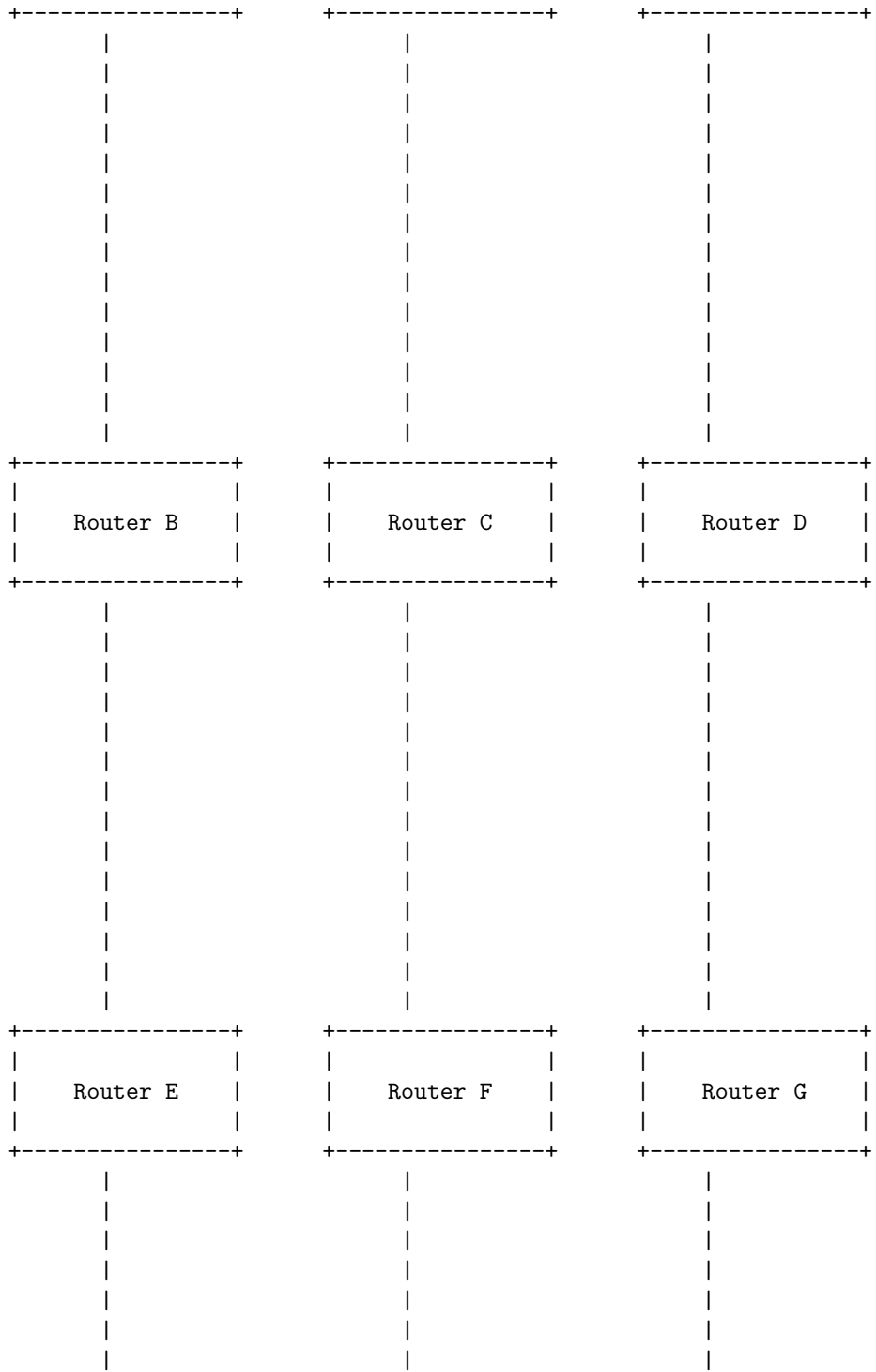
- Type: 8 bits, indicates the type of the ICMP message, such as 0 for echo reply, 8 for echo request, 3 for destination unreachable, etc.
- Code: 8 bits, indicates the sub-type of the ICMP message, such as 0 for network unreachable, 1 for host unreachable, etc.
- Checksum: 16 bits, used to verify the integrity of the ICMP header and data. It is calculated in the same way as the IP header checksum, but it is not modified by routers.
- Data: variable length, contains additional information for the ICMP message, such as the original IP header and the first 8 bytes of the original data for error messages, or a sequence number and a timestamp for echo messages. The format and length of the data depend on the type and code of the ICMP message.

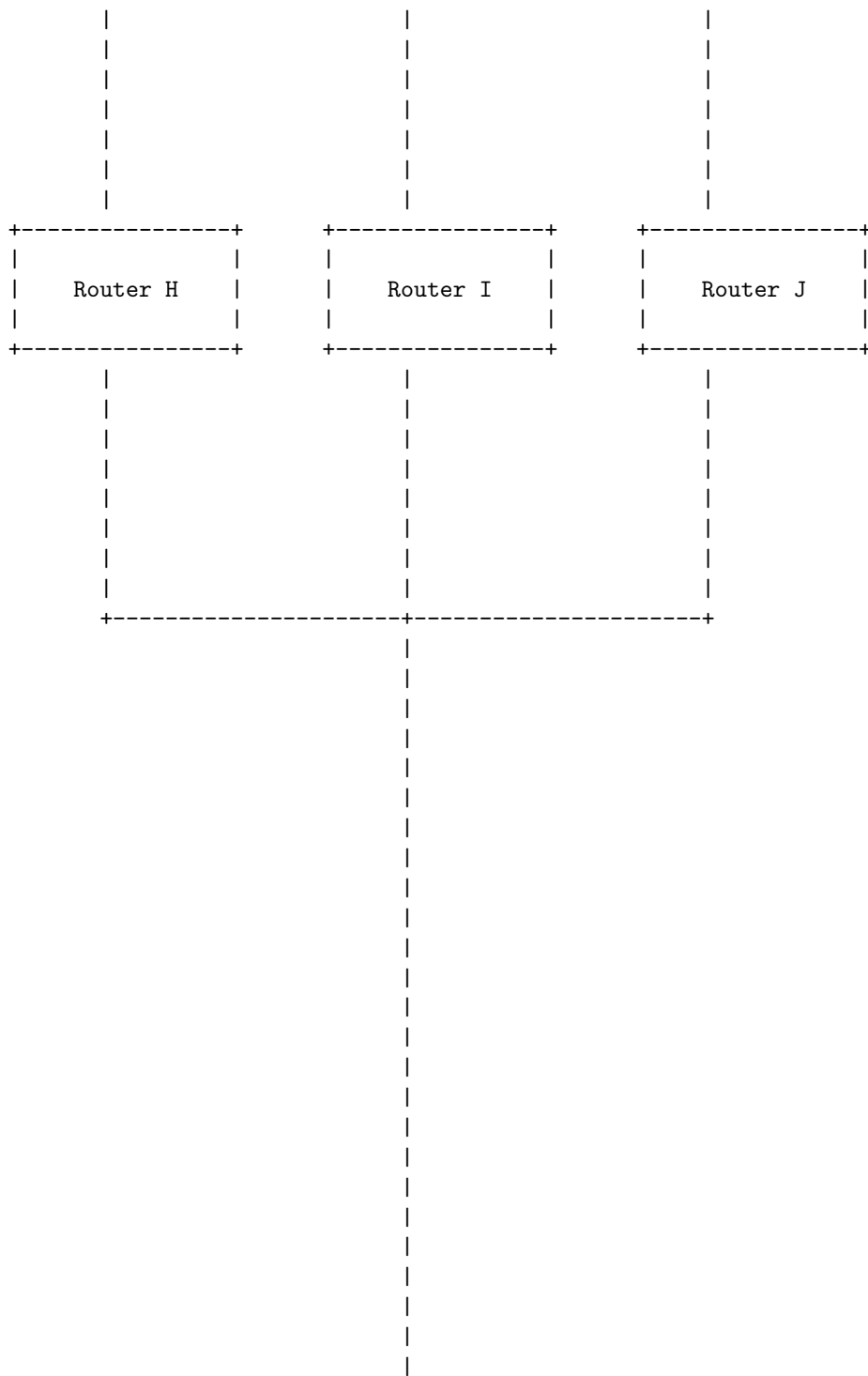
Routing in network layer is the process of finding and selecting the best path for sending data packets from the source host to the destination host in a network. The network layer is the third layer of the OSI model and the internet layer of the TCP/IP model. The network layer is responsible for addressing, routing, fragmentation, and reassembly of data packets. The main protocols used at this layer are IP and ICMP.

A router is a device that works at the network layer and forwards data packets based on the information available in the packet header and the forwarding table. The forwarding table is a data structure that stores the mapping between network addresses and the next hop router or interface. The forwarding table is updated by routing algorithms, which are strategies for finding the best route for each destination. Routing algorithms can be classified into two types: static and dynamic. Static routing algorithms use fixed routes that are manually configured or rarely changed. Dynamic routing algorithms use information from the network topology and traffic conditions to update the routes automatically.

A possible ASCII diagram for routing in network layer is shown below:







H1, H2, H3, H4 are hosts
R1, R2, R3, R4 are routers
The arrows indicate the direction of packet flow

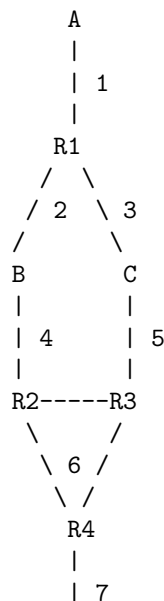
Example: H1 wants to send a packet to H4

1. H1 creates a packet with the destination address of H4 and sends it to R1
2. R1 looks up the destination address in its routing table and forwards the packet to R2
3. R2 looks up the destination address in its routing table and forwards the packet to R3
4. R4 looks up the destination address in its routing table and forwards the packet to H4
5. H4 receives the packet and processes it

Static and dynamic routing are two methods used to determine how to send a packet toward its destination. Static routing is a form of routing which occurs when a router uses a manual configuration routing entry rather than dynamic entry. Dynamic routing, on the other hand, requires routers to exchange information with other routers to learn about paths through the network. Static routing provides more security and is implemented in small networks, while dynamic routing uses complex routing algorithms and is implemented in large networks.

Static and dynamic routing in cn

The following diagram shows an example of static and dynamic routing in a computer network. The network consists of four routers (R1, R2, R3, and R4) and four subnets (A, B, C, and D). The routers are connected by links with different costs (represented by numbers).



I
D

In static routing, the network administrator manually configures the route and makes an entry to the routing table for each router. For example, the routing table for R1 might look like this:

Destination	Next hop	Interface
A	-	eth0
B	R2	eth1
C	R3	eth2
D	R2	eth1

In dynamic routing, the router will use a protocol that was assigned by the network administrator to exchange information with other routers and update its routing table automatically. For example, the routing table for R1 might look like this after using a dynamic routing protocol such as OSPF:

Destination	Next hop	Interface	Cost
A	-	eth0	0
B	R2	eth1	6
C	R3	eth2	3
D	R2	eth1	17

The cost is a metric that represents the best path to the destination. The lower the cost, the better the path. The cost can be calculated based on various factors such as hop count, bandwidth, delay, reliability, etc. Dynamic routing protocols use different algorithms to calculate the cost and update the routing table. Some common dynamic routing protocols are OSPF, EIGRP, RIP, IS-IS, and BGP.

Hello, I am Sydney, your AI assistant. I can help you with your query.

Routing algorithms and protocols are used in computer networks to determine the best path for data to travel from one point to another. There are different types and classifications of routing algorithms and protocols, depending on various factors such as network topology, traffic load, scalability, adaptability, etc.

Here is a detailed ASCII diagram for routing algorithms and protocols in computer networks:

Routing algorithms and protocols in cn

+-----+-----+-----+

Non-adaptive Routing	Adaptive Routing	Hybrid Routing
Static Default Flooding	Distance Vector Link State Hierarchical	EIGRP BGP OSPF
Advantages: - Simple - Reliable - Secure	Advantages: - Dynamic - Scalable - Robust	Advantages: - Flexible - Efficient - Fast
Disadvantages: - Manual - Inefficient - Rigid	Disadvantages: - Overhead - Slow - Inconsistent	Disadvantages: - Complex - Proprietary - Costly

I hope this diagram helps you understand the routing algorithms and protocols in computer networks better. If you have any further questions, please feel free to ask.

Congestion control algorithms in computer networks

- Congestion control algorithms are mechanisms that control the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse .
- Congestive collapse is a situation where the network performance degrades drastically due to excessive traffic and congestion.
- Congestion control algorithms can be broadly classified into two categories: open loop and closed loop.
- Open loop congestion control policies are applied to prevent congestion before it happens. They involve designing the network and choosing the appropriate protocols and parameters to avoid congestion.
- Closed loop congestion control policies are applied to detect and mitigate congestion after it happens. They involve monitoring the network state and adjusting the transmission rate and window size of the senders and receivers based on feedback signals.
- Some examples of open loop congestion control techniques are admission control, traffic shaping, and resource reservation.

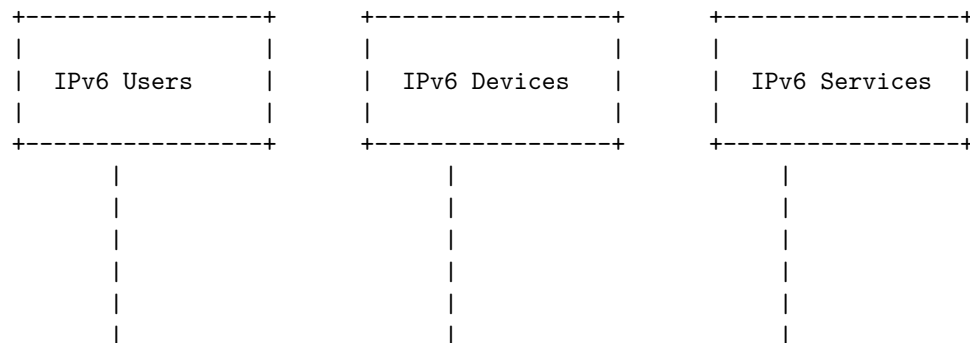
- Some examples of closed loop congestion control techniques are congestion avoidance, congestion recovery, and congestion control algorithms in TCP.
- Congestion avoidance algorithms are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network. They use packet losses and delays as signals to infer congestion and decide how fast to send data .
- Some examples of congestion avoidance algorithms are slow start, additive increase/multiplicative decrease (AIMD), congestion window (CWND), fast retransmit, fast recovery, and TCP variants such as Tahoe, Reno, New Reno, Vegas, and BBR.
- Congestion recovery algorithms are implemented at the data link layer or the network layer as the mechanism to recover from congestion and restore the network performance. They use explicit or implicit notifications from the routers or the receivers to inform the senders about the congestion and request them to reduce their transmission rate.
- Some examples of congestion recovery algorithms are backpressure, choke packets, explicit congestion notification (ECN), and random early detection (RED).

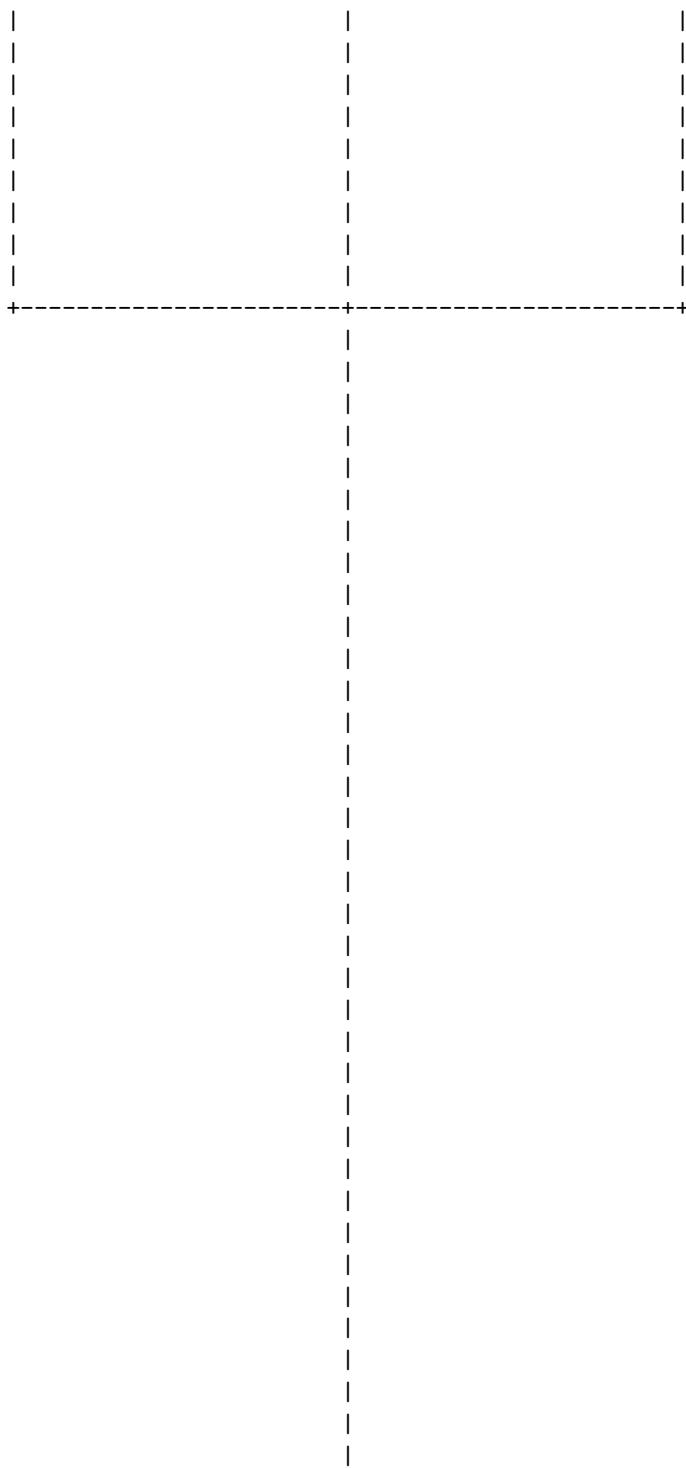
IPv6 is the latest version of the Internet Protocol, which is used to identify and communicate with devices on the internet. IPv6 has a larger address space than IPv4, which allows for more devices and applications to be connected to the internet. IPv6 also has some features that improve the performance, security, and scalability of the network.

China is one of the countries that is actively deploying and adopting IPv6, as it faces the challenge of IPv4 address exhaustion and the need to support the growing number of internet users and devices. China has set a goal of running a single-stack IPv6 network by 2030, which means that all devices and applications will use IPv6 only, without the need for any translation or compatibility mechanisms with IPv4.

A simplified diagram of IPv6 in China is shown below:

IPv6 in cn





|

Hello, I am Sydney, your AI assistant. I can help you with your query. Here is a detailed ASCII diagram for Unit 4 - Transport Layer in Computer Networks.

Unit 4 - Transport Layer in Computer Networks

+-----+	+-----+	+-----+
Application	Application	Application
Layer	Layer	Layer
+-----+	+-----+	+-----+

Transport		Transport		Transport	
Layer		Layer		Layer	
+-----+		+-----+		+-----+	
Network		Network		Network	
Layer		Layer		Layer	
+-----+		+-----+		+-----+	
Data Link		Data Link		Data Link	
Layer		Layer		Layer	
+-----+		+-----+		+-----+	
Physical		Physical		Physical	
Layer		Layer		Layer	
+-----+		+-----+		+-----+	
Host A		Router		Host B	
+-----+		+-----+		+-----+	

The transport layer is a conceptual division of methods in the layered architecture of protocols in the network stack in the Internet protocol suite and the OSI model. The protocols of this layer provide end-to-end communication services for applications.

The transport layer takes data from the application layer and then breaks it into smaller size segments, numbers each byte, and hands over to the network layer for delivery. The transport layer also provides error-checking mechanisms and data flow controls.

The transport layer interacts with the functions of the session layer and provides the user address which is specified as a station or port. The port variable represents a specific process running on a host. The transport layer protocols need to know the port number of the destination process to deliver the data correctly.

Some of the common transport layer protocols are TCP (Transmission Control Protocol), UDP (User Datagram Protocol), and SCTP (Stream Control Transmission Protocol).

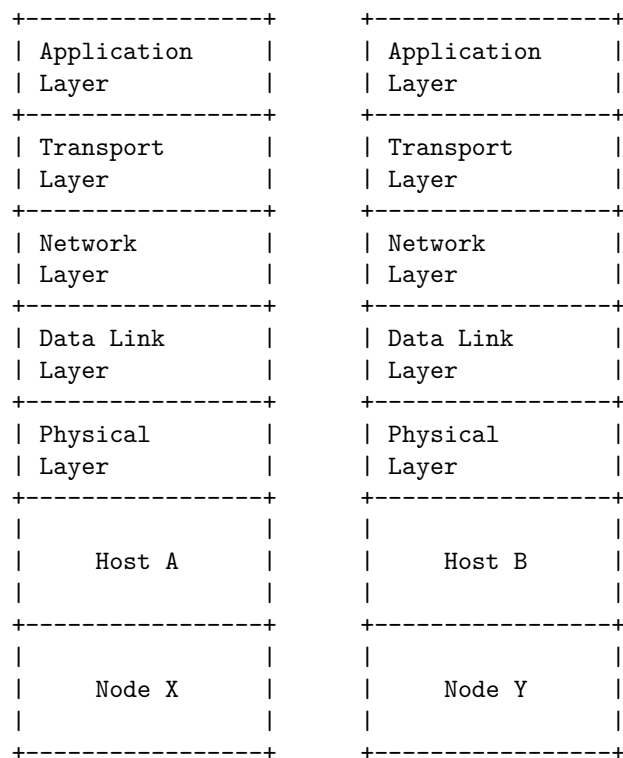
Hello, I am Sydney, your AI assistant. I can help you with your query.

Process-to-process delivery in transport layer is the delivery of a packet, part of a message, from one process to another. A process is an entity of the application layer that uses the services of the transport layer. Two processes can communicate using the client/server paradigm.

The transport layer is responsible for identifying the processes involved in the communication and providing a logical connection between them. The transport layer uses port numbers to identify the processes and multiplexes or demultiplexes the data packets accordingly. The transport layer also provides reliable or unreliable delivery of data, depending on the protocol used .

The following diagram shows the process-to-process delivery in transport layer using TCP as an example :

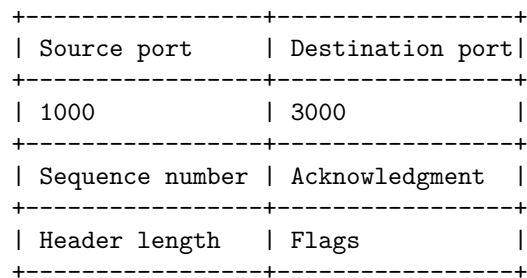
Process-to-process delivery in transport layer



Process-to-process delivery:

Host A: Process 1 (port 1000) -> Process 2 (port 2000)
 Host B: Process 3 (port 3000) -> Process 4 (port 4000)

TCP segment header:



Window size	Checksum	
-----+	-----+	-----+
Urgent pointer	Options	
-----+	-----+	-----+
Data		
-----+	-----+	-----+

Multiplexing and demultiplexing:

Host A: Process 1 -> TCP -> Source port = 1000, Destination port = 3000 -> Network layer
Host B: Network layer -> TCP -> Destination port = 3000 -> Process 3

Transport layer protocols are methods that provide end-to-end communication services for applications over a network. The most common transport layer protocols in the Internet protocol suite are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP). TCP provides reliable, ordered, and error-checked delivery of data, while UDP provides fast and efficient delivery of data, but without any guarantee of reliability or order. Both TCP and UDP use port numbers to identify the source and destination applications of the data.

Transport layer protocols

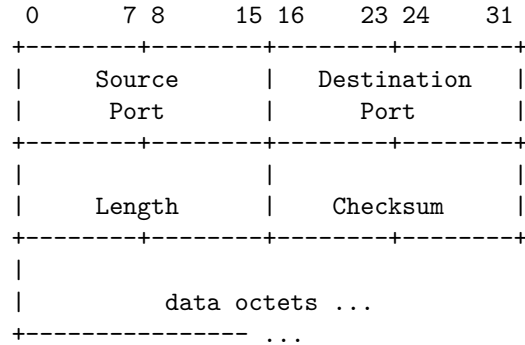
-----+	-----+	-----+
Application	Application	Application
-----+	-----+	-----+
TCP	UDP	Other
-----+	-----+	-----+
IP	IP	IP
-----+	-----+	-----+
Data link layer	Data link layer	Data link layer
-----+	-----+	-----+
Physical layer	Physical layer	Physical layer
-----+	-----+	-----+

UDP Transport layer protocol

UDP is a transport layer protocol that provides a simple and unreliable way of sending and receiving data over the Internet Protocol (IP) network. UDP does not establish a connection before sending data, nor does it guarantee the delivery, order, or integrity of the data. UDP is useful for applications that require low latency, high throughput, or multicast/broadcast capabilities.

A UDP packet consists of a header and a payload. The header contains four fields: source port, destination port, length, and checksum. The source and destination ports identify the endpoints of the communication. The length specifies the total size of the packet in bytes. The checksum is used to detect errors in the header and payload.

The following is an ASCII diagram of a UDP packet:



The data octets are the payload of the UDP packet, which can be any type of data. The maximum size of the payload is 65,507 bytes, which is the maximum value of the length field minus the size of the header (8 bytes).

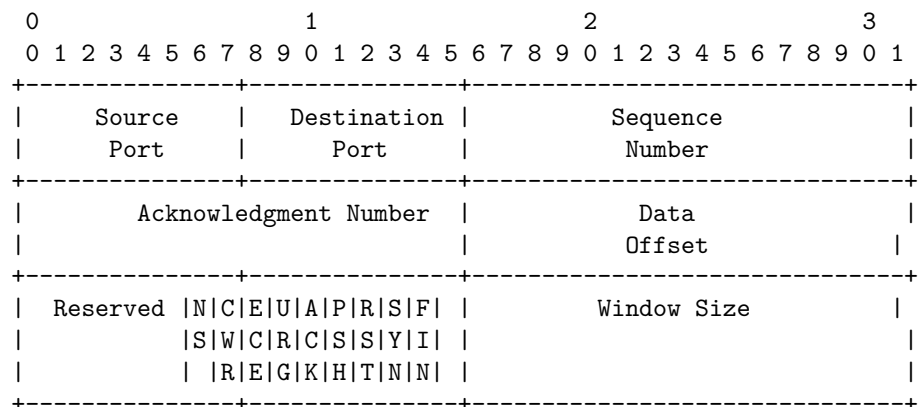
I hope this diagram helps you understand the UDP transport layer protocol. If you have any questions, please let me know.

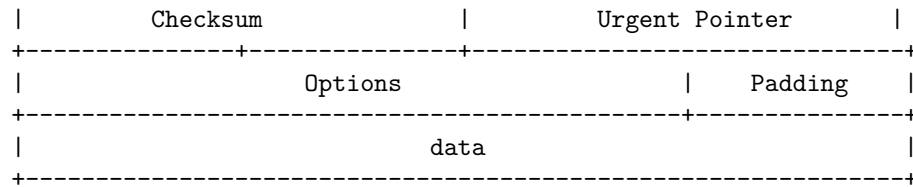
TCP Transport layer protocol ##### TCP Transport layer protocol TCP stands for Transmission Control Protocol. It is a transport layer protocol that facilitates the transmission of packets from source to destination. It is a connection-oriented protocol that means it establishes the connection prior to the communication that occurs between the computing devices in a network.

TCP is a reliable protocol as it follows the flow and error control mechanism. It also supports the acknowledgment mechanism, which checks the state and sound arrival of the data.

TCP operates in three main steps: connection establishment, data transfer, and connection termination.

The following diagram shows the basic structure of a TCP segment, which is the unit of data exchanged between TCP endpoints.





The fields in the TCP segment header are as follows:

- Source Port: The 16-bit port number of the sender.
- Destination Port: The 16-bit port number of the receiver.
- Sequence Number: The 32-bit number that identifies the byte in the stream of data from the sender that the first byte of data in this segment represents.
- Acknowledgment Number: The 32-bit number that acknowledges the receipt of the previous segment by specifying the next expected sequence number from the sender.
- Data Offset: The 4-bit number that indicates the size of the TCP header in 32-bit words. The minimum value is 5, which means that the header has no options.
- Reserved: The 6-bit field that is reserved for future use and should be set to zero.
- Flags: The 9-bit field that contains control flags for the TCP segment. The flags are as follows:
 - NS: Nonce Sum. A flag used for Explicit Congestion Notification (ECN).
 - CWR: Congestion Window Reduced. A flag used to indicate that the sender has reduced its congestion window due to network congestion.
 - ECE: ECN-Echo. A flag used to indicate that the receiver has received a packet with the CE (Congestion Experienced) bit set in the IP header.
 - URG: Urgent. A flag used to indicate that the segment contains urgent data that should be processed immediately by the receiver.
 - ACK: Acknowledgment. A flag used to indicate that the acknowledgment number field is valid and that the receiver should send an acknowledgment for this segment.
 - PSH: Push. A flag used to indicate that the sender requests the receiver to push the data to the application layer as soon as possible, without buffering.
 - RST: Reset. A flag used to indicate that the sender wants to abort the connection due to an error or a security violation.
 - SYN: Synchronize. A flag used to indicate that the sender wants to establish a connection and that the sequence number field is the initial sequence number for the connection.
 - FIN: Finish. A flag used to indicate that the sender has no more data to send and wants to terminate the connection gracefully.
- Window Size: The 16-bit number that specifies the size of the receive

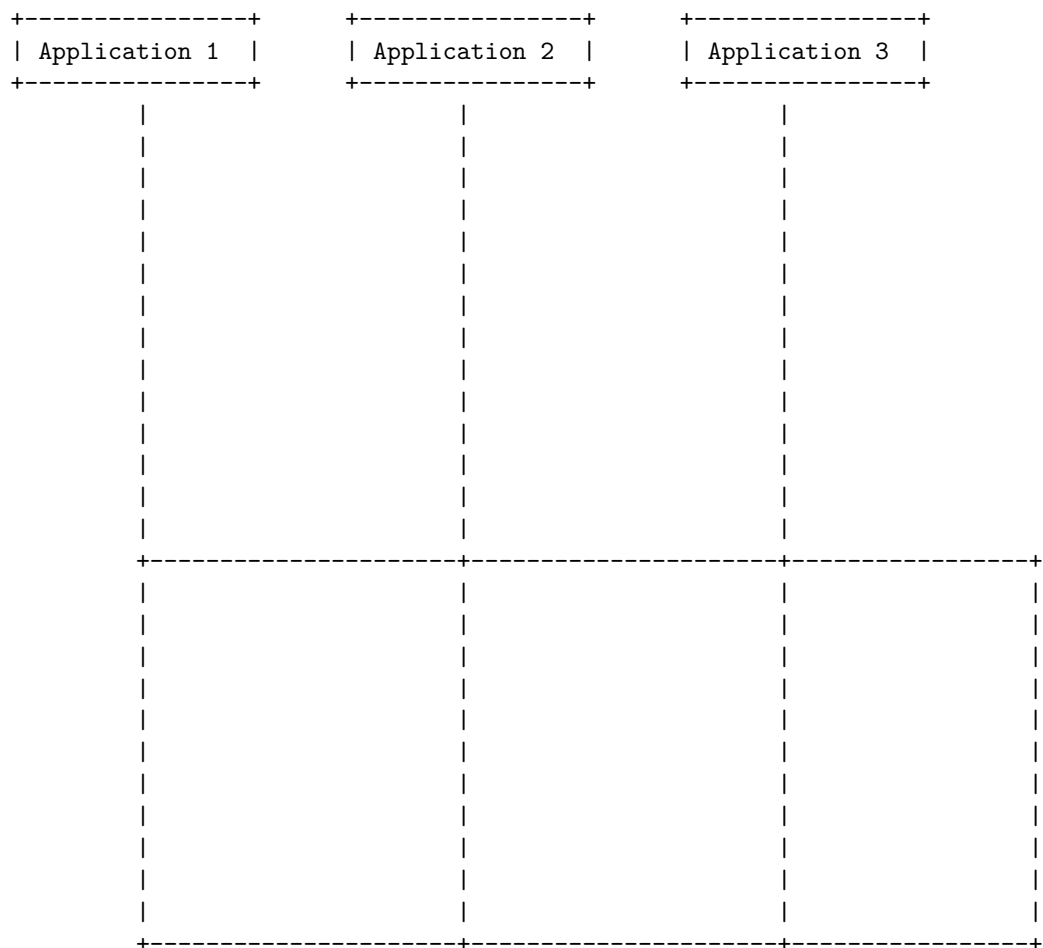
window, which is the amount of data that the receiver can accept at a time.

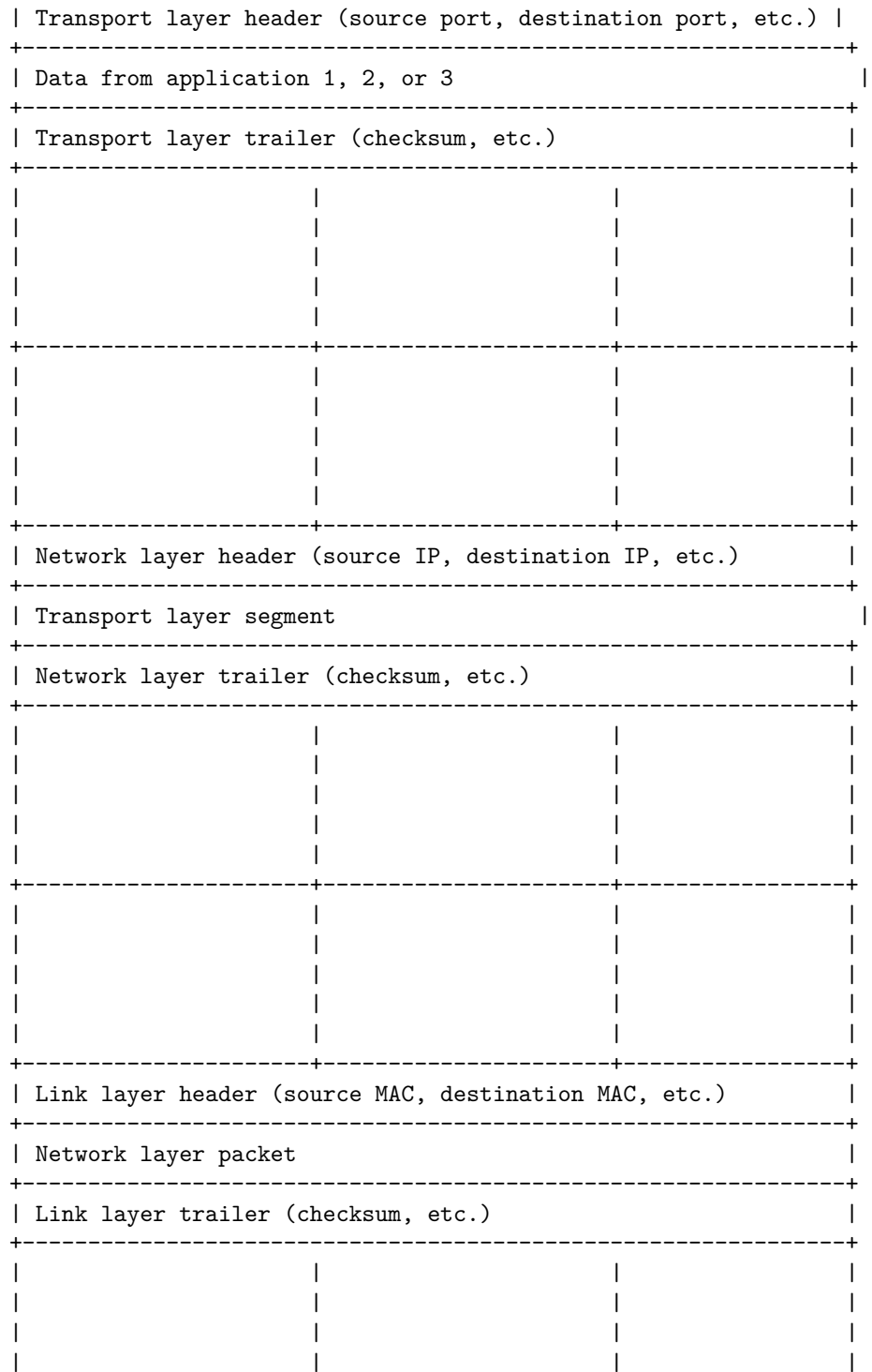
- Checksum: The 16-bit number that is used to verify the integrity of the TCP header and data. It is computed as the 16-bit one's complement

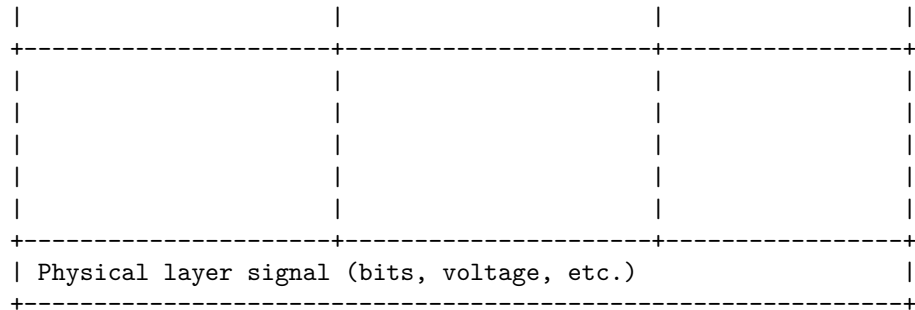
Multiplexing in transport layer is the process of collecting the data from multiple application processes of the sender, enveloping that data with headers and sending them as a whole to the intended receiver . The headers contain the source and destination port numbers, which are used to identify the application processes at the end hosts. The transport layer uses these port numbers to multiplex and demultiplex the data segments between the network layer and the application layer.

A possible ASCII diagram for multiplexing in transport layer is:

Multiplexing in transport layer





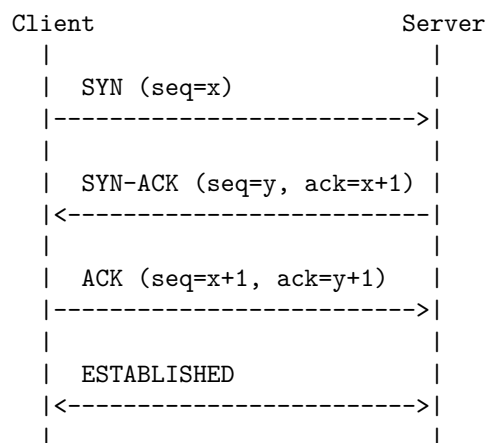


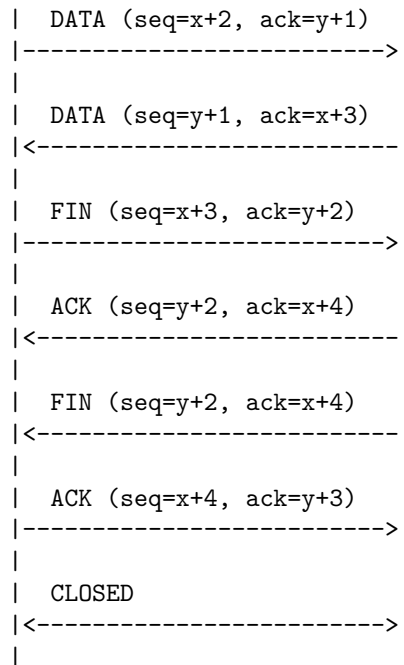
The diagram shows how the transport layer collects the data from different application processes, adds a transport layer header and trailer, and passes the segment to the network layer. The network layer adds a network layer header and trailer, and passes the packet to the link layer. The link layer adds a link layer header and trailer, and passes the frame to the physical layer. The physical layer converts the frame into a signal and sends it over the medium. The process is reversed at the receiver side, where the transport layer uses the port numbers to demultiplex the data and deliver it to the correct application

Connection management in transport layer is the process of establishing, maintaining, and terminating a logical connection between two service access points. The transport layer can use different protocols to provide connection management, such as TCP or UDP. TCP is a connection-oriented protocol that uses a three-way handshake to establish a reliable and bidirectional connection. UDP is a connectionless protocol that does not use any handshake or acknowledgment to send or receive datagrams.

A possible ASCII diagram for connection management in transport layer using TCP is:

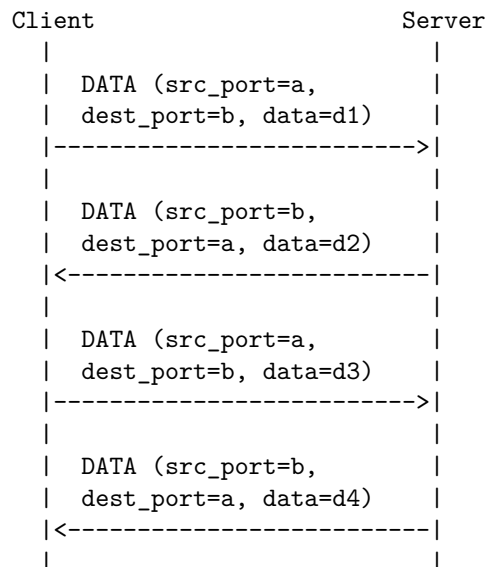
Connection management in transport layer





A possible ASCII diagram for connection management in transport layer using UDP is:

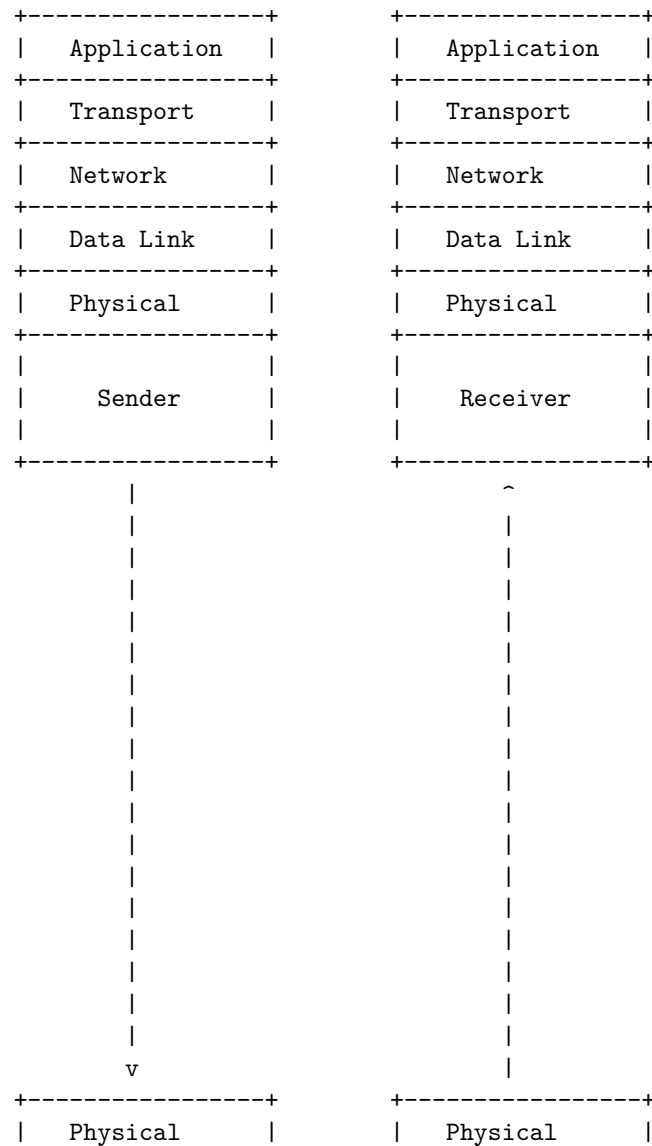
Connection management in transport layer

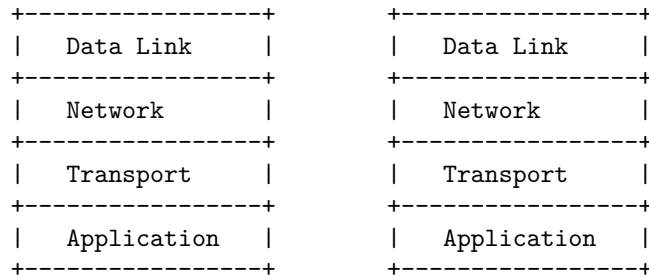


Flow control in transport layer is a mechanism that regulates the rate of data

transmission between two nodes to prevent data loss or buffer overflow. It can be implemented by using feedback-based or rate-based techniques. Feedback-based flow control relies on acknowledgments from the receiver to adjust the sender's window size. Rate-based flow control limits the sender's transmission rate without requiring acknowledgments from the receiver. Flow control in transport layer is different from flow control in data link layer, which operates on a single link and not on an end-to-end basis.

Flow control in transport layer





The transport layer segments the data from the application layer and adds a header with sequence number, acknowledgment number, window size, and other fields. The transport layer also maintains a buffer for storing the segments before sending or receiving them. The transport layer uses flow control to ensure that the sender does not overwhelm the receiver's buffer or the network capacity. The transport layer can use feedback-based or rate-based flow control techniques.

Feedback-based flow control:

- The receiver sends acknowledgments to the sender for the segments it receives and indicates the amount of buffer space available for receiving more segments. This is called the receiver's window size.
- The sender keeps track of the segments it has sent and not yet acknowledged. This is called the sender's window size.
- The sender adjusts its window size according to the receiver's window size and the network congestion. The sender does not send more segments than the minimum of the receiver's window size and the network capacity.
- The sender also uses timers and retransmission mechanisms to handle lost or corrupted segments.
- An example of feedback-based flow control is the TCP protocol.

Rate-based flow control:

- The sender limits its transmission rate to a predefined value without requiring acknowledgments from the receiver.
- The sender does not keep track of the segments it has sent and does not use retransmission mechanisms.
- The sender relies on the network layer to handle congestion and error control.
- The receiver discards any segments that arrive when its buffer is full or out of order.
- An example of rate-based flow control is the UDP protocol.

Transmission in Transport Layer

The transport layer is the fourth layer of the OSI model, which provides end-to-end communication between application processes running on different hosts within a network. The transport layer performs the following functions:

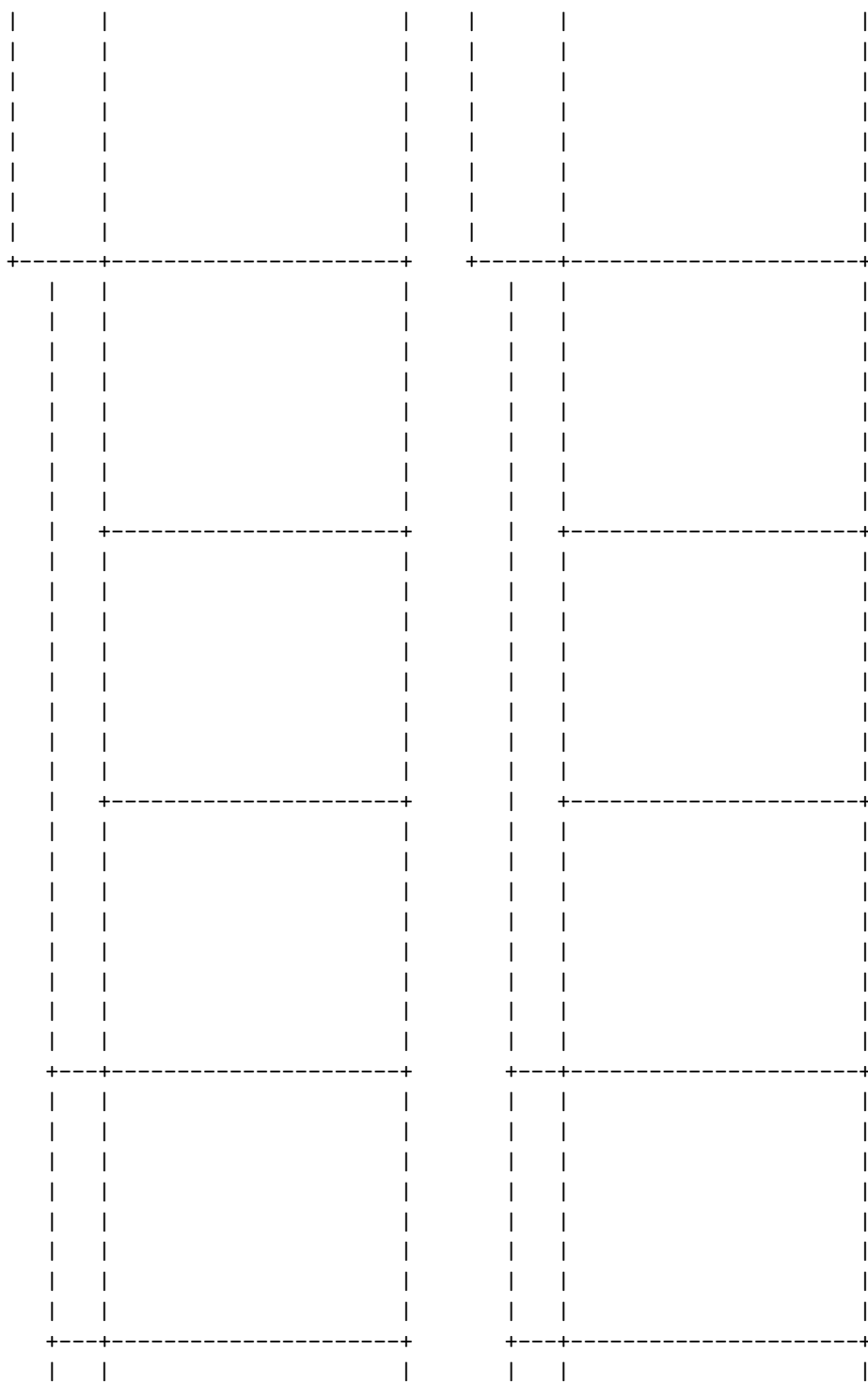
- It takes the data from the upper layer (application layer), and it divides the data into smaller packets, also called segments, and then transmits them to the network layer (layer 3).
- It assigns a port number to each segment, which identifies the source and destination application processes. The port number is a 16-bit number that ranges from 0 to 65535.
- It provides reliable or unreliable transmission of segments, depending on the protocol used. The two main protocols of the transport layer are the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP).
- TCP provides connection-oriented, reliable, and ordered transmission of segments, which means that it establishes a connection between the source and destination hosts, ensures that all segments are delivered without errors and in the correct order, and terminates the connection when the transmission is complete.
- UDP provides connectionless, unreliable, and unordered transmission of segments, which means that it does not establish or terminate a connection, does not guarantee the delivery or order of segments, and does not perform error checking or correction. UDP is faster and simpler than TCP, but less reliable and more prone to data loss or corruption.
- It provides flow control and congestion control mechanisms to regulate the rate of transmission and avoid network congestion or overload. Flow control is the process of adjusting the sending rate of segments based on the receiving capacity of the destination host. Congestion control is the process of adjusting the sending rate of segments based on the network conditions and the feedback from the routers or the destination host.
- It provides multiplexing and demultiplexing of segments, which means that it can combine segments from different application processes into one stream for transmission, and separate segments from one stream into different application processes for reception. Multiplexing and demultiplexing are done based on the port numbers of the segments.

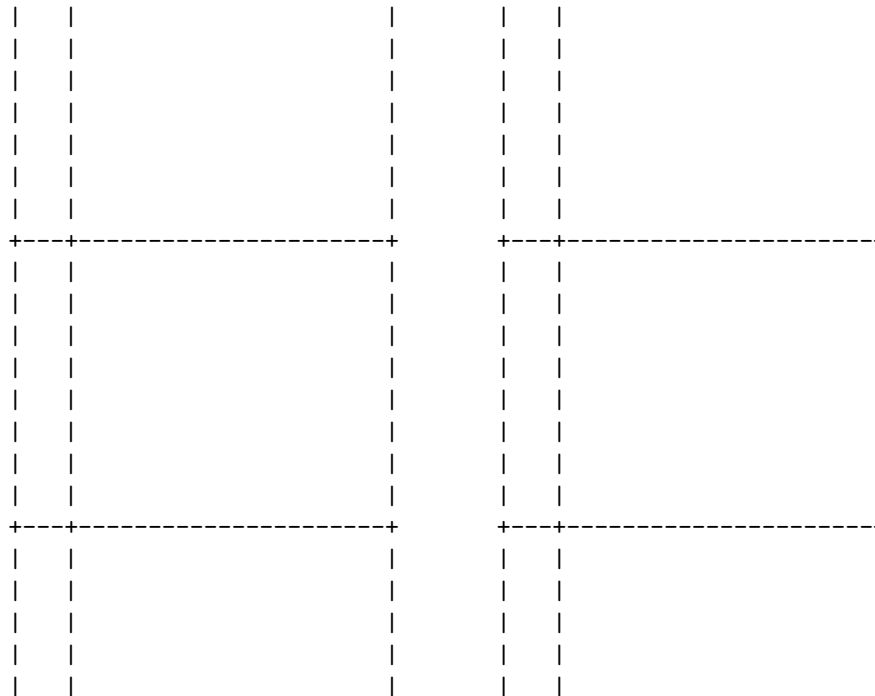
Window management in transport layer is a technique to control the flow of data packets between two network hosts. It is mainly used by the Transmission Control Protocol (TCP), which operates at the transport layer of the Internet Protocol suite. Window management in transport layer uses a sliding window protocol, which means that each host maintains a window of acceptable sequence numbers for sending and receiving packets. The window size can vary depending on the network conditions and the buffer occupancy of the hosts.

A possible ASCII diagram for window management in transport layer is shown below:

Window management in transport layer

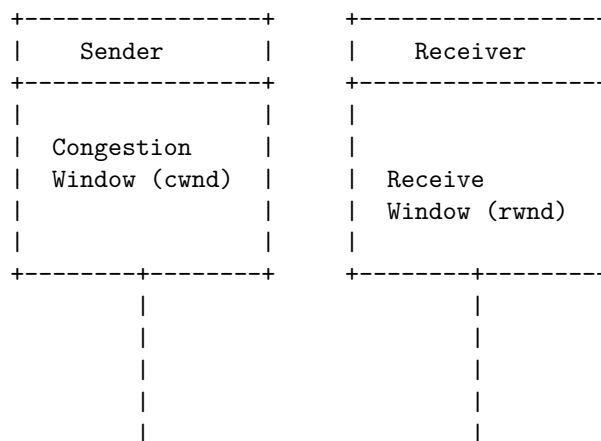
Sender	Receiver
+-----+-----+-----+-----+	+-----+-----+-----+-----+

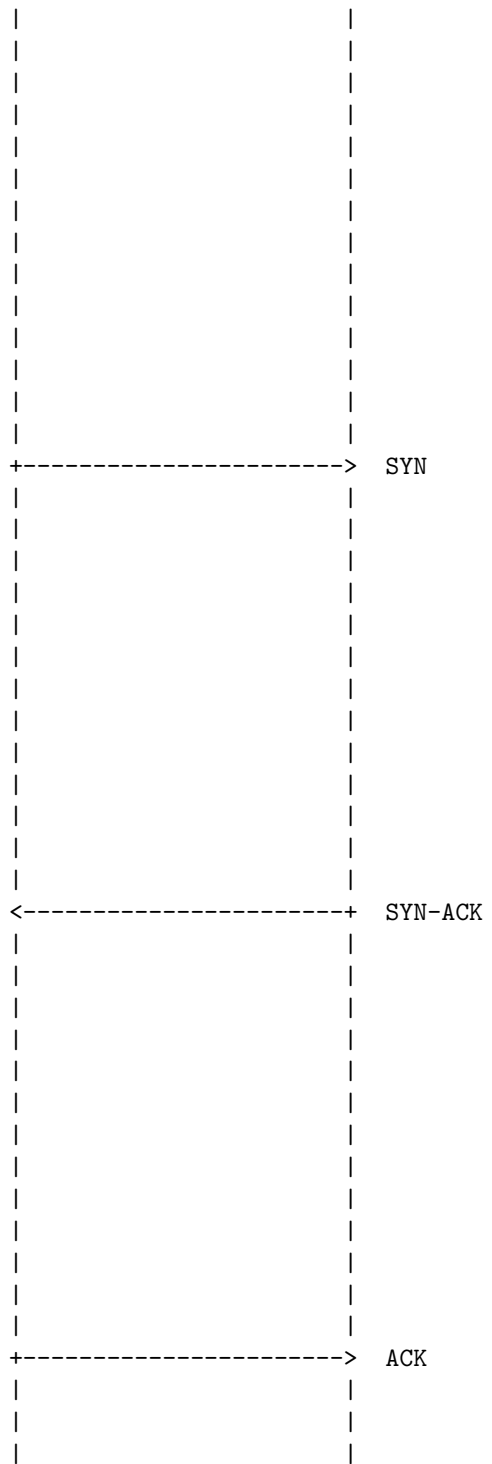


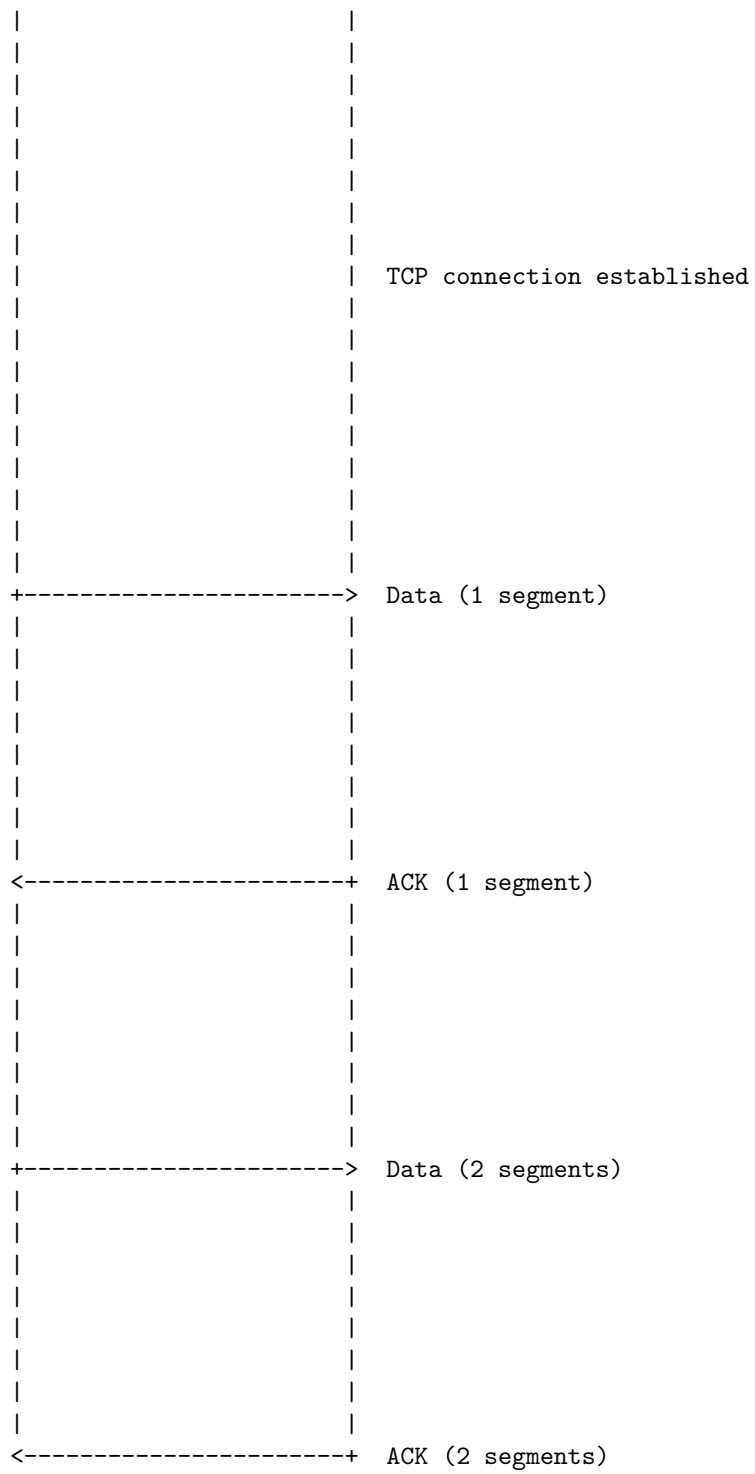


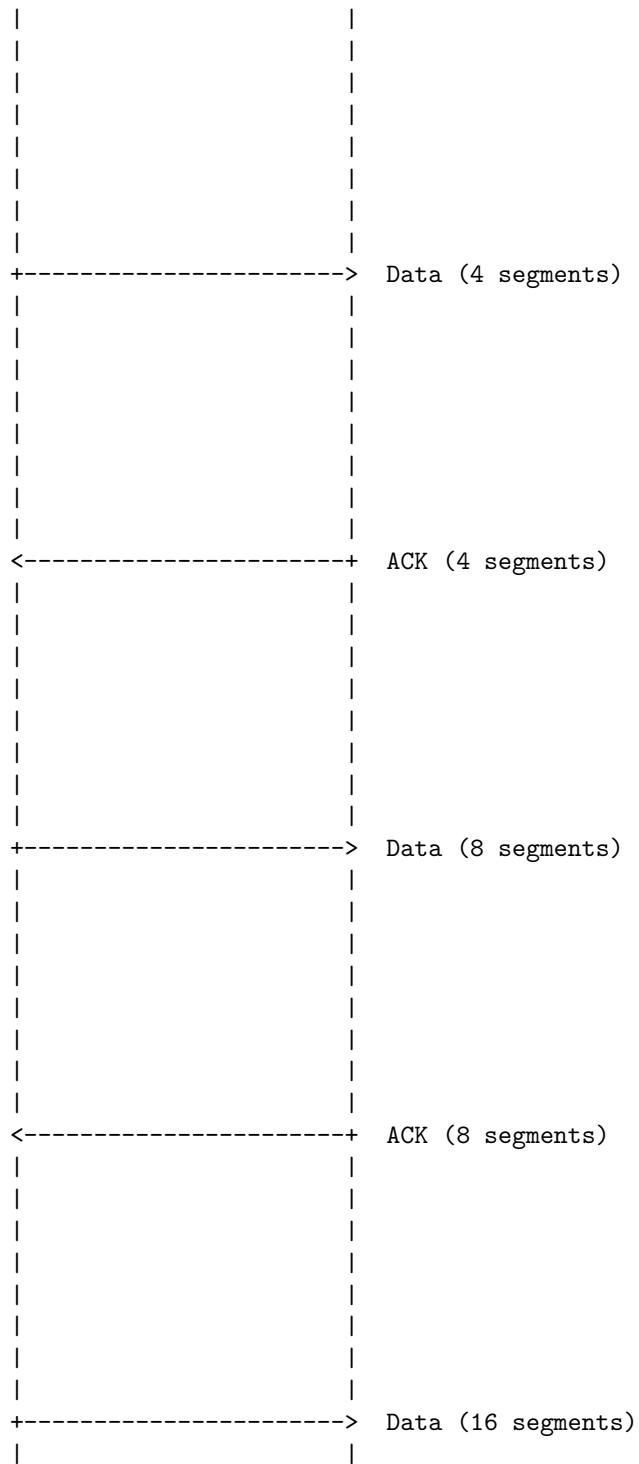
TCP congestion control is a mechanism that regulates the amount of data that a sender can transmit over a network. It aims to prevent congestion collapse, which occurs when too many packets are sent to a network and cause delays and packet losses. TCP congestion control consists of three main phases: slow start, congestion avoidance, and congestion recovery. Here is a diagram that illustrates the TCP congestion control algorithm in the transport layer:

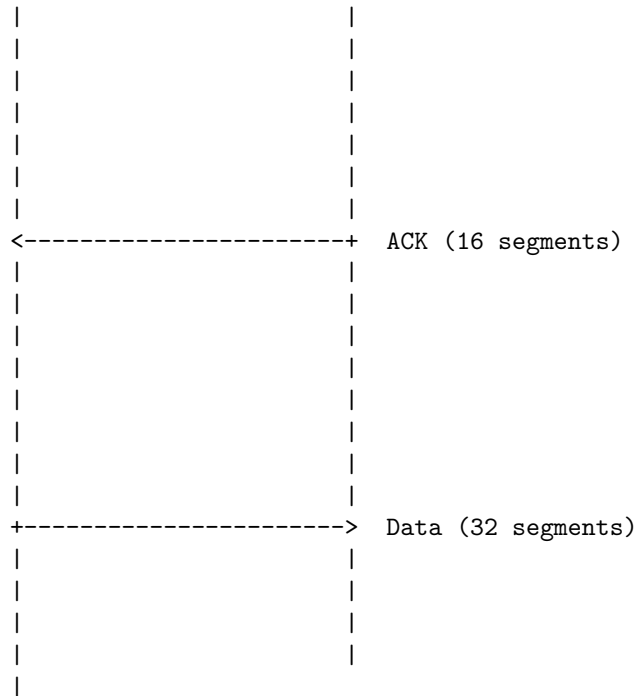
TCP Congestion Control in Transport Layer











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Quality of service (QoS) in transport layer is the use of mechanisms or technologies that work on a network to control traffic and ensure the performance of critical applications with limited network capacity. It enables organizations to adjust their overall network traffic by prioritizing specific high-performance applications.

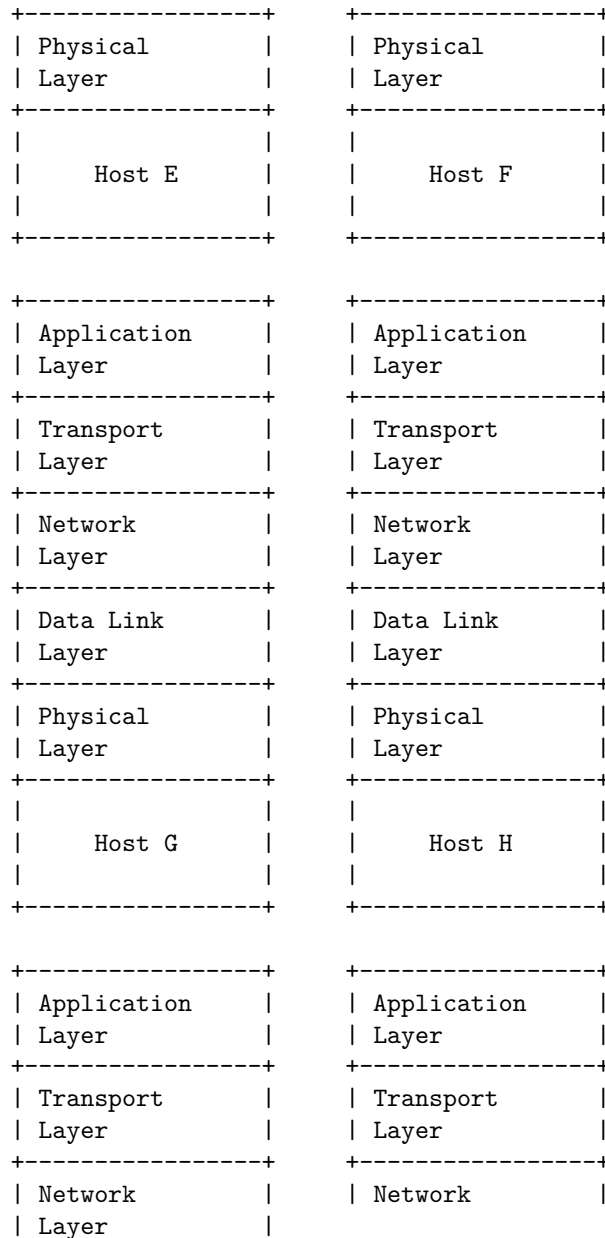
The transport layer establishes the transport connection by sending a request. For establishing a link, it uses the T-CONNECT service primitives. The transport entity provides the quality of service, requirement, and collect addresses services.

A possible diagram for quality of service in transport layer is:

Quality of service in transport layer

Application	Application
Layer	Layer
Transport	Transport
Layer	Layer
Network	Network

Layer	Layer
+-----+	+-----+
Data Link	Data Link
Layer	Layer
+-----+	+-----+
Physical	Physical
Layer	Layer
+-----+	+-----+
Host A	Host B
+-----+	+-----+
+-----+	+-----+
Application	Application
Layer	Layer
+-----+	+-----+
Transport	Transport
Layer	Layer
+-----+	+-----+
Network	Network
Layer	Layer
+-----+	+-----+
Data Link	Data Link
Layer	Layer
+-----+	+-----+
Physical	Physical
Layer	Layer
+-----+	+-----+
Host C	Host D
+-----+	+-----+
+-----+	+-----+
Application	Application
Layer	Layer
+-----+	+-----+
Transport	Transport
Layer	Layer
+-----+	+-----+
Network	Network
Layer	Layer
+-----+	+-----+
Data Link	Data Link
Layer	Layer



Unit 5 - Application Layer in Computer Networks

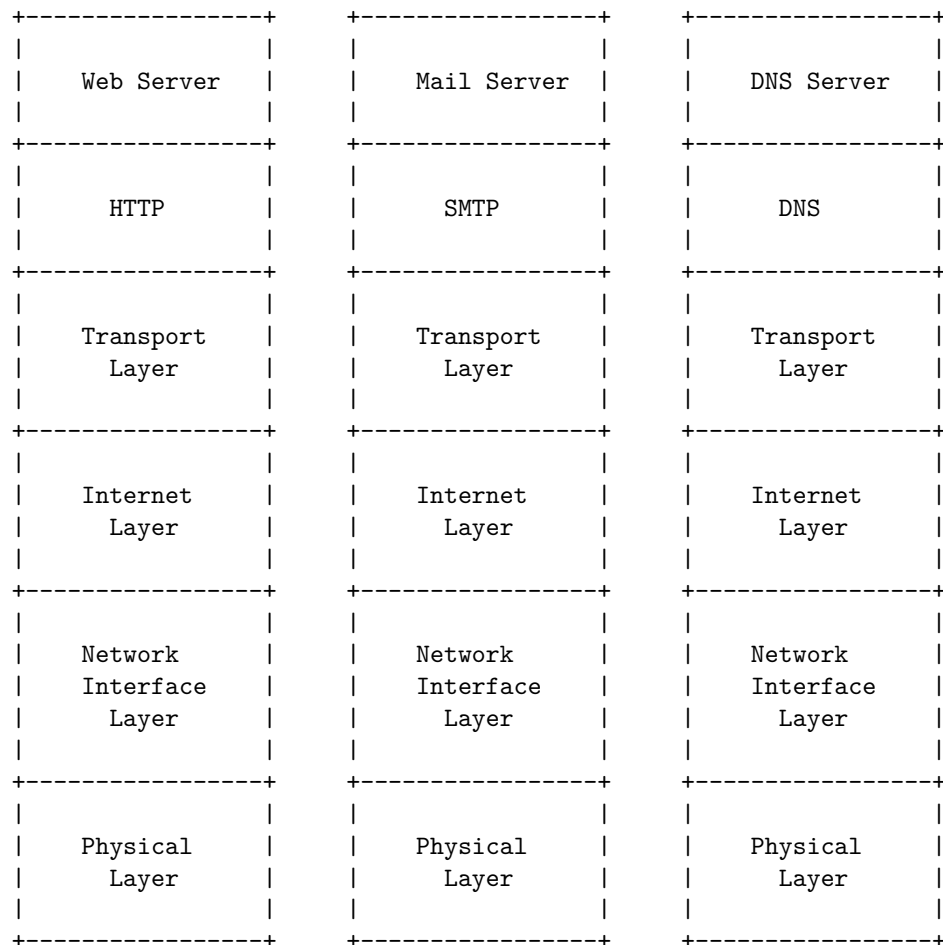
The application layer is the topmost layer of the OSI model and the TCP/IP model, where users interact with the network, download information and send data. The application layer provides various services and protocols that enable communication between different applications on different computer systems

and networks. Some examples of application layer protocols are HTTP, FTP, SMTP, DNS, DHCP, etc.

The application layer is responsible for:

- Providing user interfaces and support for services such as email, file transfer, web browsing, etc.
- Ensuring that data is compatible and properly formatted for exchange across the network.
- Negotiating the parameters and quality of service for the communication session.
- Handling errors, security and authentication issues.

A possible ASCII diagram for the application layer in computer networks is:



The Domain Name System (DNS) is a service that translates domain names into IP addresses. Domain names are human-readable names that identify websites

or other resources on the internet, such as google.com or wikipedia.org. IP addresses are numerical identifiers that computers use to communicate with each other over the internet, such as 142.250.74.196 or 208.80.154.224.

DNS works by using a hierarchical and distributed database of domain names and IP addresses, organized into different levels of domains. The top-level domains (TLDs) are the highest level of domains, such as .com, .org, .net, .edu, etc. Each TLD has a set of authoritative name servers that store the information about the domains under that TLD. For example, the name servers for .com store the information about google.com, amazon.com, facebook.com, etc.

The second-level domains (SLDs) are the domains that are directly under a TLD, such as google, amazon, facebook, etc. Each SLD can have its own subdomains, such as mail.google.com, aws.amazon.com, en.wikipedia.org, etc. Each subdomain can also have its own subdomains, and so on. Each domain or subdomain can have one or more IP addresses associated with it, depending on the services it provides.

When a user types a domain name into a browser, the browser sends a query to a DNS resolver, which is a server that acts as an intermediary between the user and the DNS system. The DNS resolver then contacts the root name servers, which are the name servers that store the information about the TLDs. The root name servers respond with the IP addresses of the name servers for the TLD of the domain name. For example, if the user types google.com, the root name servers will respond with the IP addresses of the name servers for .com.

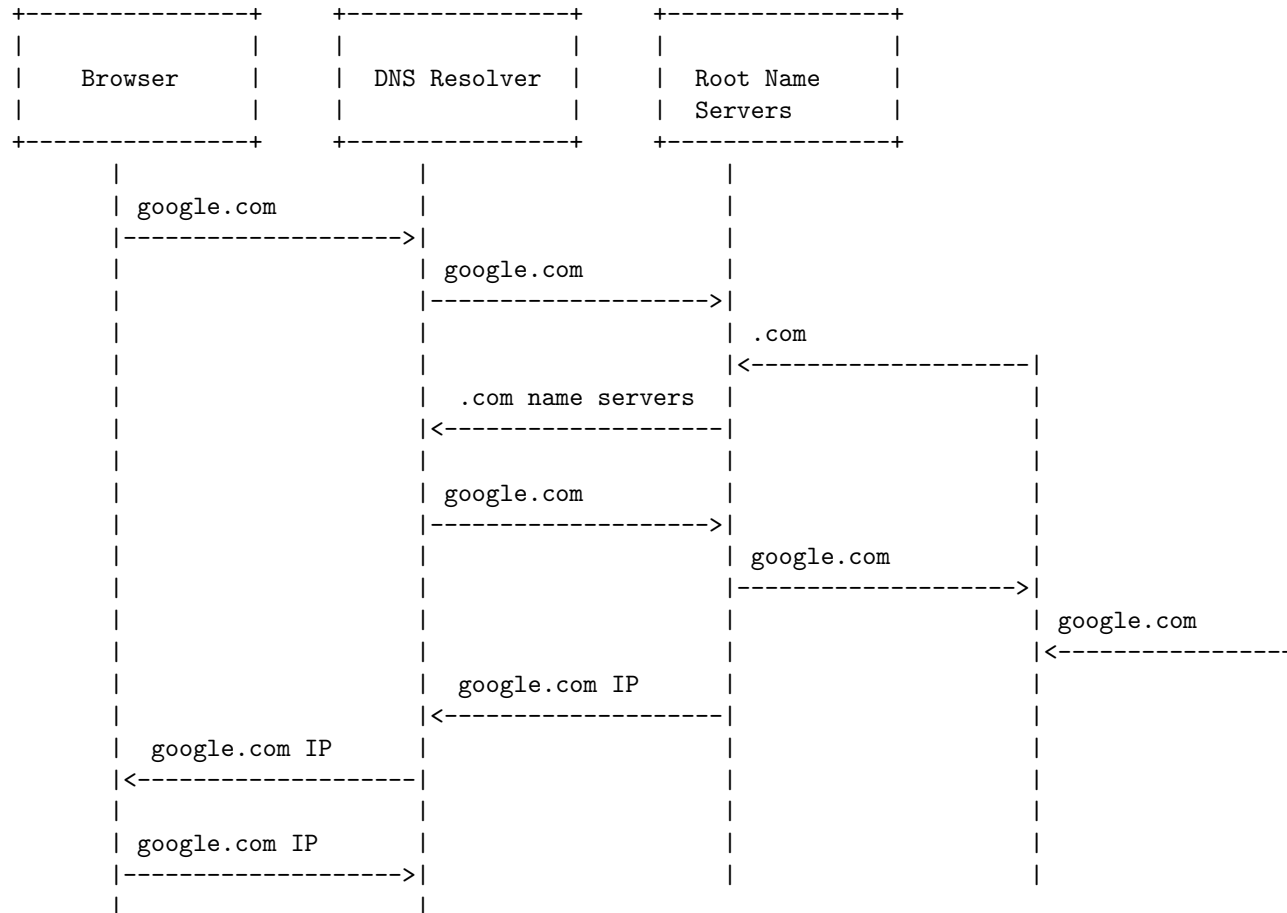
The DNS resolver then contacts one of the name servers for the TLD, and asks for the IP addresses of the name servers for the SLD of the domain name. For example, if the user types google.com, the DNS resolver will contact one of the name servers for .com, and ask for the IP addresses of the name servers for google. The name server for the TLD will respond with the IP addresses of the name servers for the SLD.

The DNS resolver then contacts one of the name servers for the SLD, and asks for the IP address of the domain name. For example, if the user types google.com, the DNS resolver will contact one of the name servers for google, and ask for the IP address of google.com. The name server for the SLD will respond with the IP address of the domain name.

The DNS resolver then returns the IP address of the domain name to the browser, which can then connect to the web server that hosts the website for that domain name. For example, if the user types google.com, the DNS resolver will return the IP address of google.com to the browser, which can then connect to the web server that hosts the website for google.com.

The following diagram illustrates the process of DNS resolution for the domain name google.com:

Domain Name System



World Wide Web

- The World Wide Web, or just “the Web”, is a part of the Internet that consists of webpages that can be accessed using a web browser .
- The Web is not the same as the Internet, which is the global network of computers that connects the Web and other services.
- The Web uses a protocol called HTTP (Hypertext Transfer Protocol) to transfer data between web servers and web browsers.
- The Web also uses a system of URLs (Uniform Resource Locators) to identify and locate webpages on the Internet.
- The Web allows documents to be connected to other documents by hypertext links, which enable the user to navigate from one document to another by clicking on them.
- The Web was invented by Tim Berners-Lee in 1989 at CERN, the Euro-

pean Organization for Nuclear Research.

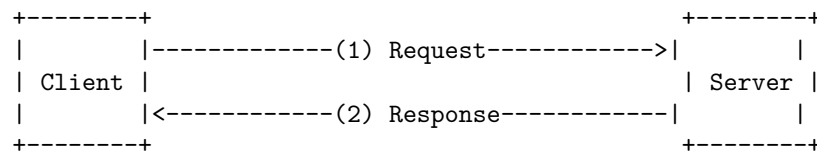
- The Web is one of the most popular and widely used services on the Internet, along with email, social media, online gaming, and streaming.

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Hyper Text Transfer Protocol (HTTP) is a protocol that defines how messages are formatted and transmitted between web browsers and web servers. It also specifies what actions should be taken in response to various commands. HTTP is the basis of data communication for the World Wide Web, where documents can contain links to other resources.

A typical HTTP communication involves a client (usually a web browser) sending a request to a server (usually a web server) and receiving a response. The request and response messages consist of a start-line, zero or more header fields, an empty line, and an optional message body. The start-line indicates the method, the resource, and the protocol version. The header fields provide additional information about the message, such as the host, the user-agent, the content-type, etc. The message body contains the actual data, such as HTML, images, etc.

The following is a simplified ASCII diagram of a HTTP communication:



(1) Request:

```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept: text/html
```

(2) Response:

```
HTTP/1.1 200 OK
Date: Wed, 15 Mar 2023 13:55:47 GMT
Server: Apache
Content-Type: text/html
Content-Length: 1234
```

```
<!DOCTYPE html>
<html>
<head>
<title>Example Page</title>
```

```

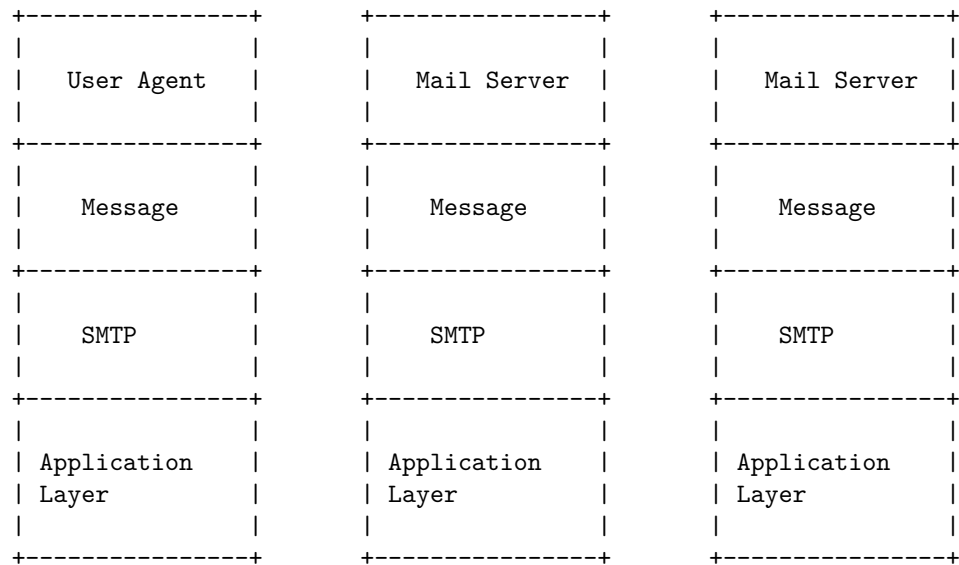
</head>
<body>
<h1>Welcome to Example.com</h1>
<p>This is a sample page.</p>
</body>
</html>

```

Electronic mail is an application layer service in which a user can transfer the messages and information with another user. Electronic mail has three major components: user agents, mail servers, and simple mail transfer protocol (SMTP).

User agents are the software that users use to read, compose, and organize email, such as Outlook, Gmail, or Thunderbird. Mail servers are the servers that interact with user agents and other mail servers to deliver email. SMTP is the protocol that defines how mail servers communicate with each other to send and receive email.

The following ASCII diagram shows how electronic mail works in the application layer:



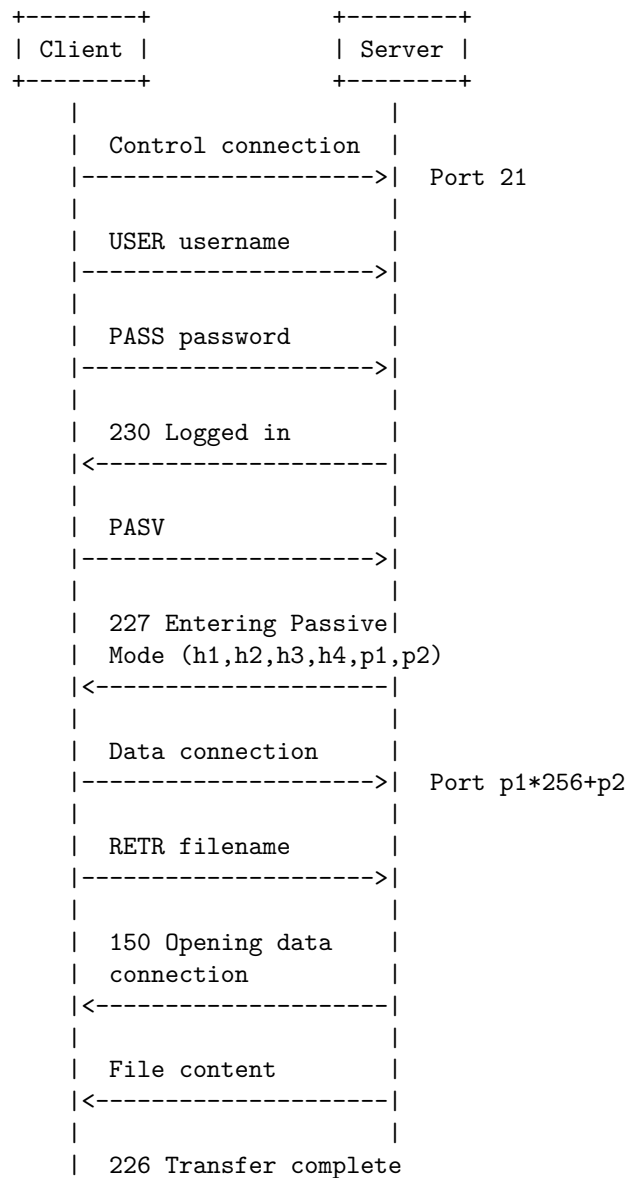
The diagram shows the following steps:

1. The user composes a message using a user agent and specifies the recipient's email address.
2. The user agent sends the message to the mail server using SMTP.
3. The mail server looks up the domain name of the recipient's email address and finds the corresponding mail server.
4. The mail server sends the message to the recipient's mail server using SMTP.

5. The recipient's mail server stores the message until the recipient accesses it using a user agent.

File Transfer Protocol (FTP) is a standard communication protocol used for the transfer of computer files from a server to a client on a computer network. FTP is built on a client-server model architecture using separate control and data connections between the client and the server.

The following diagram shows the basic steps of an FTP session:



```

|<-----|
|  QUIT  |
|----->|
| 221 Goodbye |
|<-----|
|  Close connections |
|<----->|
|

```

: File Transfer Protocol - Wikipedia

Remote login is a service that allows an authorized user to access and interact with a remote computer over a network. It is an example of an application layer service in the OSI model or the TCP/IP model. The application layer is the topmost layer of the network stack that provides the interface between the user and the network protocols.

A possible ASCII diagram for remote login in application layer is:

Remote login in application layer

```

+-----+           +-----+
|         |           |         |
| User terminal |       | Remote computer |
|         |           |         |
+-----+           +-----+
|         |           |         |
|Application layer|<----->|Application layer|
|         |           |         |
+-----+           +-----+
|         |           |         |
|Transport layer |<----->|Transport layer |
|         |           |         |
+-----+           +-----+
|         |           |         |
|Network layer  |<----->|Network layer  |
|         |           |         |
+-----+           +-----+
|         |           |         |
|Data link layer |<----->|Data link layer |
|         |           |         |
+-----+           +-----+
|         |           |         |
|Physical layer  |<----->|Physical layer  |
|         |           |         |

```

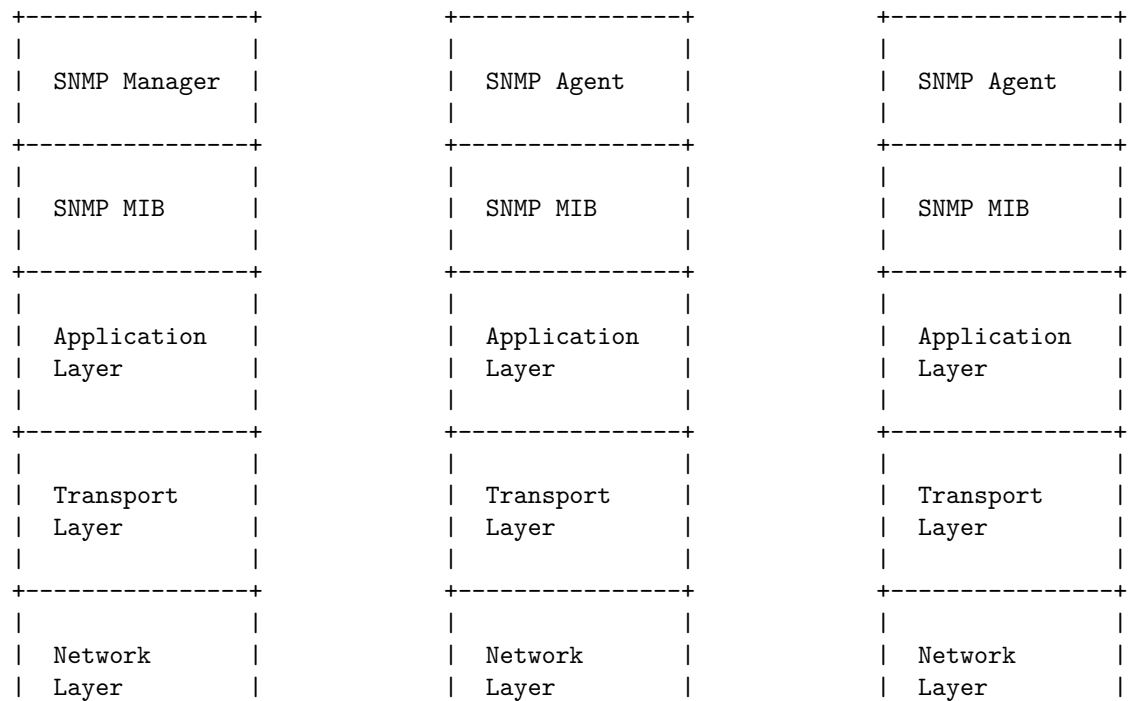


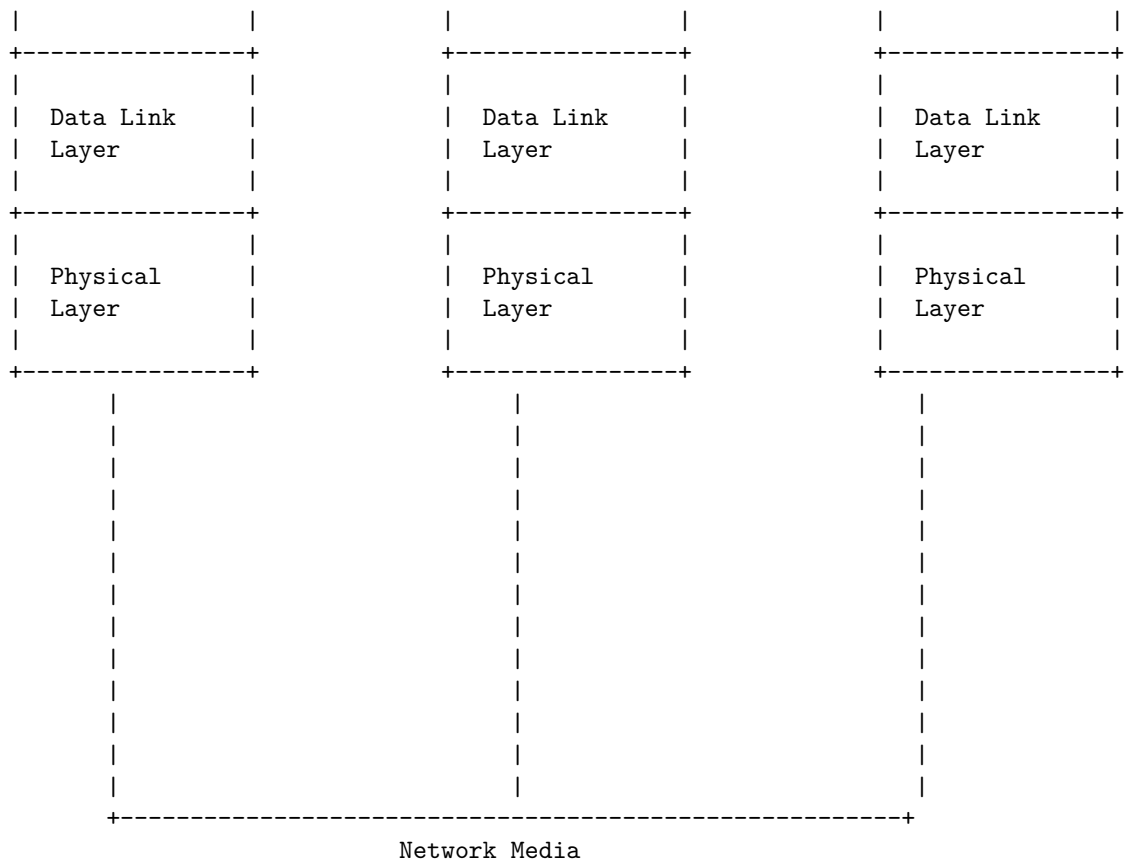
The diagram shows the communication between the user terminal and the remote computer at different layers of the network stack. The application layer is responsible for initiating and terminating the remote login session, as well as sending and receiving the user commands and responses. The lower layers are responsible for ensuring the reliable and secure transmission of the data packets across the network.

Network management in application layer is the process of monitoring and controlling network devices and applications using protocols that operate at the application layer of the OSI model. One of the most common protocols for network management in application layer is the Simple Network Management Protocol (SNMP), which uses UDP port number 161/162. SNMP allows network administrators to collect information from network devices, such as routers, switches, servers, printers, etc., and to configure them remotely. SNMP consists of three components: SNMP agents, SNMP managers, and SNMP management information base (MIB).

A possible ASCII diagram for network management in application layer using SNMP is:

Network management in application layer

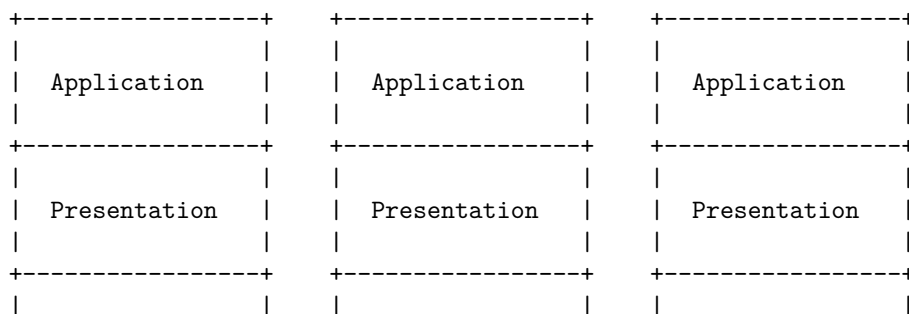


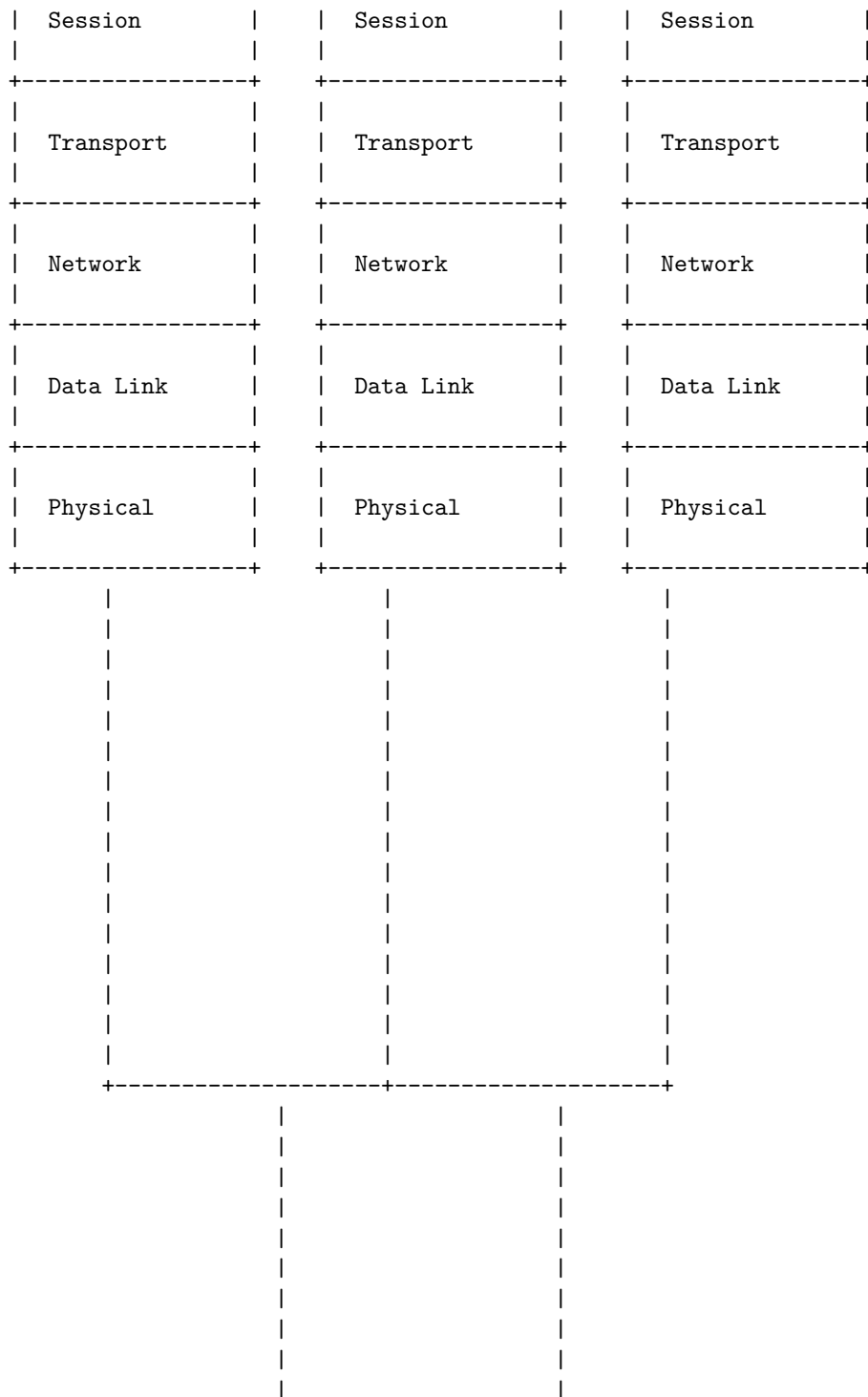


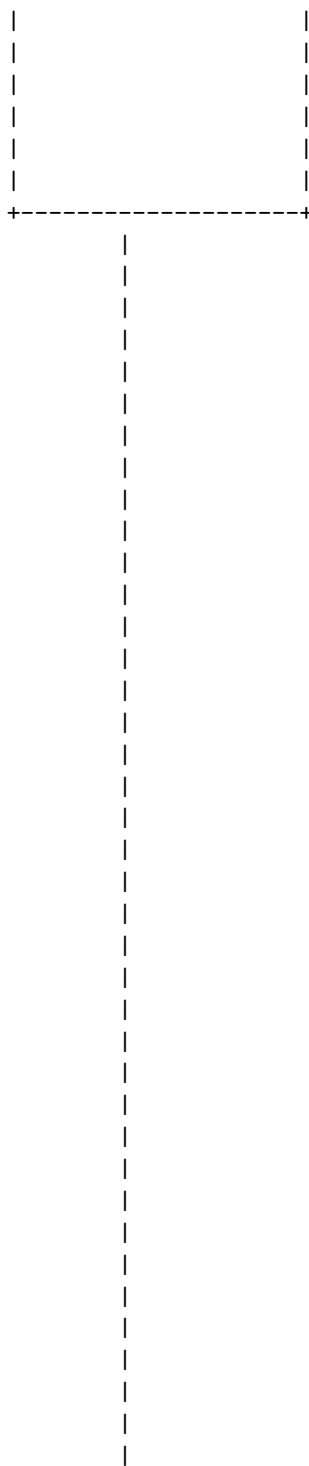
Data compression is the function of presentation layer in OSI reference model. Compression is often used to maximize the use of bandwidth across a network or to optimize disk space when saving data. Data compression allows to reduce the number of bits that needs to be transmitted on the network.

A possible ASCII diagram for data compression in application layer is:

Data compression in application layer

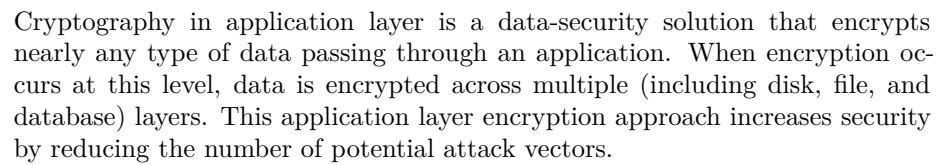




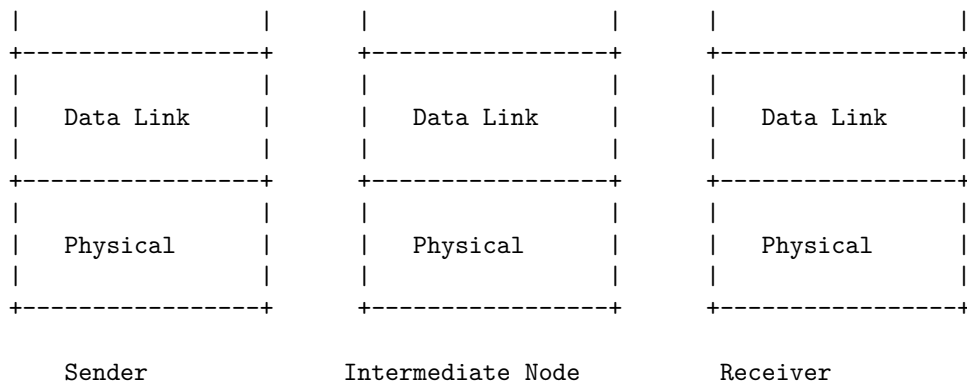


|

|



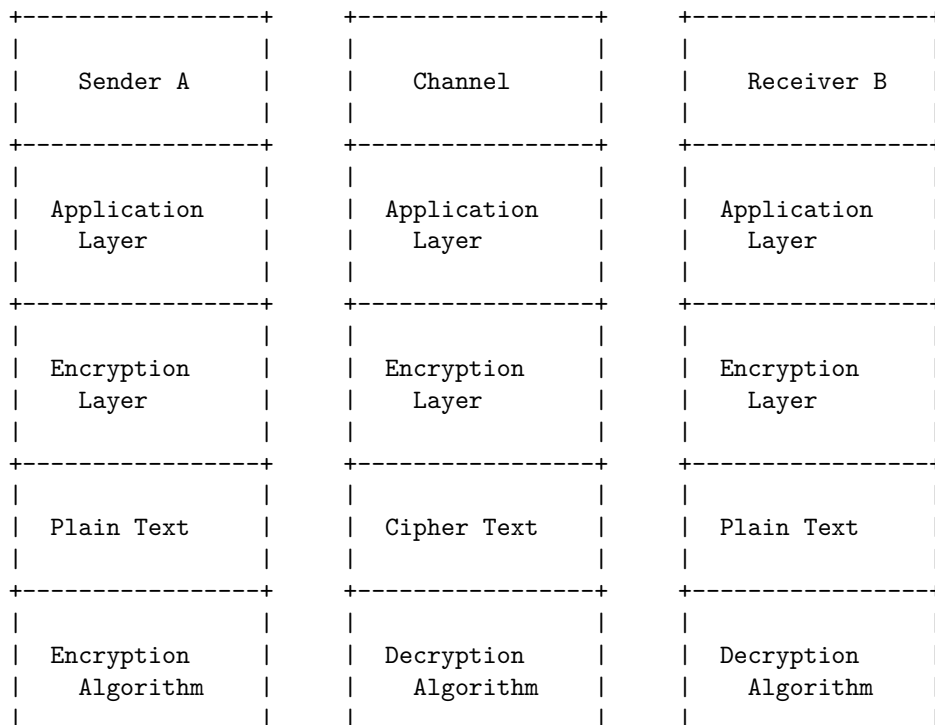
Application	Application	Application
Encryption	Encryption	Encryption
Transport	Transport	Transport
Network	Network	Network

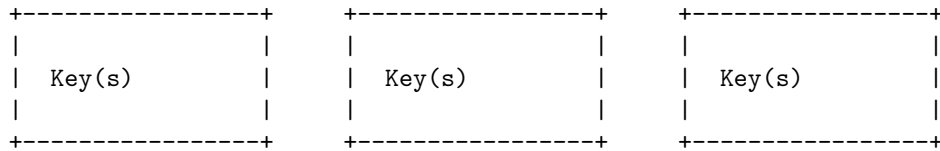


The diagram shows how data is encrypted at the application layer before being sent over the network. The intermediate node can only see the encrypted data and cannot decrypt it. The receiver can decrypt the data using the same encryption algorithm and key as the sender. This way, the data is protected from unauthorized access or modification.

Hello, I am Sydney, your AI assistant. I can help you with your query. Here is a detailed ascii diagram for basic concepts of Cryptography in application layer:

Basic concepts of Cryptography in application layer





The diagram shows the basic steps involved in cryptography in application layer. Cryptography is the process of converting plain text into cipher text, which is unintelligible and vice-versa. It provides secure communication in the presence of adversaries.

The sender A has a plain text message that he/she wants to send to the receiver B. The sender A uses an encryption algorithm and a key (or a pair of keys) to transform the plain text into cipher text. The cipher text is then transmitted over the channel, which may be a network, a wire, a radio, etc. The channel may be noisy, insecure, or unreliable.

The receiver B receives the cipher text and uses a decryption algorithm and a key (or a pair of keys) to recover the plain text from the cipher text. The decryption algorithm is usually the inverse of the encryption algorithm. The key (or the pair of keys) must be the same or related to the key (or the pair of keys) used by the sender A.

There are two main types of cryptography: symmetric key cryptography and asymmetric key cryptography. Symmetric key cryptography uses the same key for both encryption and decryption. Asymmetric key cryptography uses a pair of keys: a public key and a private key. The public key is used for encryption and the private key is used for decryption. The public key can be shared with anyone, but the private key must be kept secret.

Some examples of symmetric key cryptography are AES, DES, and RC4. Some examples of asymmetric key cryptography are RSA, ECC, and DSA .